Four Models

and a Common Fourth-Order Closure

John Henry

Physical fields, engineering tolerances,
and numerical verification

October 2026

Reading note

This monograph asks what may be reported from a numerical calculation built on a variational field model. Four actions are varied to obtain their field equations; the equations are solved numerically; and a computed state becomes a reading only once a stability estimate, a calibration, and a reporting rule are supplied. The volume computes each of these allowances in physical units and carries one scalar example through to a certified decimal.

What this volume derives. Before any reading or identification, Chapter 2 varies four actions and recovers the governing equations of each, under the stated fields and boundary sectors.

Action	Equations recovered by stationarity
First-order vacuum gravity (§2.1)	the vacuum Einstein equations, with metric compatibility
Fixed-Hamiltonian quantum evolution (§2.2)	the Schrödinger equation and its adjoint
Incompressible flow with response fields (§2.3)	Navier–Stokes momentum balance and incompressibility, with an exact stationary projection and lift
Bare Maxwell–Dirac interaction (§2.4)	the coupled Dirac and Maxwell equations, with current conservation and Gauss's constraint

In the chosen independent fields every action has degree at most three, so

$$S \xrightarrow{\text{variation}} R = DS \xrightarrow{\text{linearization}} DR = D^2 S, \qquad D^4 S = 0$$

(Theorem 2.4). Stationarity, $DS = 0$, supplies the field equations; the finite hierarchy supplies the response operators the numerical chapters solve; and the hierarchy survives a fixed affine discretization. *Proved here.* Two precisions hold throughout. *Least action* in this volume means *stationary* action, which may be a saddle and not a minimum; the fluid presentation is explicitly one. And stationarity yields these particular laws only once the action, its independent fields, and its boundary sector are given; the principle does not select them. The scalar reading of Chapter 6 and its distance from the measured constant neither establish nor erase these derivations.

The line of development. The volume follows one sequence,

$$\text{independent fields} \longrightarrow S \longrightarrow R = DS \longrightarrow DR \longrightarrow \text{computed state} \longrightarrow \text{reported reading}.$$

The four actions supply the first part (Chapter 2). Numerical solution and stability supply the middle: a residual bounds a reading only through a stability estimate. Calibration and reporting supply the last. The scalar construction is a complete finite instance, and the record of Chapter 8

places these requirements in the publication mechanism of the middle card. The shared cubic closure of the four actions removes one Taylor remainder; it does not identify one model's equations, observables, boundary conditions, or solutions with another's, and it supplies no tolerance.

Definition 0.1 (Calibrated reading)**.** A *calibrated reading* reports a quantity or statement together with the conditions under which it is obtained and compared: the admitted inputs, the operation of reading, the reference, and the allowance or criterion governing the comparison.

For a scalar numerical reading the conditions are the polynomial, the branch, the arithmetic, the scale, and the reporting cell. For an instrument they are the measurand, the response model, the reference, the operating range, and the uncertainty. For a reading of the middle card's argument they are the correspondence supplied and the mathematical properties it preserves. A tolerance is a calibrated reading's allowance, and every allowance in this volume is computed under its conditions.

The calibrated reading is the full object. Every statement of this volume is one, and its *calibration label* names the reference against which it is calibrated: *proved here*, a proof in this volume; *standard*, a cited source; *measured*, the source's declarations or a run, and the text says which; *proposed*, an identification supplied by the reader; *presence*, no reference, the item located with no fact extracted. A statement is accordingly written calibrated *proved here*, calibrated *proposed*, and so on. The labels name references; they are not a scale, and a source inspection is not an experiment. The record of Chapter 8 carries the same labels in its calibration column.

What *proposed* asserts. A proposed statement asserts that an isomorphism exists, between the numbering of the construction's descriptions by the natural numbers and the description of what it is identified with, that the machine can exploit as a computation. Both sides are finite records: the reading is a computable number, and the reference is a value with its uncertainty or a stated structure. The statement is therefore one the machine can evaluate, a case that could be false, and the comparison that would revise it is that evaluation. What no evaluation supplies is that the reference's finite record stands for what it records, as a finite halt stands for its limit.

The reporting rule. A certified interval wholly inside one reporting cell fixes the digit; a narrow interval straddling the edge does not. There is therefore a certified reading with its receipt, or there is no reading at all. This is the same rule as the companion volume's missing sign. The middle construction exhibits a record whose readings agree, checked by a proof assistant; this volume computes what any record must retain, unit by unit, for that agreement to be a calibrated reading of a stated quantity. The record chapter checks that the record carries both the reading and its receipt; its demand-level reading marks the rungs where a reading was claimed that no retained datum licenses. Every result calibrated *proved here* is also an experiment the reader can run: Appendix B gives the procedure for one run, with an excerpt of its record and its final certificate.

Prerequisites. The reader is assumed to know multivariable calculus and the calculus of variations at the level of the Euler–Lagrange equations, linear algebra including singular values, elementary numerical analysis (Newton's method, bisection, Krylov iteration), and undergraduate mechanics and electrodynamics. The field theory used in Chapter 2 is stated with its conventions where it is needed. The notation is collected after this note.

Organization. Chapter 1 separates the three derivatives the volume uses (of an action, of a beam deflection, and of a residual objective) and introduces the clamped beam as the continuing worked model. Chapter 2 derives the four field systems and proves the common closure. Chapter 3 turns a residual into an allowance on an observable through scaled coordinates, local stability, and output sensitivity. Chapter 4 gives the Newton–Krylov and multigrid solution that produces the computed state, and Chapter 5 checks the four models against analytical benchmarks. Chapter 6 carries the scalar residual through exact updates, an independent integer certificate, and its comparison with the measured constant. Chapter 7 treats calibrated instruments and the exact channels of the source construction. Chapter 8 presents the record of thirty-six replies and its press check. Chapter 9 follows the beam through dissipation, reservoir entropy, and a retained threshold count, and Chapter 10 sets out the cosmological correspondence investigated for the companion volume's shell. Appendix A gives the sources and the review correspondence, and Appendix B the reader's experiment.

Notation and conventions

Fields, units, scales, and norms are specified when introduced. The following notation collects the principal objects used throughout the monograph. Several letters carry more than one meaning in different chapters; each such letter is listed with all of its meanings, and the chapter in which a meaning applies is stated.

Symbol	Meaning
Fields, actions, and their derivatives	
$Z,\ a$	Complete list of independent fields, and its column of numerical coefficients after discretization.
$z,\ D_a$	Dimensionless solver coordinates $z = D_a^{-1}(a - a^{\mathrm{ref}})$, with diagonal component-scale matrix D_a.
$H,\ h,\ s$	Perturbation field, its coefficient direction, and the dimensionless amplitude of the straight (affine, off-shell) variation $Z + sH$.
S	Action of a selected field model, in J s (Chapters 1–2). For the other uses of S see below.
$S_0,\ \phi(s)$	Action scale and dimensionless action response $\phi(s) = S[Z + sH]/S_0$.
$R = DS$	Variational residual: the first variation of the action, whose zero is the field equation.
$DR = D^2S,\ J$	Linearized response: Newton Jacobian, tangent stiffness.
D^kS	k-th variation of the action along affine variations of the independent fields; $D^4S = 0$ for every action of Chapter 2.
$L,\ B$	Fixed linear operator and bilinear interaction in a quadratic residual $R(z) = Lz + B(z, z) - f$.
$\Phi(z),\ W$	Squared-residual objective $\Phi = \frac{1}{2}\sum_i w_i R_i^2$, with fixed positive weights $W = \mathrm{diag}(w_i)$.
$D_Z,\ D_R,\ r(z; d)$	State and residual scale matrices, and the scaled residual vector of Chapter 3, with data d.
$\sigma,\ K$	In Chapter 3: smallest singular value of the scaled Jacobian, and stability constant $K = 1/\sigma$.
The beam (the continuing worked model)	
$x,\ \xi,\ L$	Position along the beam, $\xi = x/L$, and beam length.
$q(x),\ q(x, t)$	Transverse displacement in metres; $q_{xxxx} = q^{(4)}$ is its fourth spatial derivative.
$E,\ I_b,\ EI$	Young's modulus, second moment of area (written I where no identity matrix occurs), and flexural rigidity in $\mathrm{N\,m^2}$.
$U[q],\ \Pi[q],\ E[q]$	Bending energy (J), total potential (J), and normalized bending functional U/EI ($\mathrm{m^{-1}}$).
$M_b,\ f,\ d$	Bending moment EIq'', distributed load, and imposed end displacement.

Symbol	Meaning
$K_e,\ N_i$	Element stiffness matrix and cubic Hermite shape functions.
$\zeta,\ G,\ \kappa$	Drag coefficient, energy gap above the equilibrium cubic, and its decay rate (Chapter 9).
The scalar reading	
$r(y)$	Scalar residual $13y^2 - 36y + 18$ in the normalized coordinate y (Chapter 6).
$\alpha,\ \beta,\ d$	Its larger and smaller roots $(18 \pm 3\sqrt{10})/13$, and their separation $d = \alpha - \beta$.
$x = g_x y,\ g_x$	Reported-output coordinate and the declared gain $g_x = 324/5$.
$q(x)$	Output-coordinate residual $325x^2 - 58320x + 1889568 = 324^2 r(5x/324)$.
$N,\ H$	Newton and Halley update maps.
$L_k,\ U_k$	Lower and upper walls of the bisection bracket after k halvings.
$S = 10^{24},\ X$	Reporting scale of the integer arithmetic, and the stored integer $X = Sx$.
K	In Chapter 6: the certified reporting integer, $\lfloor Sg_x L_{108} \rfloor = \lfloor Sg_x U_{108} \rfloor$.
$t,\ c,\ r$	The three source readings 5, 18, 18 of the four-model quartic $t^2(c-t)x^2 - 2ctr^2x + cr^4 = 0$.
$M,\ N_n,\ \lambda_\pm$	Click (multiplicity) matrix $\left(\begin{smallmatrix} 18 & 5 \\ 18 & 18 \end{smallmatrix}\right)$, closed-walk counts $N_n = \operatorname{tr}(M^n)$, and its eigenvalues $18 \pm 3\sqrt{10} = 13\alpha, 13\beta$.
Counts and entropies	
$\log \lambda_+$	Transition-count growth rate, in nats per clock step; no energy or temperature is attached.
$S_{\text{reservoir}},\ \Delta S$	Thermodynamic entropy of a heat reservoir, in $\mathrm{J\,K^{-1}}$, and its change over an interval (Chapter 9).
$\Sigma(t),\ \sigma$	Entropy transferred by the relaxing beam, and the entropy increment assigned to one registered pulse (Chapter 9).
$g_\sigma,\ g_n$	Threshold count $\lfloor \Sigma/\sigma \rfloor$, and the record's retained counter $g_{n+1} = g_n + r_n$.
$A/4G$	Horizon entropy of the de Sitter static patch, with G Newton's constant (Chapter 10).
Other recurring symbols	
e	An error $y - \alpha$, $q_h - q_*$, or $z - z_*$, as stated locally; in Chapter 2, the electromagnetic coupling.
λ	Eigenvalue $(\lambda_\pm,\ \lambda_j)$, adjoint solution $(J^\mathsf{T}\lambda = g)$, or line-search step, as stated locally; $\lambda_C = \hbar/(mc)$ is the reduced Compton wavelength.
γ^μ	Dirac matrices, $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I$.

Three fourth derivatives. Three different fourth derivatives occur, and they are never interchanged.

Expression	Variable changed	Object described
$D^4 S$	independent fields along affine variations	action degree
q_{xxxx}	spatial position	beam balance
$\dfrac{\mathrm{d}^4}{\mathrm{d}s^4}\Phi(z + sh)$	solver-state variation	residual objective

For the actions of Chapter 2 the first vanishes. The second equals f/EI at equilibrium for constant rigidity, and vanishes there only between loads. The third equals $12\sum_i w_i B_i(h,h)^2$ for a quadratic residual and is generally nonzero.

The several scales. *Component scales* a_{0j}, D_Z, D_R convert coefficients and residuals with units into dimensionless coordinates. The *action scale* S_0 and *energy scale* $U_0 = EId^2/L^3$ normalize an action or an energy. *Physical scales* such as L/U, ρU^2, and the de Sitter radius ℓ belong to a model. The *gain* g_x converts the normalized scalar coordinate to the reported coordinate, and the *reporting scale* $S = 10^{24}$ fixes the reporting cell of the integer arithmetic. A reported number carries the scales under which it was computed.

The several entropies. Three quantities are kept apart: a *count* or its logarithmic growth rate (dimensionless, or nats per step), the *Shannon uncertainty* of a message (bits or nats, defined only with a probability distribution), and *thermodynamic entropy* ($\mathrm{J\,K^{-1}}$). A count becomes an entropy only through an explicit calibration, such as the one of Chapter 9.

Euclidean vector norms and their induced operator norms are used unless a weighted norm is declared. Angles are in radians. A star denotes an adjoint for the stated pairing, and T a transpose.

Contents

Reading note **i**

Notation and conventions **iv**

1 Physical variables, action, and engineering response **1**
1.1 Field components and calibrated perturbations . . . 1
1.2 Why the perturbation expansion terminates . . . 2
1.3 Force residuals, tangent response, and least squares . . . 3
1.4 The beam model . . . 4
1.5 The clamped beam: energy gap and dimensional reactions . . . 5
1.6 Cubic beam elements and spline reactions . . . 6
1.7 Converting a beam force residual into deflection error . . . 7

2 Four field models, their actions, and physical scales **9**
2.1 First-order vacuum gravity . . . 10
2.2 Fixed-Hamiltonian Schrödinger evolution . . . 13
2.3 Incompressible Navier–Stokes with response fields . . . 16
2.4 Bare QED on a fixed background . . . 18
2.5 The common conclusion and its domain . . . 21
2.6 The derivative of the modified field equations . . . 21

3 From closure to an engineering tolerance **24**
3.1 The framework of a calibrated numerical reading . . . 24
3.2 Put the equations and their tolerances in compatible units . . . 25
3.3 Sensitivity through the smallest Jacobian singular value . . . 26
3.4 Pressure and multiplier constraints are part of the solve . . . 26
3.5 Exact remainder and Jacobian-free evaluation . . . 27
3.6 A sharp cubic interpolation certificate . . . 28
3.7 An error budget for a physical reading . . . 28
3.8 Two readers, decimal decisions, and cycle closure . . . 29

4 Newton–Krylov solution and multigrid correction **31**
4.1 Start with the momentum and volume balances . . . 31
4.2 A matrix-free tangent has a finite error floor . . . 32
4.3 Inexact Newton: spend accuracy where it changes the answer . . . 32
4.4 Krylov vectors approximate a correction, not a field theory . . . 33
4.5 An exact two-step GMRES check . . . 34
4.6 Why multiple grid scales help . . . 35
4.7 An exact three-unknown coarse-grid calculation . . . 36
4.8 A finite solution and acceptance procedure . . . 37
4.9 Verification by a prescribed exact flow . . . 38

5 Verification against physical field equations **40**

5.1 Norms and reference quantities . . . 40
5.2 Gravity: a nonconstant flat-coordinate benchmark . . . 41
5.3 Schrödinger: phase, spectral truncation, and detector error . . . 41
5.4 Navier–Stokes: an exact viscous-decay experiment . . . 43
5.5 QED: electromagnetic constraints and fermion block checks . . . 44
5.6 Protocol for runs . . . 45

6 A worked scalar reading **47**
6.1 The residual, its branch, and its units . . . 47
6.2 What the exact updates prove . . . 49
6.3 The integer updates actually compared . . . 50
6.4 A certificate obtained without either update . . . 51
6.5 Residual size, sensitivity, and reproducibility . . . 52
6.6 Coefficient uncertainty and calibrated parameter sensitivity . . . 53
6.7 Where the coefficients come from, and their neighbours . . . 53
6.8 Resolution, stopping, and computational work . . . 55

7 Calibrated instruments and common readings **57**
7.1 The measurement chain and the quantity being measured . . . 57
7.2 A quantitative calibration and digitization budget . . . 58
7.3 A common indication and an independent reference . . . 59
7.4 Exact count channels and storage consistency . . . 59
7.5 An erased channel and the auxiliary record . . . 60
7.6 A finite transition network and the scalar root . . . 61
7.7 Software instruments, blanks, and independent checks . . . 61

8 The record, the thirty-six replies, and the press check **63**
8.1 The record . . . 63
8.2 The thirty-six replies . . . 65
8.3 The lifts . . . 73
8.4 The press check . . . 75
8.5 The hypothesis the apparatus tests . . . 77

9 Dissipation, entropy, and a retained count **78**
9.1 The physical beam and its dissipative equation . . . 78
9.2 Power, boundary work, and the decay estimate . . . 79
9.3 A mode calculation with declared units . . . 80
9.4 What a time discretization dissipates . . . 80
9.5 Heat transfer and an entropy calibration . . . 81
9.6 A threshold reader and its retained count . . . 81
9.7 The physical thesis and its controlling estimate . . . 82

10 The shell, the counter, and the two holographies **84**
10.1 The correspondence investigated . . . 84
10.2 Standard facts . . . 85
10.3 What is exact . . . 86
10.4 The proposed correspondence . . . 86

A Sources and the review correspondence **88**
A.1 Scope and snapshot . . . 88
A.2 The scalar reading (Chapter 6) . . . 88
A.3 Count channels, the erased channel, and the transition matrix (Chapter 7) . . . 89
A.4 Work measurements (Section 7.7) . . . 89
A.5 The census and register names behind the count channels . . . 90
A.6 The review: page to reply map . . . 90

A.7 Provenance of the readings handed over by name . . . 92
A.8 The typeset record: field map . . . 92
A.9 The floor built by hand and the press check . . . 93
A.10 The hypothesis . . . 93
A.11 Reproduction . . . 93
A.12 Stanzas and results . . . 94

B The reader's experiment **96**
B.1 Setup: the certified reading . . . 96
B.2 Procedure . . . 97
B.3 The exhibited record . . . 97
B.4 What the run detects . . . 98
B.5 A bench counterpart . . . 98
B.6 Every proof as a run . . . 98

Bibliography **100**

Chapter 1

Physical variables, action, and engineering response

A tolerance belongs to a quantity, a unit, an operating range, and a measurement procedure. A deflection tolerance of one micrometre and a relative pressure residual of one part per million cannot be compared by looking at their printed exponents. The same requirement applies to a fourth derivative: the independent variable, the dependent quantity, and its normalization have to be stated before a zero or a small number has an engineering interpretation.

Three fourth derivatives occur in this volume. The first is taken along a controlled change of independent fields in a variational formulation; its vanishing is an exact termination of a perturbation expansion. The second is the spatial fourth derivative of a beam deflection, which expresses a force balance. The third is taken along a change of solver state in the squared residual left by a numerical solution (page v). The first and third give the derivative hierarchy $S \to R = DS \to DR$ and its objective Φ; the second is carried by a clamped Euler–Bernoulli beam, the continuing worked model of the volume, whose parameters are fixed in Section 1.4.

1.1 Field components and calibrated perturbations

For a fluid calculation the unknowns may be velocity components in metres per second and pressure in pascals. For an elastic calculation they may be displacement in metres and rotations in radians. For a wave calculation they include complex amplitudes, with real and imaginary components stored separately. Gravity and electrodynamics add metric, connection, potential, and spinor components with their chosen coordinate and unit conventions. Chapter 2 specifies the actual independent variables in each selected action.

After a mesh or a mode expansion is chosen, collect the independent coefficients in a column $a = (a_1, \dots, a_N)^T$. Before differentiating, fix the component scales $a_{0j} > 0$ and the reference coefficients a_j^{ref}. The dimensionless solver coordinates are

$$z_j = \frac{a_j - a_j^{\mathrm{ref}}}{a_{0j}}, \qquad a = a^{\mathrm{ref}} + D_a z, \qquad D_a = \operatorname{diag}(a_{01}, \dots, a_{0N}).$$

The scales are fixed for the perturbation being tested. A velocity increment and a pressure increment therefore enter the same numerical column only after their separate dimensional scales have been

applied. Changing the unit from metres to millimetres changes D_a and the printed coordinates, while preserving the physical field.

Choose a trial field Z and a perturbation field H with the same units and boundary restrictions. Here Z denotes the complete list of independent fields; its numerical coefficients are the entries of a. The straight perturbation is $Z + sH$, with dimensionless amplitude s. The word straight means that the independent coefficients change linearly in s; the perturbed fields need not solve the equations of motion. Such a variation is called off shell in physics. A perturbation of a metric must stay nondegenerate over the interval used. A perturbation of a clamped displacement has zero displacement and zero slope at its fixed endpoints.

Calibration here fixes the fields, component scales, sign conventions, boundary data, and output conversion used for the comparison; it inserts no term in a field equation. In the fermion calculation the anticommutation convention is also specified explicitly, and a calibration cannot turn anticommuting coefficients into ordinary commuting real numbers.

For a dynamical action S, choose an action scale $S_0 > 0$ with units of joule seconds and define

$$\phi(s) = \frac{S[Z + sH]}{S_0}. \tag{1.1}$$

Then ϕ and s are dimensionless, so their fourth derivative is dimensionless. The scale S_0 may be a declared characteristic energy times a characteristic time; a quantum calculation may use $\hbar$. A static elastic energy, such as the beam energy of Section 1.4, instead has units of joules and requires its own energy scale. A static energy and a dynamical action have different units even when both are minimized or made stationary by variation.

1.2 Why the perturbation expansion terminates

Consider a real-coefficient discrete action consisting of constant, linear, quadratic, and cubic terms,

$$S(a) = S_c + \sum_i b_i a_i + \frac{1}{2}\sum_{ij} K_{ij} a_i a_j + \frac{1}{6}\sum_{ijk} C_{ijk} a_i a_j a_k.$$

The coefficients include the quadrature weights, differential matrices, material constants, and fixed backgrounds. In this displayed real example they may be symmetrized without changing S. Substitution of $a + s\delta a$ gives

$$S(a + s\delta a) = c_0 + c_1 s + c_2 s^2 + c_3 s^3.$$

Each original term has at most three field factors. Selecting either the unperturbed value or the perturbation in each factor cannot produce a fourth power of s. A fixed spatial derivative is linear and preserves this count. Integration with fixed weights also preserves it.

Theorem 1.1 (Termination of the action perturbation expansion)**.** *If the selected action contains at most three independent-field factors in every term, with fixed coefficients and linear differential operators, then $D^4 S = 0$ and $\phi^{(4)} = 0$. Its variational equations, expressed with fixed coefficient scales, contain at most quadratic field terms, so their third perturbation derivative vanishes: $D^3 R = 0$.*

Proof. The substitution above gives the complete amplitude expansion. Differentiating it four times gives zero. The same count holds when four independent perturbation amplitudes are introduced:

there is no term containing their product. Differentiating the action once to form the variational equations removes one field factor. Multiplication by fixed scaling matrices does not add one. The resulting residual is at most quadratic, and three further amplitude derivatives give zero. □

For the real coefficient model, the residual is explicitly

$$R_i(a) = b_i + \sum_j K_{ij} a_j + \frac{1}{2} \sum_{jk} C_{ijk} a_j a_k, \qquad J_{ij}(a) = \frac{\partial R_i}{\partial a_j}.$$

The matrix J is the tangent stiffness or Newton Jacobian. In a continuum calculation the corresponding residual is the coefficient of an arbitrary virtual displacement or field perturbation after integration by parts. Its boundary terms must be included or eliminated by the stated boundary conditions. In QED the expansion is performed with the fermion factors in their retained order. The zero fourth variation is then an ordered-coefficient identity in the declared fermion formulation, rather than a norm estimate for quantum states.

The hypothesis of Theorem 1.1 is a property of a formulation. Eliminating an independent connection, solving a constraint through nonlinear coordinates, introducing a state-dependent normalization, or changing material coefficients with the unknown field can introduce further powers or denominators. For instance, a displacement-dependent tangent stiffness may still be a useful engineering model, but it requires its own perturbation count.

Example 1.2 (Changing the perturbation coordinate)**.** The dimensionless response $\phi(s) = s^3$ has zero fourth derivative. If the controller instead reports $s = t^2$, the reported response is t^6 and its fourth derivative with respect to t is $360t^2$. The independent perturbation amplitude is part of the test. Physical time is another variable: a harmonic displacement $q(t) = q_0 \cos \omega t$ satisfies $q^{(4)} = \omega^4 q$, even when its mechanical action is quadratic.

1.3 Force residuals, tangent response, and least squares

Many discretized field balances have linear and quadratic terms. In calibrated coefficient coordinates write

$$R(z) = Lz + B(z, z) - f, \tag{1.2}$$

where L is a fixed matrix or linear differential operator and $B_i(z, w) = \sum_{jk} B_{ijk} z_j w_k$ is a bilinear interaction. The fluid convective term is an example, with $B(u, w) = (u \cdot \nabla)w$ before pressure and other terms are assembled. A direct expansion gives

$$DR(z)h = Lh + B(h, z) + B(z, h), \tag{1.3}$$

$$D^2 R(z)[h, k] = B(h, k) + B(k, h), \qquad D^3 R = 0. \tag{1.4}$$

Both interaction terms are part of the usual Newton linearization. For fluid velocity they are $(h \cdot \nabla)u$ and $(u \cdot \nabla)h$. Retaining the second perturbation derivative is useful for an error estimate or a higher-order update; the exact Jacobian already contains both convective contributions.

Residual components must be scaled before their squares are added. A force imbalance in newtons, a divergence defect in inverse seconds, and a boundary displacement error in metres are separate channels. After division by their declared reference values, use fixed positive weights w_i and define

$$\Phi(z) = \frac{1}{2} \sum_i w_i R_i(z)^2, \qquad \|R\|_W^2 = \sum_i w_i R_i^2.$$

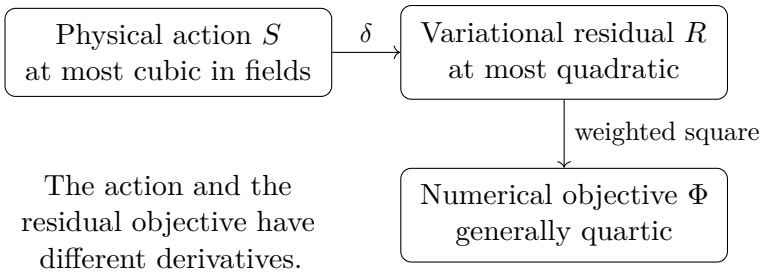


Figure 1.1: Action, field balance, and numerical objective. The field units, residual scales, and fixed weights determine the meaning of each derivative.

The weights can include cell-volume fractions or sensor weighting. For the unit-weight convention used in the displayed equation below, $R(z + sh) = R(z) + sJ(z)h + s^2B(h, h)$ implies

$$\frac{\mathrm{d}^4}{\mathrm{d}s^4}\Phi(z + sh) = 12\,\|B(h, h)\|_2^2\,. \tag{1.5}$$

With fixed weights the right side is $12\sum_i w_i B_i(h, h)^2$. Thus a squared residual can be quartic while the physical action is cubic. The distinction affects a numerical minimization: stationarity of Φ gives $J^T W R = 0$, whereas a solved balance requires $R = 0$. The elementary residual $R(z) = z^2 + 1$ has a least-squares stationary point at $z = 0$ and no real zero. A solver must report the balance defect as well as a change in its objective.

Figure 1.1 summarizes the hierarchy.

1.4 The beam model

Let x measure distance along a straight beam of length L, and let $q(x)$ be transverse displacement; both are in metres. In the Euler–Bernoulli model the slope is small, material response is linear elastic, and transverse shear deformation is neglected. Young's modulus E has units $\mathrm{N\,m^{-2}}$; the cross-section's second moment of area I_b has units $\mathrm{m^4}$. Write $EI = EI_b$ for the constant flexural rigidity, with units $\mathrm{N\,m^2}$. The approximate curvature is q'', with units $\mathrm{m^{-1}}$.

The bending energy and total potential for a distributed transverse load $f(x)$ are

$$U[q] = \frac{EI}{2}\int_0^L (q'')^2\,\mathrm{d}x, \qquad \Pi[q] = U[q] - \int_0^L fq\,\mathrm{d}x.$$

Here f has units $\mathrm{N\,m^{-1}}$ and both terms in Π are joules. The total potential is quadratic in q with fixed coefficients, so Theorem 1.1 applies to it, and its residual is linear. Choose the energy-conjugate bending moment $M_b = EIq''$; its sign fixes the following moment and shear convention. Twice integrating the virtual work gives

$$\delta\Pi = [M_b\delta q' - M_b'\delta q]_0^L + \int_0^L (M_b'' - f)\delta q\,\mathrm{d}x.$$

For clamped ends, $\delta q = \delta q' = 0$ at both supports. The interior balance is $(EIq'')'' = f$, or $EIq^{(4)} = f$ for constant rigidity. At an unloaded free end the independent virtual displacement and rotation instead give $M_b' = 0$ and $M_b = 0$. These force, moment, constitutive, and boundary relations are

the engineering content of the variational calculation [41]. With zero distributed load and constant rigidity, $q^{(4)} = 0$ between loads.

The normalized bending functional is

$$E[q] = \frac{U[q]}{EI} = \frac{1}{2}\int_0^L (q'')^2 \,\mathrm{d}x. \tag{1.6}$$

This normalization removes the stiffness; unless position and displacement are also nondimensionalized, $E[q]$ has units of inverse metres. It is distinct from Young's modulus, which always occurs in EI in the beam formulas. A fully dimensionless energy uses $\xi = x/L$, $q = d\widehat{q}$, and scale $U_0 = EId^2/L^3$, where d is a declared displacement scale.

Declared parameters. The numerical examples of this volume use

$$EI = 1\,\mathrm{N\,m^2}, \qquad L = 1\,\mathrm{m}, \qquad d = 10^{-3}\,\mathrm{m},$$

with d the imposed end displacement of Section 1.5. These are design inputs of the stated linear model, not measured material data.

1.5 The clamped beam: energy gap and dimensional reactions

Proposition 1.3 (Exact clamped-beam energy gap)**.** *Let q_* be the unloaded cubic satisfying four fixed endpoint displacements and slopes. For a perturbation v with $v = v' = 0$ at both ends and finite integrated squared curvature,*

$$U[q_* + v] - U[q_*] = \frac{EI}{2}\int_0^L (v'')^2 \,\mathrm{d}x, \qquad E[q_* + v] - E[q_*] = \frac{1}{2}\left\|v''\right\|_{L^2}^2.$$

Consequently q_* is the unique minimum-energy deflection for those data.

Proof. Expanding the bending energy leaves the cross term $EI\int q_*''v''$. Its boundary terms vanish, and its interior term is $EI\int q_*^{(4)}v = 0$. The remaining squared-curvature term is nonnegative. If it vanishes, $v'' = 0$, so v is a straight displacement and rotation. The homogeneous clamp conditions force both to zero. □

Impose $q(0) = q'(0) = q'(L) = 0$ and $q(L) = d$. Solving the four coefficient equations gives

$$q_*(x) = d(3\xi^2 - 2\xi^3), \qquad \xi = x/L, \qquad U[q_*] = \frac{6EId^2}{L^3}.$$

The curvature is $(6d/L^2)(1 - 2\xi)$ and the third derivative is $-12d/L^3$. The support moments have magnitude $6EI|d|/L^2$ and the transverse reaction magnitude is $12EI|d|/L^3$. Differentiating the energy with respect to the imposed displacement gives that same reaction: $\partial U/\partial d = 12EId/L^3$. Quasistatic work from zero displacement to d is the integral of this linear reaction, $6EId^2/L^3$, which equals the stored bending energy.

With the declared parameters of Section 1.4, the stored energy is $6 \times 10^{-6}\,\mathrm{J}$, the reaction magnitude is $0.012\,\mathrm{N}$, and each end-moment magnitude is $0.006\,\mathrm{N\,m}$. The maximum slope is $1.5d/L = 0.0015$. In dimensionless coordinates $d = L = 1$, the normalized energy is the previous scalar value $E[q_*] = 6$.

An applied distributed load changes the solution order. For example, a simply supported beam with constant load f_0 has

$$q(x) = \frac{f_0}{24EI}(L^3x - 2Lx^3 + x^4), \qquad q(L/2) = \frac{5f_0L^4}{384EI}.$$

Direct differentiation gives $EIq^{(4)} = f_0$; both endpoint displacements and endpoint bending moments vanish. The quartic term is the required load response. A variable stiffness likewise requires $(EI(x)q'')'' = f$. Neither case permits replacing that balance by $q^{(4)} = 0$ without retaining its load or material terms.

1.6 Cubic beam elements and spline reactions

A two-node Euler–Bernoulli element uses the physical degrees of freedom $d_e = (q_0, \theta_0, q_L, \theta_L)^T$. Its displacement is

$$q = N_1q_0 + LN_2\theta_0 + N_3q_L + LN_4\theta_L,$$

where

$$N_1 = 1 - 3\xi^2 + 2\xi^3, \quad N_2 = \xi - 2\xi^2 + \xi^3,$$
$$N_3 = 3\xi^2 - 2\xi^3, \qquad N_4 = -\xi^2 + \xi^3.$$

The factors of L are essential because rotations and displacements have different units. Inserting the interpolation into U gives $U_e = \frac{1}{2}d_e^T K_e d_e$, with

$$K_e = \frac{EI}{L^3}\begin{pmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{pmatrix}.$$

The force entries of $K_e d_e$ have units of newtons and its moment entries have units of newton metres. A free element has two zero-energy motions, translation and rigid rotation. Suitable fixed supports remove these motions and make the assembled elastic solve uniquely determined. The same element exactly represents the unloaded clamped cubic of Section 1.5. Exact representation of this case is a useful verification test, while a loaded quartic still needs mesh refinement or an enriched interpolation.

For several interpolation knots, the deflection and slope are retained at adjoining elements. Minimization with free interior slopes gives continuous bending moment when there is no applied concentrated couple. For uniform stiffness this is continuity of q''. A jump in EIq''' represents a concentrated transverse reaction. Consequently a curve can be cubic on every open segment while its global fourth spatial derivative includes point-load terms. Natural spline endpoint curvature is zero when the endpoint slopes are free and no endpoint moment is applied. These are the classical spline minimum-energy conditions [10, 9]; their engineering interpretation is moment and shear balance at the joins.

The Galerkin reading. The element matrix above is a Galerkin matrix. With $a(u, v) = \int EI\, u''v''\, dx$ and a load functional ℓ, the method seeks q_h in the span V_h of the shape functions with

$$a(q_h, v) = \ell(v) \qquad \text{for every } v \in V_h,$$

so the residual is orthogonal to the same space the answer is built from: trial space equals test space [34]. *Standard.* Every entry of K_e is $a(N_j, N_i)$ up to the stated factors of L, the integral over $[0,1]$ of a product of two linear polynomials in ξ, and therefore a rational number: 12, 6, 4, 2 and their signs. *Proved here*, by integrating the products. No limit is taken. The continuous problem enters only through the bilinear form evaluated on a finite basis, and the solve holds nothing but those rationals. When the loads are concentrated forces and couples at the nodes, the exact deflection is cubic on each element with the same nodal values and slopes, so it lies in V_h and the Galerkin solution equals it. *Proved here*, by uniqueness of the supported solve.

The multiplicity matrix of Section 7.6, $M = \begin{pmatrix} 18 & 5 \\ 18 & 18 \end{pmatrix}$, admits the same reading on a two-function basis, one function per state. It is not symmetric, and a nonsymmetric bilinear form on one space already gives a nonsymmetric matrix; what M carries is an exact similarity. With $D = \operatorname{diag}(1, \sqrt{18/5})$,

$$D^{-1} M D = \begin{pmatrix} 18 & 3\sqrt{10} \\ 3\sqrt{10} & 18 \end{pmatrix},$$

equivalently $WM = M^T W$ for $W = D^{-2} = \operatorname{diag}(1, 5/18)$: M is self-adjoint for the weighted pairing $\langle x, y \rangle_W = x^T W y$, and its spectrum $18 \pm 3\sqrt{10}$ is real. *Proved here*, by multiplication. The similarity $M \mapsto D^{-1} M D$ is not the congruence $A \mapsto P^T A P$ of a bilinear form under a common basis change. The variational reading is separate: that the two states are basis functions of a problem whose form, in the pairing W, has matrix M, with trial and test spaces that the transposed orientation exchanges (Petrov–Galerkin). *Proposed.*

1.7 Converting a beam force residual into deflection error

For one explicit tolerance calculation, suppose the true and computed beam deflections have identical fixed clamps and enough smoothness for integration by parts. Put $e = q_h - q_*$ and let $r = EIe^{(4)}$ be the remaining distributed force imbalance. A force residual is useful only if this equation, the material coefficient, and its homogeneous boundary conditions are the ones being tested. Multiplying by e gives

$$EI \, \|e''\|_{L^2}^2 = \int_0^L re \, \mathrm{d}x.$$

Two one-dimensional Poincaré estimates, or a sine-series estimate with these homogeneous endpoints, give $\|e\|_{L^2} \le (L/\pi)^2 \, \|e''\|_{L^2}$. Hence

$$\|e''\|_{L^2} \le \frac{L^2}{EI\pi^2} \, \|r\|_{L^2}, \qquad \|e\|_{L^2} \le \frac{L^4}{EI\pi^4} \, \|r\|_{L^2}.$$

A point displacement follows from $e(x) = \int_0^x (x - s) e''(s) \, \mathrm{d}s$ and Cauchy–Schwarz. Writing $r_{\mathrm{rms}} = (L^{-1} \int_0^L r^2 \, \mathrm{d}x)^{1/2}$ yields the conservative engineering bound

$$\max_x |e(x)| \le \frac{L^4}{\sqrt{3}\, EI\pi^2} \, r_{\mathrm{rms}}.$$

Its units are metres. With the declared parameters of Section 1.4, $r_{\mathrm{rms}} \le 10^{-3} \, \mathrm{N\,m^{-1}}$ gives a displacement bound of approximately $5.85 \times 10^{-5} \, \mathrm{m}$. A one-micrometre specification would require a proportionally smaller residual or a sharper, verified response estimate.

This calculation assumes that concentrated reactions and boundary errors have already been accounted for. An element join with an unbalanced shear can have zero open-element fourth derivative and a nonzero point force. Ignoring that force would invalidate the load residual used above. The engineering tolerance therefore records the interior balance, interface forces and moments, support data, material uncertainty, and the displacement output separately.

The beam calculation has every element the field models will need: an action, a residual obtained by variation, a response operator, and a stability estimate converting the residual into an output allowance. For the beam the action was given and its degree was evident. For a field model Theorem 1.1 applies only once an action, its independent fields, and its boundary sector are chosen with at most three field factors in every term; the next chapter supplies four such actions.

Chapter 2

Four field models, their actions, and physical scales

Chapter 1 set up the derivative hierarchy $S \to R = DS \to DR = D^2S$ and showed that it terminates when the action has degree at most three in its independent fields. A residual is the first variation of an action, so the hierarchy needs actions to act on. This chapter supplies four, derives their stationary equations, and counts their degree.

The fields represent spacetime geometry, a quantum wavefunction, an incompressible viscous fluid, and electromagnetic interaction with a charged spinor. Each realization depends on a choice of independent variables. Retaining a connection makes the gravitational action cubic; retaining response fields gives a cubic stationary formulation of viscous flow. The fixed-Hamiltonian Schrödinger action is quadratic, while the bare Maxwell–Dirac action has a cubic interaction. These statements concern the actions before solving constraints, eliminating fields, or passing to an effective theory. The derivations distinguish three operations: adjoining auxiliary fields, projecting a stationary system onto its physical fields, and eliminating those auxiliaries inside an action. Projection can preserve the solution set without preserving polynomial degree, a norm, or a variational minimum. Throughout, derivatives of fields count linearly: replacing Z by $Z + sH$ also replaces ∂Z by $\partial Z + s\partial H$.

Presentation	Independent fields	Action degree
Vacuum gravity	Densitized inverse metric, torsion-free connection	≤ 3
Schrödinger	Complex wavefunction and its conjugate variation	≤ 2
Navier–Stokes	Velocity, pressure, two response fields	≤ 3
Bare QED	Gauge potential, spinor, independent adjoint spinor	≤ 3

The boundary sectors are part of the presentations. Compactly supported variations suffice for the local field equations. Whenever an endpoint is free, its boundary term is retained explicitly. A new boundary action changes the variational problem and receives its own degree count.

The middle card's two reading directions name gravity and fluid response *covariant*, and the wavefunction and QED presentations *contravariant*. Raising or lowering an index does not turn one stationary system into another.

2.1 First-order vacuum gravity

2.1.1 Independent fields and admissible variations

Work on a four-dimensional spacetime region admitting a smooth Lorentzian metric. In a coordinate chart write $g = \det(g_{\mu\nu}) < 0$ and introduce the symmetric contravariant tensor density

$$h^{\mu\nu} = \sqrt{-g}\, g^{\mu\nu}.$$

The independent fields are h and a torsion-free affine connection $\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$. Nondegeneracy and Lorentzian signature define the admissible open sector for h; affine variations remain in it for sufficiently small parameter intervals. The independently varied connection is not assumed to be metric compatible.

The difference C of two connections is a tensor, so $(h, \Gamma) + s(k, C)$ is a legitimate path of independent fields, with a coordinate-independent meaning. Here s is a dimensionless perturbation amplitude, not proper time, coordinate distance, or an additional spacetime coordinate. For example, one may choose k and LC to have unit size in a declared component norm, for a reference length L, so that $|s|$ measures the imposed change relative to those field scales.

2.1.2 Action and boundary sector

With a fixed nonzero normalization suppressed, take

$$S_{\mathrm{GR}}[h, \Gamma] = \int h^{\mu\nu} R_{\mu\nu}(\Gamma)\, \mathrm{d}^4 x, \tag{2.1}$$

where our curvature convention is

$$R_{\mu\nu}(\Gamma) = \partial_\lambda \Gamma^{\lambda}_{\mu\nu} - \partial_\nu \Gamma^{\lambda}_{\mu\lambda} + \Gamma^{\lambda}_{\lambda\rho}\Gamma^{\rho}_{\mu\nu} - \Gamma^{\lambda}_{\nu\rho}\Gamma^{\rho}_{\mu\lambda}.$$

The density makes the coordinate integral invariant. Its two kinds of monomial are $h\,\partial\Gamma$ and $h\Gamma\Gamma$, of degrees two and three. This is the first-order polynomial mechanism used in Deser's derivation; the compatibility calculation below fixes the precise torsion-free sector used here [11].

For the local field equations the variations have compact support. Fixing the independent connection at a boundary instead removes the corresponding surface variation.

2.1.3 First variation and stationary equations

For a symmetric variation $k^{\mu\nu} = \delta h^{\mu\nu}$, stationarity immediately gives

$$R_{(\mu\nu)}(\Gamma) = 0. \tag{2.2}$$

For the connection variation use

$$\delta R_{\mu\nu} = \nabla_\lambda \delta\Gamma^{\lambda}_{\mu\nu} - \nabla_\nu \delta\Gamma^{\lambda}_{\mu\lambda}.$$

Integration by parts with compact support is legitimate for the weight-one density h. Symmetrizing in the two lower indices of $\delta\Gamma$ yields

$$-\nabla_\lambda h^{\mu\nu} + \frac{1}{2}\delta^{\nu}_{\lambda} D^{\mu} + \frac{1}{2}\delta^{\mu}_{\lambda} D^{\nu} = 0, \qquad D^{\mu} = \nabla_\rho h^{\mu\rho}. \tag{2.3}$$

Contracting λ with ν gives $3D^{\mu}/2 = 0$ in four dimensions. Thus the connection equation is exactly $\nabla_\lambda h^{\mu\nu} = 0$.

2.1.4 Projection, lift, or equivalence

To recover ordinary metric compatibility, put $Q_{\lambda\mu\nu} = \nabla_\lambda g_{\mu\nu}$ and $q_\lambda = g^{\mu\nu}Q_{\lambda\mu\nu}$. The density derivative gives

$$0 = \nabla_\lambda h^{\mu\nu} = \sqrt{-g}\left(-g^{\mu\alpha}g^{\nu\beta}Q_{\lambda\alpha\beta} + \frac{1}{2}q_\lambda g^{\mu\nu}\right).$$

Lowering the free indices gives $Q_{\lambda\mu\nu} = q_\lambda g_{\mu\nu}/2$. Its trace is $q_\lambda = 2q_\lambda$, so $q = Q = 0$. The unique torsion-free metric-compatible connection is therefore the Levi–Civita connection of g. Its Ricci tensor is symmetric, and (2.2) becomes $R_{\mu\nu}(g) = 0$. Taking the trace shows this is equivalent to $R_{\mu\nu} - g_{\mu\nu}R/2 = 0$ in this dimension.

Proposition 2.1 (Vacuum stationary projection)**.** *For smooth nondegenerate Lorentzian h and torsion-free Γ, the interior stationary equations of* (2.1) *are equivalent to the vacuum Einstein equations with $\Lambda = 0$, together with $\Gamma = \Gamma(g)$. In these independent variables, $D^4 S_{\mathrm{GR}} = 0$.*

Proof. The two variations and contractions above give the forward implication. Conversely, a vacuum metric and its Levi–Civita connection satisfy both Euler–Lagrange equations. The degree conclusion follows from the displayed monomials and Theorem 1.1. □

The proposition is a local stationary equivalence; it neither fixes coordinates nor supplies an evolution estimate. Matter fields, torsion, or an unrestricted connection change the hypotheses and can change both the degree and the auxiliary equations. The cosmological term is absorbed instead by restricting the admissible variations.

The cosmological term as a residue

De Sitter space is the vacuum solution with a positive cosmological constant, and the slice read in Chapter 10 needs one. In the densitized variables the cosmological volume term is $\Lambda\sqrt{-g} = \Lambda(-\det h)^{1/2}$, the square root of a quartic polynomial in the components of h, so adding it to (2.1) would take the action outside the degree count of Theorem 2.4. Fixing the volume element instead of varying it makes the cosmological constant an integration constant of the stationary equations, which is the unimodular presentation of gravity [39, 21, 40], and keeps the polynomial unchanged.

Since $\det(h^{\mu\nu}) = (\sqrt{-g})^4 \det(g^{\mu\nu}) = g$, the condition $\sqrt{-g} = 1$ in the chosen chart is $\det h = -1$. Call the set of nondegenerate Lorentzian h with $\det h = -1$ the unimodular sector. It is a smooth hypersurface of the field space, and its tangent directions at h are the densities k with $h_{\mu\nu}k^{\mu\nu} = 0$, where $h_{\mu\nu}$ denotes the matrix inverse of $h^{\mu\nu}$; on the sector $h_{\mu\nu} = g_{\mu\nu}$. The action (2.1) is the same cubic polynomial in (h, Γ) as before. Only the admitted variations change: Γ is varied freely, h is varied along the sector's tangent directions. The fourth variation of Proposition 2.1 concerns that polynomial along affine paths of the ambient field space and still vanishes.

Proposition 2.2 (The cosmological constant as an integration constant)**.** *On the unimodular sector, with Γ varied freely and h varied along tangent directions, the interior stationary equations of* (2.1) *are*

$$\Gamma = \Gamma(g), \qquad R_{\mu\nu}(g) - \tfrac{1}{4}\, g_{\mu\nu}R(g) = 0.$$

The contracted Bianchi identity then forces $\partial_\nu R = 0$, so there is a constant Λ with $R = 4\Lambda$, and the equations are

$$R_{\mu\nu}(g) = \Lambda\, g_{\mu\nu}.$$

The constant is not a coefficient of the action; it is a residue of the integration, fixed by data and carried unchanged by the solution. The fourth variation of the action is zero as in Proposition 2.1.

Proof. The variation in Γ is unrestricted and gives, exactly as before, $\nabla_\lambda h^{\mu\nu} = 0$, hence $\Gamma = \Gamma(g)$ and a symmetric Ricci tensor. The variation in h along a tangent direction k gives $\int k^{\mu\nu} R_{\mu\nu} = 0$ for every k with $g_{\mu\nu} k^{\mu\nu} = 0$, which is the vanishing of the traceless part of $R_{\mu\nu}$. For the Levi–Civita connection the contracted Bianchi identity reads $\nabla^\mu (R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = 0$; subtracting the divergence of the traceless equation gives $\frac{1}{4} \partial_\nu R = 0$ on a connected region, so R is a constant, written 4Λ. Substituting into the traceless equation gives $R_{\mu\nu} = \Lambda g_{\mu\nu}$. The converse is immediate. The degree statement is the one of Proposition 2.1: the polynomial was not changed. □

De Sitter space, $R_{\mu\nu} = \Lambda g_{\mu\nu}$ with $\Lambda > 0$, is therefore a solution of this gravity model on the unimodular sector, with Λ the residue of the Bianchi integration. The unimodular restriction admits only volume-preserving coordinates, and the constant is what the integration leaves over once the volume has been fixed, carried unchanged, as the record of Chapter 8 asks of a register. Two other presentations remain open and are recorded as such in the ledger: enforcing $\det h = -1$ by a multiplier term $\lambda(\det h + 1)$ raises the degree to five, and an auxiliary density that linearizes the volume term changes the field space. Neither is needed for the slice read in Chapter 10.

2.1.5 Polynomial degree

Along the affine path $(h, \Gamma) + s(k, C)$, the cubic coefficient in the chosen chart is

$$\int k^{\mu\nu} \left(C^\lambda_{\lambda\rho} C^\rho_{\mu\nu} - C^\lambda_{\nu\rho} C^\rho_{\mu\lambda} \right) \mathrm{d}^4 x.$$

No term of fourth degree in s occurs. The same argument with several independent variation parameters establishes vanishing of the full fourth multilinear variation, rather than only one selected diagonal derivative. Nondegeneracy restricts the admissible interval of the path; it does not insert a denominator into this independent-field action. The fourth derivative with respect to the amplitude s is zero even when the resulting gravitational field has arbitrarily complicated coordinate dependence.

Eliminating Γ substitutes an expression containing g^{-1} and derivatives of g; the resulting metric action need not be polynomial in an affine metric variation.

2.1.6 Scales and the observable used later

For dimensions, use coordinates $x^0 = ct, x^1, x^2, x^3$ measured in length, with dimensionless metric components. The dimensional action is $c^3 S_{\mathrm{GR}}/(16\pi G)$: Γ has units of inverse length, $R_{\mu\nu}$ of inverse length squared, and the integral has units of length squared. The prefactor therefore gives the action units of energy times time. Suppressing a nonzero constant changes neither stationarity nor the number of field factors; c and Newton's constant return in predictions expressed in SI units.

A reference length L gives a useful numerical form. Put $\widehat{x} = x/L$, $\widehat{\Gamma} = L\Gamma$, and $\widehat{R} = L^2 R$. Then

$$S^{\mathrm{SI}}_{\mathrm{GR}} = \frac{c^3 L^2}{16\pi G} \int h^{\mu\nu} \widehat{R}_{\mu\nu}(\widehat{\Gamma}) \, \mathrm{d}^4 \widehat{x}.$$

The dimensionless integral is the quantity a code may assemble. The length L must still accompany a quoted curvature tolerance: a componentwise residual of 10^{-8} in this integral corresponds to $10^{-8}/L^2$ in dimensional Ricci curvature. A tidal acceleration measurement requires a further conversion using the selected observer's frame and the separation of its test masses. In an orthonormal frame, curvature components $R_{\hat{0}\hat{i}\hat{0}\hat{j}}$ enter geodesic deviation with an acceleration scale $c^2\ell/L^2$ for separation ℓ. A small coordinate component has a physical interpretation only after that frame and length scale have been supplied.

A finite evolution calculation must in addition prescribe compatible initial data, a coordinate condition, and boundary treatment for incoming information. The light cones follow from the recovered Lorentzian metric. The present action does not impose the Lorentz transformation as an extra scalar multiplier, and a choice of coordinates is not itself a source of stress-energy.

Vacuum is an explicit physical assumption. This action contains no fluid stress, electromagnetic stress, cosmological term, or material constitutive law. The dimensionless compactness $GM/(Lc^2)$ is useful when a mass M appears through the exterior solution or boundary data, but it is not an independently varied matter density in the action written here. A weak-field calculation can expand about Minkowski data; a strong-field calculation must additionally monitor nondegeneracy and coordinate conditioning. Both may use the same cubic first-order action. The observable used later is a bound on a detector displacement, which is assembled from curvature, coordinate constraints, and the selected detector response. A flat but nonconstant coordinate example in Chapter 5 makes this distinction directly testable.

2.2 Fixed-Hamiltonian Schrödinger evolution

2.2.1 Independent fields and admissible variations

Use complex wavefunctions with finite integrated probability, $\mathcal{H} = L^2(\Omega;\mathbb{C})$, and the overlap integral $\langle\psi,\phi\rangle = \int_\Omega \bar{\psi}\phi\,\mathrm{d}x$. Fix a time-independent self-adjoint Hamiltonian H, with the field regularity and boundary conditions specified by $D(H)$. For example,

$$H = -\frac{\hbar^2}{2m}\Delta + V(x), \qquad m > 0,$$

with real V and boundary conditions that make this realization self-adjoint. The domain is part of the model: a differential expression alone does not determine H. The physical equation and its probability interpretation are the ones in Tong's Sections 1.2–1.2.2 [36].

Use paths regular enough that $\dot{\psi}$ and $H\psi$ belong to $\mathcal{H}$ and the following integrations exist. For an unbounded Hamiltonian, require both ψ and $H\psi$ to have finite squared integrals, choose time dependence regular enough for the stated integration by parts, and admit perturbations preserving those conditions. This supplies the regularity needed for a strong solution, rather than merely differentiating a formal differential expression. Spatial boundary conditions enter through $D(H)$ and are not free variations. The mass m, Planck constant $\hbar$, potential V, and spatial boundary conditions are prescribed physical inputs; they do not become additional field variations because an implementation stores them alongside the wavefunction.

Varying real and imaginary parts independently is equivalent to treating ψ and $\bar{\psi}$ as independent coordinates during the formal variation and then imposing complex conjugacy.

2.2.2 Action and boundary sector

On $[0, T]$ define the real action

$$S_{\mathrm{Sch}}[\psi] = \int_0^T \left[\frac{i\hbar}{2}(\langle \psi, \dot{\psi} \rangle - \langle \dot{\psi}, \psi \rangle) - \langle \psi, H\psi \rangle \right] \mathrm{d}t. \tag{2.4}$$

Before imposing endpoint restrictions, its variation carries the boundary contribution

$$\frac{i\hbar}{2}(\langle \psi, \delta\psi \rangle - \langle \delta\psi, \psi \rangle)\Big|_0^T.$$

The boundary sector used here takes variations vanishing at both time endpoints; unrestricted endpoint variation would define a different stationary problem. The endpoint restriction is a device for deriving the equation, not a requirement that a physical initial-value problem prescribe both initial and final wavefunctions.

2.2.3 First variation and stationary equations

For variations vanishing at both time endpoints, integration by parts gives

$$\delta S_{\mathrm{Sch}} = 2\,\mathrm{Re} \int_0^T \langle \delta\psi, i\hbar\dot{\psi} - H\psi \rangle \, \mathrm{d}t.$$

Hence stationarity gives the evolution and its adjoint equation,

$$i\hbar\partial_t\psi = H\psi, \qquad -i\hbar\partial_t\langle\psi| = \langle\psi|H. \tag{2.5}$$

In the real-potential differential example the second equation is $-i\hbar\partial_t\bar{\psi} = H\bar{\psi}$.

2.2.4 Projection, lift, or equivalence

Conversely, the equations (2.5) make the interior variation zero, so the stationary points are exactly the strong solutions. For a strong solution, self-adjointness also gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\psi\|^2 = 2\,\mathrm{Re}\langle \psi, -iH\psi/\hbar \rangle = 0.$$

The spectral calculus supplies the unitary initial-value solution $\psi(t) = e^{-itH/\hbar}\psi(0)$, with strong differentiability for initial data in $D(H)$. Unitarity bounds state perturbations in this probability-amplitude norm.

For the real scalar potential used above, direct substitution of the Schrödinger equation gives the local probability balance

$$\partial_t|\psi|^2 + \nabla \cdot j = 0, \qquad j = \frac{\hbar}{m}\,\mathrm{Im}(\bar{\psi}\nabla\psi).$$

Periodicity cancels the integrated boundary flux. An impermeable quantum boundary must instead make its normal probability current vanish; Dirichlet data or homogeneous Neumann data are standard examples for the free kinetic operator on a suitable domain. Absorbing boundary conditions change this balance and cannot be included in a claim of exact norm preservation without also accounting for the absorbed flux.

2.2.5 Polynomial degree

All terms in (2.4) are quadratic in the independent complex coordinates. Thus $D^3 S_{\mathrm{Sch}} = 0$, and its Euler–Lagrange residual is linear. In particular its fourth variation vanishes with one order to spare. This affine degree statement is made off shell: a variation $\psi + s\eta$ need not stay on the unit sphere. Solving a normalization constraint first would instead introduce a nonlinear parametrization of that sphere. The degree concerns the declared variation parameter s: one energy eigenstate already has time dependence $e^{-iEt/\hbar}$, whose fourth time derivative generally does not vanish.

2.2.6 Scales and the observable used later

In d spatial dimensions a probability-normalized wavefunction has units length$^{-d/2}$, because $\int_\Omega |\psi|^2 \,\mathrm{d}x = 1$. The expectation $\langle \psi, H\psi \rangle$ is an energy, and both terms of the action integrate to energy times time.

For the free-particle kinetic term, take a length L and define

$$t_* = \frac{mL^2}{\hbar}, \qquad E_* = \frac{\hbar^2}{mL^2}, \qquad \widehat{\psi} = L^{d/2}\psi, \quad \tau = t/t_*, \quad \widehat{x} = x/L.$$

The equation becomes $i\partial_\tau \widehat{\psi} = [-\Delta_{\widehat{x}}/2 + V/E_*]\widehat{\psi}$. Consequently a dimensionless phase tolerance has a definite laboratory time scale, and a potential error must be divided by E_* before it is combined with a normalized kinetic residual. A different reference length changes these reported dimensionless numbers without changing the dimensional physical prediction. For an unbounded Hamiltonian the dimensionally compatible size of a state is $\|\psi\| + \|H\psi\| / E_*$.

The readout must be named with equal care. For a bounded self-adjoint observable A and normalized states ψ, ϕ,

$$|\langle \psi, A\psi \rangle - \langle \phi, A\phi \rangle| \le 2 \|A\| \, \|\psi - \phi\| .$$

To see this, insert $\langle \phi, A\psi \rangle$ between the two expectations and apply Cauchy–Schwarz to each difference. A detector probability for a spatial region is the expectation of its projection operator, so $\|A\| = 1$ supplies an immediate probability-error bound. Energy and momentum are generally unbounded operators. An L^2 state error alone does not bound their errors; the energy-domain or derivative control required by those observables must be included in the numerical contract. Norm preservation is consequently a valuable conservation test, but does not certify the phase of an interference pattern or the energy content of an unresolved high-frequency tail.

This distinction also identifies the source of stiffness. For a Fourier mode with wave number k, the kinetic angular frequency is $\omega_k = \hbar|k|^2/(2m)$. Refining the spatial mesh introduces faster oscillations and tighter phase requirements, even though the action is still quadratic. A self-adjoint spectral discretization preserves the free mode frequencies exactly; a finite-difference discretization changes them through its discrete Laplacian. A variable potential also couples Fourier modes, so projection and multiplication must be assessed with the actual cutoff and quadrature. A numerical Fourier truncation has its own projection and time-stepping errors, which unitarity does not remove, and using periodic modes is a physical and numerical boundary decision, rather than an innocuous change of basis. The computed phase example in Section 5.3 separates these spatial choices from the error of the time integrator.

2.3 Incompressible Navier–Stokes with response fields

2.3.1 Independent fields and admissible variations

Take a periodic spatial domain $\Omega = \mathbb{R}^d/\mathbb{Z}^d$, $d = 2$ or 3, and a finite time interval. Fix $\nu > 0$, a force f, and a divergence-free initial velocity u_0. Density has been normalized to one, so p is pressure divided by that density. The physical equations are the incompressible equations stated by Fefferman [14]; the response-field construction below is an explicit stationary representation of them.

Introduce independent real fields u, p, v, q. All fields are sufficiently smooth for the displayed products and integrations by parts. Pressure is normalized to spatial mean zero. Response pressure q can receive the same normalization. On a periodic domain this loses no divergence equation, since a divergence already has zero spatial integral. The response velocity v and response pressure q are independent multipliers for momentum and incompressibility. They are not a second physical fluid, a measured negative viscosity, or hidden turbulent fluctuations.

2.3.2 Action and boundary sector

The momentum residual and action are

$$R_m(u,p) = \partial_t u - \nu\Delta u + (u\cdot\nabla)u + \nabla p - f, \tag{2.6}$$

$$S_{\mathrm{NS}}[u,p,v,q] = \int_0^T \int_\Omega \big[v\cdot R_m(u,p) + q\,\nabla\cdot u\big]\,\mathrm{d}x\,\mathrm{d}t. \tag{2.7}$$

The response fields are multiplier amplitudes enforcing the balances; they have not been assigned an elastic energy or a material constitutive law.

For initial-value data impose $u(0) = u_0$, so $\delta u(0) = 0$. The time integration produces the terminal term $\int_\Omega v(T)\cdot\delta u(T)\,\mathrm{d}x$. Allowing the final velocity to vary freely gives the natural condition $v(T) = 0$. Periodicity removes all spatial boundary terms. These choices completely specify the boundary sector used in the following result.

2.3.3 First variation and stationary equations

Variations in v and q give, respectively, $R_m = 0$ and $\nabla\cdot u = 0$. Variation in p gives $\nabla\cdot v = 0$. To obtain the remaining equation, vary u before imposing incompressibility. The nonlinear derivative is

$$\delta\big((u\cdot\nabla)u\big) = (\delta u\cdot\nabla)u + (u\cdot\nabla)\delta u.$$

The second term transfers a derivative onto uv; its divergence includes an off-shell contribution $-(\nabla\cdot u)v$. The complete interior stationary system is consequently

$$\begin{aligned} &R_m(u,p) = 0, \qquad \nabla\cdot u = 0, \qquad \nabla\cdot v = 0,\\ &-\partial_t v - \nu\Delta v - (u\cdot\nabla)v - (\nabla\cdot u)v + (\nabla u)^{\mathsf{T}} v - \nabla q = 0. \end{aligned} \tag{2.8}$$

Here $(\nabla u)_{ij} = \partial_j u_i$. On the physical constraint surface the divergence contribution vanishes, leaving the usual formal adjoint of the momentum linearization, coupled to response pressure.

2.3.4 Projection, lift, or equivalence

Proposition 2.3 (Exact stationary projection and lift)**.** *Every sufficiently regular stationary point of* (2.7)*, with these data, projects to a solution* (u, p) *of incompressible Navier–Stokes. Every sufficiently regular such solution has a stationary lift* $(u, p, 0, 0)$*. The enlarged action has* $D^4 S_{\mathrm{NS}} = 0$*.*

Proof. The response variations impose the physical equations. Conversely zero response satisfies its divergence, adjoint, and terminal equations, while the physical solution satisfies the other two variations. The advective term $v_i u_j \partial_j u_i$ is cubic. Time, viscosity, pressure, and divergence terms are quadratic, and the fixed-force term is linear. Theorem 1.1 therefore applies. □

The response fields impose equations by stationarity; they do not turn viscous flow into a positive action minimum. At zero response the action is zero even for an off-shell velocity. When a constraint residual pairs nontrivially with a response variation, scaling that variation changes the action with either sign: the stationary point is a saddle. In adjoint-based sensitivity calculations, nonzero response fields are instead driven by a selected objective and its terminal data; the zero-terminal-data stationary lift proved here has no such objective term.

2.3.5 Polynomial degree

The count in the proof of Proposition 2.3 is a count in the independent fields u, p, v, q. Eliminating responses changes it. In a dimensionless finite realization with a fixed inner product, replace the action by

$$S_\epsilon = \langle v, R_m \rangle + \langle q, \nabla \cdot u \rangle - \frac{\epsilon}{2}(\|v\|^2 + \|q\|^2), \qquad \epsilon > 0.$$

Its response equations give $v = R_m/\epsilon$ and $q = (\nabla \cdot u)/\epsilon$. Substitution yields

$$S_\epsilon^{\mathrm{red}}(u, p) = \frac{1}{2\epsilon}\left(\|R_m(u, p)\|^2 + \|\nabla \cdot u\|^2\right).$$

The enlarged penalized action remains cubic, but this reduced objective is generally quartic. Its stationary equations impose an adjoint residual condition, not automatically a zero residual. Consequently a penalized stationary point is not interchangeable with the exact stationary projection of Proposition 2.3. This finite calculation identifies the precise operation responsible for the change of derivative order; it makes no limiting assertion as ϵ tends to zero.

2.3.6 Scales and the observable used later

Before normalizing density, the dimensional momentum equation is

$$\rho(\partial_t u + (u \cdot \nabla)u) = \nabla \cdot \sigma + \rho f, \qquad \sigma = -p_{\mathrm{phys}} I + 2\mu D(u), \quad D(u) = \tfrac{1}{2}(\nabla u + \nabla u^{\mathsf{T}}).$$

The constitutive assumptions are a Newtonian fluid, constant density $\rho > 0$, and constant dynamic viscosity $\mu > 0$, with kinematic viscosity $\nu = \mu/\rho$. Incompressibility then reduces the viscous force per unit mass to $\nu\Delta u$. In the equations used here, $p = p_{\mathrm{phys}}/\rho$ has units of velocity squared and f has units of acceleration. A non-Newtonian stress, variable density, compressibility, temperature-dependent transport, or relativistic stress requires a different model; the response fields do not carry such extensions.

For a characteristic speed U and length L, use $t_* = L/U$, $p_* = U^2$, and $f_* = U^2/L$. The dimensionless physical momentum equation is

$$\partial_\tau \widehat{u} + (\widehat{u} \cdot \widehat{\nabla})\widehat{u} + \widehat{\nabla}\widehat{p} - \frac{1}{\mathrm{Re}}\widehat{\Delta}\widehat{u} = \widehat{f}, \qquad \mathrm{Re} = \frac{UL}{\nu}.$$

All momentum terms now have the same acceleration scale U^2/L. The divergence residual has scale U/L instead. Adding their raw numerical magnitudes would mix unlike units; both must be normalized before a single stopping threshold is meaningful. The Reynolds number compares advective transport and viscous diffusion. The periodic unit domain of this section can be understood as these already scaled coordinates.

For the response-field action use these dimensionless coordinates and physical fields, and take v, q dimensionless as well. The integral (2.7) is then dimensionless. Multiplication by a fixed action scale S_0 restores joule seconds without changing its stationary equations. Rescaling a multiplier by a fixed nonzero factor does not alter the equation it enforces, but it does change the numerical weighting of its assembled residual. That weighting must therefore be reported when a code compares momentum and divergence errors.

The observable used later is the kinetic energy. For a smooth periodic solution, multiplying the momentum equation by u and integrating gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{L^2}^2 + \nu\|\nabla u\|_{L^2}^2 = \int_\Omega f \cdot u \,\mathrm{d}x. \tag{2.9}$$

Incompressibility cancels advection and pressure work. This is a kinetic energy balance, distinct from minimization of the response action. It does not control every higher derivative; its use in an error estimate requires the regularity and stability assumptions of that estimate.

For numerical work the energy balance is a specific diagnostic. In dimensional form the fluid kinetic energy is $\rho\int|u|^2\,\mathrm{d}x/2$, viscous conversion is $\mu\int|\nabla u|^2\,\mathrm{d}x$ on this periodic incompressible domain, and forcing contributes $\rho\int f\cdot u\,\mathrm{d}x$. A discrete convective operator that does not preserve the required skew-adjoint cancellation can create an energy defect even when its assembled nonlinear residual is small. Pressure requires a compatible velocity–pressure discretization or an exact divergence-free representation; an arbitrary pressure mode can be invisible to the momentum test. These issues are controlled by the discretization and its stability constants.

The two familiar time scales also remain: diffusion acts over L^2/ν, while transport acts over L/U. On a resolved Fourier mode, viscosity contributes decay rate $\nu|k|^2$, and advection about a velocity of order U contributes frequencies of order $U|k|$. An explicit integrator must respect the stability restrictions of the actual spatial operator. An implicit Newton solve trades those restrictions for a conditioned nonlinear and linear algebra problem; it does not delete the physical decay or phase error. The periodic Taylor–Green calculation in Section 5.4 gives exact velocity, pressure, decay, and energy values against which these parts can be checked.

2.4 Bare QED on a fixed background

2.4.1 Independent fields and admissible variations

Choose Minkowski spacetime with $\eta = \mathrm{diag}(1,-1,-1,-1)$, natural units $\hbar = c = 1$, and fixed gamma matrices satisfying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}I$ and $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$. The gauge potential A_μ is real

and commuting; ψ and $\bar{\psi}$ are independent anticommuting spinor variables in the independent-field variation. The physical conjugacy is $\bar{\psi} = \psi^\dagger\gamma^0$. We use a trivial gauge bundle and compactly supported variations, so a single potential and the following integrations by parts are available. Affine variations $(A, \psi, \bar{\psi}) + s(a, \chi, \bar{\chi})$ use a commuting parameter s. Tong's Sections 6.1 and 6.3.1 fix the Maxwell and charged-fermion construction [37].

2.4.2 Action and boundary sector

Define

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad D_\mu\psi = (\partial_\mu + ieA_\mu)\psi,$$

and take the bare classical action

$$S_{\mathrm{QED}} = \int \left[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi\right] \mathrm{d}^4x. \tag{2.10}$$

The boundary sector is that of compactly supported variations. The convention also fixes the gauge action: $A_\mu \mapsto A_\mu + \partial_\mu\alpha$, $\psi \mapsto e^{-ie\alpha}\psi$, and $\bar{\psi} \mapsto \bar{\psi}e^{ie\alpha}$. These transformations preserve the action and current. Gauge-equivalent potentials describe the same electromagnetic field; their redundant coefficients leave zero modes in the unconstrained tangent matrix.

2.4.3 First variation and stationary equations

Using left and right fermionic variations consistently gives the two Dirac equations

$$(i\gamma^\mu D_\mu - m)\psi = 0, \qquad i(D^{\leftarrow}_\mu\bar{\psi})\gamma^\mu + m\bar{\psi} = 0, \tag{2.11}$$

where $D^{\leftarrow}_\mu\bar{\psi} = \partial_\mu\bar{\psi} - ieA_\mu\bar{\psi}$. For the gauge variation, antisymmetry gives $\delta(-F^2/4) = -F^{\mu\nu}\partial_\mu\delta A_\nu$. After integration by parts its coefficient is therefore

$$\partial_\mu F^{\mu\nu} = e\bar{\psi}\gamma^\nu\psi. \tag{2.12}$$

The homogeneous Maxwell equation $\partial_{[\lambda}F_{\mu\nu]} = 0$ follows from $F = \mathrm{d}A$, without another multiplier.

2.4.4 Projection, lift, or equivalence

Equations (2.11) imply conservation of $j^\mu = \bar{\psi}\gamma^\mu\psi$: substituting the two Dirac equations in $\partial_\mu j^\mu$ cancels both the mass and potential terms. Thus the source in (2.12) satisfies the compatibility condition obtained by taking its divergence, and the stationary system is the coupled Maxwell–Dirac system with a conserved current.

The $\nu = 0$ Maxwell equation is Gauss's constraint $\partial_i F^{i0} = e\psi^\dagger\psi$. It remains part of the stationary system when the spatial components are used as evolution equations. In a Lorenz gauge, $\partial_\mu A^\mu = 0$, the potential equation takes the form $\Box A^\nu = ej^\nu$; that gauge condition is an additional choice of representative, not another equation obtained by varying an independent multiplier in the displayed action.

2.4.5 Polynomial degree

Maxwell and free Dirac terms are quadratic; the ordered product $\bar{\psi}A\psi$ is cubic. With a commuting affine parameter s, substitution of $(A, \psi, \bar{\psi}) + s(a, \chi, \bar{\chi})$ produces a polynomial of degree at most three, coefficient by coefficient in the Grassmann algebra. Hence $D^4 S_{\mathrm{QED}} = 0$ formally, and the Euler–Lagrange residual in this independent-field presentation is at most quadratic. This calculation does not assign an ordinary positive norm to Grassmann numbers.

One can check the coefficient count without commuting any fermion factors. Keep the order $\bar{\psi}A\psi$ fixed while expanding the three affine factors. The highest power is $-es^3\bar{\chi}\gamma^\mu a_\mu\chi$; every other contribution has lower degree. Anticommutation signs matter when differentiating with respect to fermionic coordinates, but they cannot create a fourth factor or a fourth power of the commuting parameter.

The bare degree identity persists for the displayed polynomial; an effective action obtained by integrating out fields need not be cubic. A fixed curved spin background retains this count when its connection and volume element are coefficients. Making that background dynamical, using nonlinear link variables, or changing the boundary sector requires a fresh count in the newly declared fields. The Abelian qualification also matters: here F is linear in A. In a non-Abelian potential formulation the additional $[A_\mu, A_\nu]$ term makes F^2 contain quartic monomials, and the argument for bare QED does not transfer to that presentation.

2.4.6 Scales and the observable used later

In these rationalized natural units the action is dimensionless, energy and inverse length have the same units, and the local Lagrangian density has units of energy to the fourth power. Dimensional counting gives $[A_\mu] = \text{energy}$, $[\psi] = [\bar{\psi}] = \text{energy}^{3/2}$, $[m] = \text{energy}$, and dimensionless coupling e. A reference length L therefore permits $\widehat{A} = LA$, $\widehat{\psi} = L^{3/2}\psi$ and $\widehat{x} = x/L$. The parameters of the scaled bare equations are e and mL. Restoring $\hbar, c$ makes the latter L/λ_C, where $\lambda_C = \hbar/(mc)$ is the reduced Compton wavelength. In the usual SI convention the fine-structure coupling is $\alpha = e_{\mathrm{SI}}^2/(4\pi\epsilon_0\hbar c)$. This conversion is necessary before comparing a dimensionless spinor calculation with an energy or cross-section measurement.

The words *bare QED* delimit what this model controls. The Maxwell–Dirac action is the classical polynomial used to define the interactions; QED predictions also involve a regulator, quantum states, and a prescription for determining renormalized parameters from physical measurements. The bare mass symbol m need not equal a measured pole mass before that prescription is supplied. A tree-level interaction term contains three fields, but a scattering probability contains products and sums of amplitudes, and loop corrections introduce integrations over internal momenta, with their own cutoff dependence and physical uncertainties. Quantization changes the objects being compared: quantum states, correlation functions, and renormalized observables require a specified construction and regulator.

A numerical state for the electromagnetic sector may contain the potential, the electric and magnetic fields, or both. The physical fields are derivatives of the potential; checking only a potential residual can miss a derivative error unless the chosen norm controls it. Gauss's law is a constraint on the initial fields and charge, and charge continuity makes its propagation compatible with Maxwell evolution. A discrete current deposition that violates continuity can therefore produce a Gauss-law error whatever the action degree. A chosen gauge and constraint treatment are consequently part of any numerical stability claim. A boundary with an incoming electromagnetic wave, a conducting

wall, and a periodic boundary impose different energy fluxes. Their treatment must be fixed before declaring energy drift to be a solver defect.

For a regulated coefficient calculation one may choose a finite ordered basis of fermion monomials and compare the resulting coefficient arrays. This supplies ordinary numerical norms for a particular finite representation while retaining the anticommutation rule. For a quantum calculation, state normalization, positive probabilities, and the observable's error estimate belong to the quantum construction itself. The vacuum Maxwell mode and free Dirac operator checks of Section 5.5 test parts of a QED implementation, each separately.

2.5 The common conclusion and its domain

Theorem 2.4 (Shared closure of the selected presentations)**.** *In the regular boundary sectors specified above, the independent-field actions satisfy* $D^4 S_i = 0$ *and their Euler–Lagrange residuals satisfy* $D^3 R_i = 0$*. For QED these are coefficient identities in the independent-field fermion presentation. For Schrödinger the stronger identities* $D^3 S = 0$ *and* $D^2 R = 0$ *hold.*

Proof. Each action has the displayed polynomial degree, and variation lowers that degree by one. Apply Theorem 1.1 in the respective field coordinates; the same coefficient argument works with anticommuting fermion coefficients and a commuting affine parameter s. □

The result survives a fixed affine discretization. If $Z = Z_0 + E_h z$, with E_h linear and independent of z, then $S_{i,h}(z) = S_i(Z_0 + E_h z)$ still has degree at most three. Fixed linear quadrature preserves the count. With the inherited dual pairing its residual is $E_h^* R_i(Z_0 + E_h z)$, of degree at most two. This gives an exact discrete algebraic conclusion without assuming that the discrete stationary fields approximate the continuum fields. State-dependent quadrature, a nonlinear coordinate map, or elimination of discrete auxiliaries falls outside this particular preservation argument.

The theorem supplies an exact shared derivative hierarchy; the field spaces remain different. Lorentzian metrics occupy a nondegenerate sector, wavefunctions live in a complex Hilbert space, incompressible velocities satisfy a differential constraint, and QED includes gauge and spinor structure, and their initial data, boundary conditions, and solutions are not identified by the common degree. The domain of the theorem is the declared variation parameter in the declared independent fields: a gravitational field keeps the fourth spatial derivatives of its metric components, and an energy eigenstate keeps its phase $e^{-iEt/\hbar}$. Positivity, regularity, stability, and the quantum construction of bare QED are supplied, where they are supplied, by the estimate or construction named in the corresponding section.

2.6 The derivative of the modified field equations

The hierarchy has two readings, the pointwise first variation and the Sobolev derivative, and for the modified gravity equations of Section 2.1 they are one object. Its first derivative is also the subject of a proposed identification with the electrodynamic model of Section 2.4.

2.6.1 Two readings of one derivative

Proposition 2.5 (Gateaux and Fréchet derivatives coincide for bounded polynomials of degree at most three)**.** *Let X be a Banach space and $S(Z) = a + b(Z) + c(Z,Z) + d(Z,Z,Z)$ with b, c, d bounded symmetric multilinear forms on X (a bounded multilinear form may be replaced by its symmetrization without changing S). Then S is Fréchet differentiable at every Z, its Fréchet derivative equals its Gateaux derivative*

$$DS(Z)H = \frac{\mathrm{d}}{\mathrm{d}s} S(Z+sH)\Big|_{s=0} = b(H) + 2c(Z,H) + 3d(Z,Z,H),$$

and the same holds for the residual $R = DS$ and its derivative DR, which are bounded polynomial maps of degrees two and one.

Proof. Expanding $S(Z+H) - S(Z) - DS(Z)H$ leaves the terms $c(H,H) + 3d(Z,H,H) + d(H,H,H)$, each at least quadratic in H and bounded by $(\|c\| + 3\,\|d\|\,\|Z\| + \|d\|\,\|H\|)\,\|H\|^2$, which is $o(\|H\|)$. Symmetry of the forms gives the displayed formula. The residual and its derivative are polynomial maps whose coefficients are the same bounded forms, so the argument repeats. □

For the gravity model take, on a bounded region, h in the nondegenerate sector of $L^\infty \cap H^1$ and Γ in H^1, with the norm $\|h\|_\infty + \|h\|_{H^1} + \|\Gamma\|_{H^1}$. The two monomials of (2.1) are then bounded forms: $|\int k\,\partial C| \le \|k\|_2\,\|\partial C\|_2$ and $|\int k\,C\,C'| \le \|k\|_\infty\,\|C\|_2\,\|C'\|_2$. The proposition applies, and it applies verbatim to the other three actions on the spaces their sections declare, with the Grassmann coefficient identity of Section 2.4 standing in for a norm estimate where the fields anticommute.

Remark 2.6 (The pointwise first variation determines the Sobolev derivative)*.* The classical first variation is computed along smooth compactly supported directions and read pointwise. Let V be the closure of those directions in the declared norm. On an H^1 component, $V = H^1_0$, the directions with zero boundary trace; on the bounded metric component the L^∞ part of the norm is not approximated by smooth functions, and V is the closure in the full norm, a proper subspace. By continuity of $DS(Z)$ (Proposition 2.5), the first variation on smooth compactly supported directions determines $DS(Z)$ on V. Directions with nonzero boundary trace belong to the boundary sector and are governed by the boundary terms of the action. On an H^1 component, the Meyers–Serrin theorem [30] gives density of $C^\infty \cap H^1$ in H^1, so smooth directions, not compactly supported, determine $DS(Z)$ on the whole of H^1. On V the pointwise reading and the Sobolev reading of the first derivative are one bounded linear functional, and a press check on the pair reports not slipped by construction. The equality of functionals says nothing about work; work belongs to a representation and an algorithm. The device's reading of the distinction is its pure and branched traversals, whose operations Section 7.7 measures; that this is the pointwise and the Sobolev evaluation is a correspondence, *proposed*. The record of Chapter 8 keeps work in the velocity and not in the value.

2.6.2 A proposed identification

Remark (Electrodynamics as the derivative of the modified field equations)*.* With the cosmological constant supplied as a residue by Proposition 2.2, the modified field equations have the residual $R_{\mathrm{GR}} = DS_{\mathrm{GR}}$, quadratic, and the derivative $DR_{\mathrm{GR}} = D^2 S_{\mathrm{GR}}$, linear in the perturbation and bounded by Proposition 2.5. The electrodynamic model of Section 2.4 has the same hierarchy. The identification proposed here compares the linearized electrodynamic residual with that derivative, pulled back along a map of field spaces. Stated with its domains: let X_{QED} and X_{GR} be the field

spaces of Sections 2.4 and 2.1 with their declared norms, so that $R_{\mathrm{QED}} : X_{\mathrm{QED}} \to X^*_{\mathrm{QED}}$ and $DR_{\mathrm{GR}}(g) : X_{\mathrm{GR}} \to X^*_{\mathrm{GR}}$. The proposal is a background $\psi_0 \in X_{\mathrm{QED}}$ and a differentiable map $\Phi : X_{\mathrm{QED}} \supset U \to X_{\mathrm{GR}}$ for which the residual values are compared by the pullback $D\Phi(\psi_0)^* : X^*_{\mathrm{GR}} \to X^*_{\mathrm{QED}}$, and the identity

$$DR_{\mathrm{QED}}(\psi_0) = D\Phi(\psi_0)^* \circ DR_{\mathrm{GR}}\big(\Phi(\psi_0)\big) \circ D\Phi(\psi_0)$$

holds on X_{QED}. Its realization is that identity, with the spaces, the background, and Φ exhibited. Linearity in the perturbation is a property of each side at fixed background; dependence on the background, affine or otherwise, is a separate property and is not asserted. The identity is calibrated *proposed*; it is not derived here. The literature supplies context and not this identity: the double copy constructs gravity amplitudes from gauge-theory kinematic numerators satisfying color–kinematics duality, by replacing color factors with a second copy of the numerators, and generalized double copies add correction terms where the duality is not manifest [4, 5, 7, 6]. Discharging the proposal would require Φ, the operator identity above with the Sobolev spaces matched, and, for the quantum theory, the regulator that Section 2.4 already requires. The common derivative hierarchy, Theorem 2.4, is what the two models are proved to share; the identification is what the record of Chapter 8 lists among the rungs where a reading handed over by name says more than the derived one.

Each of the four actions now supplies a residual $R = DS$ and a linearization DR on a declared field space. A residual measures how far a computed state is from stationarity; it does not yet measure how far a reported reading is from the stationary one. That second distance is controlled only through a stability estimate, which relates the size of the residual, in declared units, to the size of the error in a declared observable. Chapter 3 supplies those estimates.

Chapter 3

From closure to an engineering tolerance

Chapter 2 supplies, for each selected action, a residual $R = DS$ and its tangent $DR = D^2S$. A residual bounds a reading only through a stability estimate. An engineering tolerance is an allowable change in a stated quantity: a length, a force, a phase, a mass-flow rate, or a reported number, with units, an operating range, a reference state, and a method of comparison. This chapter turns a residual into a finite allowance on an observable and states the result as one framework, Definition 3.1.

3.1 The framework of a calibrated numerical reading

Let Z collect the unknown coefficients of a selected spatial and temporal discretization, let d contain the prescribed loads, material coefficients, boundary values, and initial data, and let $R(Z; d) = 0$ be the assembled equation. Let Q be a selected output, Z_h a computed state, and $\widehat{Q}$ the reported value.

Definition 3.1 (Calibrated numerical reading)**.** A *calibrated numerical reading* of Q is the reported value $\widehat{Q}$ together with the following five items.

(i) *Scaled residual.* Positive diagonal scales D_Z, D_R and a reference state Z_0 giving dimensionless coordinates z, a residual r, and a Jacobian J as in (3.1). Every norm, threshold, and singular value refers to these declared coordinates (Section 3.2).

(ii) *Local stability.* An established solution z_* of $r(z_*) = 0$, a lower bound $\sigma > 0$ on the smallest singular value of $Dr(z_*)$, and a neighbourhood on which Proposition 3.2 converts a residual into a state error; for constrained systems, the corresponding lower bound on the scaled constraint operator (Sections 3.3 and 3.4).

(iii) *Output sensitivity.* A bound L_Q on the change in Q per unit state change in that neighbourhood, or an adjoint solution λ with $J_*^{\mathsf{T}}\lambda = \nabla_z Q$ evaluating the first-order output defect as $\lambda^{\mathsf{T}} r$ (Section 3.3).

(iv) *Deterministic allowances.* Enclosures ϵ_{alg}, ϵ_{con}, ϵ_{disc}, ϵ_{data} in residual units, a readout allowance ϵ_Q, and, for comparison with a physical experiment, a model-discrepancy allowance ϵ_{model},

combined by (3.6) and (3.7) into one bound on $\left|\widehat{Q} - Q(Z_*)\right|$ (Sections 3.5–3.7).

(v) *Reporting cell.* A resolution $\Delta > 0$, a truncation or rounding rule with its tie rule, and the requirement that the certified interval for Q lie wholly inside one cell (Section 3.8).

This is Definition 0.1 specialized to a numerical calculation. The admitted inputs are the scaled equations, data, and scales of item (i); the operation of reading is the solve together with the evaluation of Q, controlled by items (ii) and (iii); the reference is $Q(Z_*)$, or a physical value once ϵ_{model} is included; the allowance is the bound of item (iv); and the criterion is the cell test of item (v). The items are independent hypotheses. When items (i)–(iv) hold, (3.7) bounds the reported output; when the resulting interval lies in one cell, item (v) fixes the reported integer. The scalar reading of Chapter 6 supplies all five items in exact rational arithmetic, and Chapter 7 adds the instrument chain to item (iv).

3.2 Put the equations and their tolerances in compatible units

Some coefficients of Z can represent displacements, others velocities, pressures, potentials, or multiplier reactions. The components of $R(Z; d)$ can have different dimensions. Adding a force residual in newtons to a continuity residual in cubic metres per second has no physical meaning.

Choose positive diagonal scale matrices D_Z and D_R, and a reference state Z_0. Define dimensionless state and residual coordinates by

$$z = D_Z^{-1}(Z - Z_0), \qquad r(z; d) = D_R^{-1} R(Z_0 + D_Z z; d). \tag{3.1}$$

Every entry of D_Z has the units of its state component; every entry of D_R has the units of its residual component. The dimensionless Jacobian is

$$J(z) = D_R^{-1} \frac{\partial R}{\partial Z}(Z_0 + D_Z z) D_Z.$$

A threshold on $\|r\|_2$ is consequently a statement about specified component scales. It is reproducible only if those scales are recorded. For example, dividing force imbalance by a one-newton allowable imbalance and volume imbalance by a $10^{-8}\,\mathrm{m^3\,s^{-1}}$ allowance gives both equations a dimensionless interpretation. An order-one residual then has a clear meaning relative to the selected allowances.

A raw Euclidean norm also changes when cells or finite-element nodes are added. For values of a scalar field sampled on cells with volumes V_i, a useful normalized root-mean-square norm is

$$\|e\|_{\text{rms}}^2 = \frac{\sum_i V_i (e_i / e_{\text{scale}})^2}{\sum_i V_i}.$$

This approximates a volume-averaged continuum quantity under mesh refinement. A mass matrix supplies the corresponding finite-element quadrature. A maximum norm serves a different requirement: it protects the largest component rather than an average. Thus an RMS temperature tolerance cannot certify the hottest point. Below, the norms and singular values refer to the declared scaled coordinates; a weighted Euclidean norm can be converted to ordinary Euclidean coordinates by its positive square root. This is numerical scaling with an explicit physical interpretation, not a new physical law [17, 24].

For incompressible flow, characteristic speed U, length L, density ρ, and viscosity μ give time scale L/U and inertial pressure scale ρU^2. Dividing momentum by $\rho U^2/L$ produces a viscous coefficient

$1/\mathrm{Re}$, where $\mathrm{Re} = \rho U L/\mu$. These are physical nondimensionalization choices. They are distinct from selecting a residual threshold to meet, for example, a ten-micrometre displacement allowance. At very small Reynolds number a viscous pressure scale can be more useful numerically. Changing a scale changes a matrix condition number and the interpretation of its norm; it leaves the underlying dimensional balance unchanged and authorizes no different tolerance.

3.3 Sensitivity through the smallest Jacobian singular value

A small force imbalance implies a small displacement only for a system with sufficient stiffness. The numerical version of that statement is a lower singular-value bound on the scaled tangent matrix. A nearly free mechanism can move a large distance while producing a tiny restoring force.

Proposition 3.2 (Conditional residual-to-state bound)**.** *Let z_* solve the finite scaled equations $r(z_*) = 0$. Suppose the smallest singular value of $J_* = Dr(z_*)$ is at least $\sigma > 0$, and along the straight segment from z_* to z,*

$$\|Dr(z_* + te) - J_*\|_2 \le Lt \|e\|_2 , \qquad e = z - z_*, \quad 0 \le t \le 1.$$

If $L\|e\|_2 \le \sigma$, then

$$\|z - z_*\|_2 \le \frac{2}{\sigma} \|r(z)\|_2 .$$

Equivalently the stability constant is $K = 1/\sigma$ and the localization condition is $KL\|e\|_2 \le 1$.

Proof. The fundamental theorem of calculus along the segment gives $r(z) = J_* e + \rho$ with $\|\rho\|_2 \le L\|e\|_2^2/2$. The defining singular-value inequality gives $\sigma\|e\|_2 \le \|J_* e\|_2 \le \|r(z)\|_2 + L\|e\|_2^2/2$. Under the stated localization condition the quadratic term is at most $\sigma\|e\|_2/2$, which can be moved to the left. □

This is a local estimate about an established equilibrium and an identified region. A validated Newton calculation, a direct solution, or a separate existence argument supplies those premises. The proposition carries its witness, the equilibrium z_* and the region $L\|e\|_2 \le \sigma$, and the record of Chapter 8 files it with that witness. In the one-component spring equation $r_\epsilon(z) = \epsilon z$, the trial displacement $z = 1$ has residual ϵ and error one. The smallest tangent singular value is exactly ϵ, so the estimate correctly deteriorates as the spring becomes soft.

For a measured output with local gradient $g = \nabla_z Q$, linear sensitivity is $g^{\mathsf{T}} J_*^{-1} r$. This can be far smaller than the worst-case state estimate $\|g\|_2 \|J_*^{-1}\|_2 \|r\|_2$. Solving the adjoint system $J_*^{\mathsf{T}} \lambda = g$ evaluates the first-order output defect as $\lambda^{\mathsf{T}} r$. Any use as a rigorous error bound still needs the nonlinear remainder and localization allowance. An adjoint estimate is particularly useful when a pressure drop is required accurately but every velocity coefficient need not have the same absolute accuracy.

3.4 Pressure and multiplier constraints are part of the solve

A linearized constrained equilibrium often has the block form

$$\begin{bmatrix} A & B^{\mathsf{T}} \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix} . \tag{3.2}$$

Here u is the primary field, p is a multiplier, and $Bu = g$ is a constraint such as discrete volume conservation. The sign convention for p has been selected so the upper off-diagonal block is B^{T}. When A is invertible, elimination gives

$$Sp = BA^{-1}f - g, \qquad S = BA^{-1}B^{\mathsf{T}}, \qquad u = A^{-1}(f - B^{\mathsf{T}}p).$$

The Schur complement S expresses how a pressure correction changes the constraint after the velocity response. If A is symmetric positive definite and B has full row rank, then $p^{\mathsf{T}}Sp = (B^{\mathsf{T}}p)^{\mathsf{T}}A^{-1}(B^{\mathsf{T}}p) > 0$ for nonzero p. A small eigenvalue of S signals a pressure pattern that hardly changes the constraint and therefore can be poorly determined.

An elementary example makes the reaction explicit. Take $A = \operatorname{diag}(2, 2)$, $B = [1 \;\; 1]$, $f = (2, 0)^{\mathsf{T}}$, and $g = 0$. Then $S = 1$, $p = 1$, and $u = (1/2, -1/2)^{\mathsf{T}}$. Ignoring pressure would give $u = (1, 0)^{\mathsf{T}}$ and violate $Bu = 0$. Accurate momentum equations alone would consequently accept the wrong constrained state.

For incompressible flow with closed or periodic boundary conditions, pressure can be determined only up to a constant. Fix its mean or one reference value before asking for an invertible matrix. Compatibility of the prescribed boundary flux is also required: an enclosed volume cannot have a nonzero net prescribed inflow in an incompressible steady problem. Discrete velocity and pressure spaces must retain a positive inf-sup constant in their selected mass or energy norms. That condition prevents an undetected pressure pattern. In finite coordinates it amounts to a lower singular-value bound for the appropriately scaled constraint operator on the pressure space after its constant mode has been removed; this is the constrained form of item (ii) of Definition 3.1. A multiplier variable alone supplies neither that bound nor flux compatibility [8].

3.5 Exact remainder and Jacobian-free evaluation

For the quadratic residual in (1.2), Taylor expansion terminates exactly:

$$R(Z + H) - R(Z) - DR(Z)H = B(H, H) = \tfrac{1}{2}D^2R[H, H]. \tag{3.3}$$

If $\|D^2R\| \le M_2$ in specified norms, the linearization defect is at most $M_2 \|H\|^2 /2$. The missing fourth action term gives a quadratic residual defect. A Newton correction still has a finite defect because a quadratic equation is not a linear equation.

For $\delta \neq 0$,

$$\frac{R(Z + \delta H) - R(Z)}{\delta} = DR(Z)H + \frac{\delta}{2}D^2R[H, H]. \tag{3.4}$$

Both bilinear slots in (1.3) are present. A centered difference gives exactly $DR(Z)H$ in exact arithmetic because the quadratic terms cancel. If each residual evaluation has error at most e_R, the forward-difference error is bounded by

$$\tfrac{1}{2}\,|\delta|\, M_2 \|H\|^2 + \frac{2e_R}{|\delta|}.$$

The two terms have opposite responses to decreasing δ: truncation falls while cancellation and evaluation error are amplified. A matrix-free calculation must budget both. Chapter 4 uses this bound to choose a differencing scale and an achievable inner linear tolerance [25].

Suppose a linear solver returns $DR(Z)H = -R(Z) + \ell$. Equation (3.3) gives

$$\|R(Z+H)\| \le \|\ell\| + \tfrac{1}{2}M_2 \|H\|^2 .$$

The first term is an incomplete linear solve; the second is nonlinear curvature over the proposed step. Neither is a mesh error; both belong to ϵ_{alg}. Preconditioning changes a third quantity: if P is invertible, a reported $\|P^{-1}R\| \le \eta$ implies only $\|R\| \le \|P\|\,\eta$ in compatible norms. Always convert the monitored norm to the residual norm used by the physical acceptance test.

3.6 A sharp cubic interpolation certificate

A discretization allowance ϵ_{disc} needs an explicit interpolation estimate. Let a real $\phi \in C^4([a,b])$ be interpolated by the cubic $H_3\phi$ matching value and first derivative at both endpoints; write $h = b-a$. The Hermite remainder is

$$\phi(x) - H_3\phi(x) = \frac{\phi^{(4)}(\xi_x)}{4!}(x-a)^2(x-b)^2.$$

To obtain it, subtract a suitable multiple of $(t-a)^2(t-b)^2$ from the interpolation error, make the result vanish at x, and apply the repeated Rolle argument, counting both endpoint zeros twice. Since the maximum of the product is $h^4/16$,

$$\|\phi - H_3\phi\|_\infty \le \frac{h^4}{384}\left\|\phi^{(4)}\right\|_\infty . \tag{3.5}$$

For $\phi(x) = x^4$ on $[0,1]$, the error is $x^2(1-x)^2$ and reaches $1/16 = 24/384$. The constant is sharp for this class.

This estimate has a direct dimensional reading. If ϕ is a transverse beam displacement measured in metres and x is measured in metres, then $\phi^{(4)}$ has units m^{-3}. Multiplication by h^4 restores a displacement. For a prismatic Euler–Bernoulli beam with constant bending stiffness EI and distributed transverse load f, the static equation is $EI\phi^{(4)} = f$. On an interval where $|f| \le f_{\max}$, a certified bound is

$$\|\phi - H_3\phi\|_\infty \le \frac{h^4 f_{\max}}{384EI}.$$

Here EI is measured in force times length squared. A target interpolation allowance ϵ_b therefore admits $h \le (384EI\epsilon_b/f_{\max})^{1/4}$ when $f_{\max} > 0$. An unloaded constant-stiffness span is cubic and has zero interpolation error under these endpoint data. Variable stiffness instead gives $(EI\phi'')'' = f$ and requires its own derivative bound. Point loads and stiffness jumps require intervalwise estimates and junction conditions.

The same remainder applies to a smooth calibration trace plotted against a dimensionless sweep parameter. It concerns a physical displacement curve or a scalar readout only when its own fourth derivative is bounded. A nonlinear readout can turn a cubic state trace into a noncubic output, and the off-shell action's zero fourth variation is not a bound on a measured displacement derivative [9, 10].

3.7 An error budget for a physical reading

Suppose a selected model, its mesh transfers, and its stability estimates supply

$$\|Z_h - Z_*\|_X \le K_s(\epsilon_{\text{alg}} + \epsilon_{\text{con}} + \epsilon_{\text{disc}} + \epsilon_{\text{data}}). \tag{3.6}$$

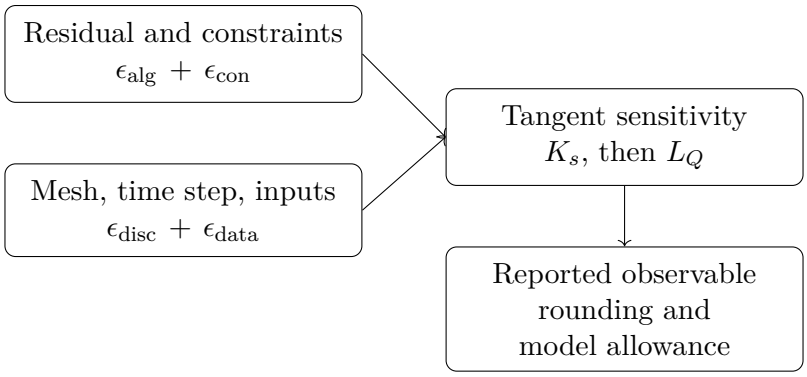


Figure 3.1: A physical tolerance passes through the actual operators and scales. Polynomial closure controls a remainder; the remaining allowances retain separate measurements or estimates.

Here the state is compared in one declared normalized norm. ϵ_{alg} includes the incomplete nonlinear and linear solves and residual evaluation error; ϵ_{con} measures constraint defects after compatible scaling. ϵ_{disc} measures spatial and temporal consistency defects in the same residual units. ϵ_{data} measures changes in equations induced by calibrated input allowances, including loads and constitutive parameters. Writing the sum assumes a deterministic enclosure for each contribution. A statistical standard uncertainty is a different object and is not an absolute error bound without additional coverage assumptions [22, 24].

If Q changes by at most L_Q times the state change in this region and its evaluation and reporting error is at most ϵ_Q, then

$$\left|\widehat{Q} - Q(Z_*)\right| \leq L_Q K_s \sum_j \epsilon_j + \epsilon_Q. \tag{3.7}$$

Insert $Q(Z_h)$ between the two outputs and apply the triangle inequality. For $K_s = 2$, $L_Q = 3$, residual-unit allowances totaling 10^{-6}, and readout allowance 10^{-7}, the output bound is $6.1 \cdot 10^{-6}$ in the chosen output units.

The selected equation can itself disagree with a physical experiment. If a separately supported model-discrepancy allowance satisfies $|Q(Z_*) - Q_{\text{physical}}| \leq \epsilon_{\text{model}}$, then add ϵ_{model} to the right side. Refining a mesh does not calibrate this term. Nor does driving an algebraic residual to machine precision establish an uncertain material coefficient. Calibration connects an indication to a reference quantity through a documented measurement model [23].

3.8 Two readers, decimal decisions, and cycle closure

If two instruments or two numerical representations estimate the same quantity q with certified allowances E_1, E_2, then

$$|\widehat{q}_1 - \widehat{q}_2| \leq E_1 + E_2.$$

If they estimate different quantities q_1, q_2, add a bound on $|q_1 - q_2|$. Agreement within the combined allowance is a consistency test of the declared common measurement. It cannot establish the measurement model without that declaration and its calibration evidence. Conversely disagreement larger than the sum rejects at least one allowance, the common-quantity assumption, or the implementation.

A reporting cell adds a discontinuous final decision. For truncation with resolution $\Delta > 0$, cell k is $[k\Delta, (k+1)\Delta)$. A certified closed interval wholly within one cell fixes the reported integer k. A narrow interval crossing its upper edge does not. A nearest-rounding rule has shifted boundaries and needs a declared tie rule. Chapter 6 checks the relevant endpoints by exact integer arithmetic instead of treating a repeated display as a certificate.

Convergence also needs its target. A Cauchy sequence of residuals need not converge to zero: $1+1/n$ is an elementary counterexample. If residual and consistency allowances ρ_n tend to zero, a common stable reference state exists, and one uniform constant gives $\|Z_n - Z_*\| \le K\rho_n$, then the computed states converge to that state. On changing meshes the comparison requires explicit transfers to one physical norm. A decreasing relative residual is insufficient if its normalizing load or the stability constant is changing without control.

Finally, cycle closure is a measured endpoint condition. For an exact reading $q(t)$ and computed reading $\widehat{q}(t)$ whose two endpoint errors are at most ϵ, the triangle inequality gives

$$\|\widehat{q}(1) - \widehat{q}(0)\| \le \|q(1) - q(0)\| + 2\epsilon.$$

An exact closed cycle therefore admits the finite closure allowance 2ϵ. A cubic trace need not close: $\phi(s) = s^3$ has $\phi^{(4)} = 0$ and endpoint difference one. Cycle closure, residual convergence, and the resolution of successive recorded events consequently require separate engineering acceptance tests.

The framework takes the linear-solve defect ℓ, the tangent-product error, and the algebraic allowance ϵ_{alg} as given. Chapter 4 computes the Newton correction and bounds the floors that set them.

Chapter 4

Newton–Krylov solution and multigrid correction

Definition 3.1 takes as given the correction a solver returns and the algebraic allowance ϵ_{alg} that remains after it. A nonlinear equation, its linearized correction, and the algorithm used to compute that correction are three different engineering objects. This chapter follows an incompressible-flow time step through all three. The same numerical organization can be used for other residual equations, provided their fields, constraints, boundary conditions, and tangent operators are supplied explicitly [25, 2]. Its stopping rule (Section 4.8) rests on the reevaluated nonlinear residual and an output sensitivity, not on an iteration count or a small update.

4.1 Start with the momentum and volume balances

For density $\rho > 0$, viscosity $\mu > 0$, and force per unit volume f, backward Euler applied to incompressible momentum gives the residual

$$R_m(u, p) = \rho \frac{u - u^{\text{old}}}{\Delta t} + \rho(u \cdot \nabla)u + \nabla p - \mu \Delta u - f, \tag{4.1}$$

$$R_c(u) = \nabla \cdot u. \tag{4.2}$$

The spatial derivatives here denote the declared discrete operators when the equations are assembled. Momentum components have units of force per volume; divergence has units of inverse time. They require separate residual scales as in (3.1). Boundary fluxes, velocity values, the previous time state, and a pressure reference are part of the equation specification. They cannot be recovered from a Krylov iteration history.

For a velocity correction v and pressure correction π, the tangent is

$$DR_m(u, p)[v, \pi] = \rho v/\Delta t + \rho(v \cdot \nabla)u + \rho(u \cdot \nabla)v + \nabla \pi - \mu \Delta v, \tag{4.3}$$

$$DR_c(u)[v] = \nabla \cdot v.$$

Both convection terms are required. Expanding $(u + v) \cdot \nabla(u + v)$ leaves the exact nonlinear defect $\rho(v \cdot \nabla)v$. Freezing the advecting velocity retains only $(u \cdot \nabla)v$; that produces a useful Picard or Oseen iteration, but a different tangent from Newton's method. For a fixed discrete bilinear convection operator the same expansion is an exact coefficient identity. Limiters, changing turbulence

closures, contact switches, or state-dependent mesh transformations can change the polynomial and differentiability assumptions and must be included in the residual actually differentiated.

Write all scaled equations as $r(z) = 0$. At nonlinear iteration k, Newton asks for a correction s_k satisfying

$$J_k s_k = -r_k, \qquad J_k = Dr(z_k), \quad r_k = r(z_k).$$

4.2 A matrix-free tangent has a finite error floor

A Jacobian-free product uses residual evaluations in place of storing J_k. For a unit direction v in the scaled state norm, define

$$\widehat{Jv} = \frac{\widehat{r}(z + \delta v) - \widehat{r}(z)}{\delta}.$$

Assume each evaluated residual differs from its exact discrete value by at most e_R, and $\|D^2 r[v, v]\| \leq M_2$ over the step. Then

$$\left\|\widehat{Jv} - Jv\right\| \leq \frac{M_2 \left|\delta\right|}{2} + \frac{2e_R}{\left|\delta\right|}. \tag{4.4}$$

For a quadratic residual this truncation expression follows from an exact Taylor identity. For a general smooth residual it follows by integrating the second derivative. The residual-evaluation allowance includes floating-point roundoff and any inexact subsidiary calculation. It must be supplied or estimated for the implementation; setting it equal to machine precision without accounting for residual magnitudes is not a bound.

When $M_2, e_R > 0$, minimizing the right side gives

$$|\delta_{\text{opt}}| = 2\sqrt{e_R/M_2}, \qquad \epsilon_{Jv,\min} = 2\sqrt{M_2 e_R}.$$

Thus an arbitrarily small differencing displacement does not give an arbitrarily accurate tangent product. For a nonunit direction, apply the formula to $v/\left\|v\right\|$ and multiply the result by $\|v\|$, keeping the state displacement at the selected size. Component scales in D_Z make that displacement dimensionally meaningful. A direction dominated by pressure otherwise could select a completely different physical perturbation from one dominated by velocity.

For a quadratic residual the centered quotient has no truncation defect in exact arithmetic, but its evaluation-error bound is $e_R/\left|\delta\right|$. For a general residual with third-derivative bound M_3, the corresponding bound is $M_3\delta^2/6 + e_R/\left|\delta\right|$. Neither estimate licenses a step outside the admissible material range or across a nonsmooth switch. An analytic tangent, automatic differentiation of the actual residual, or a carefully validated difference quotient can reduce this uncertainty; all still require finite arithmetic analysis [25].

4.3 Inexact Newton: spend accuracy where it changes the answer

Solving every early linear system to the final reporting tolerance wastes work while the nonlinear state is still far away. An inexact Newton step requires instead

$$\left\|J_k s_k + r_k\right\| \leq \eta_k \left\|r_k\right\|, \qquad 0 \leq \eta_k < 1. \tag{4.5}$$

The dimensionless forcing number η_k is the fraction of the current residual left by the linearized solve. Combining this inequality with the second-derivative remainder gives

$$\|r(z_k + s_k)\| \le \eta_k \|r_k\| + \tfrac{1}{2} M_2 \|s_k\|^2 . \tag{4.6}$$

If $\left\|J_k^{-1}\right\| \le K$, then $\|s_k\| \le K(1+\eta_k)\|r_k\|$, so the right side is at most

$$\eta_k \|r_k\| + \tfrac{1}{2} M_2 K^2 (1+\eta_k)^2 \|r_k\|^2 .$$

A fixed forcing fraction retains a linear term. Choosing $\eta_k \le c\|r_k\|$ in a sufficiently small stable neighborhood makes both terms quadratic in the residual. Such a choice is useful only above the arithmetic and tangent-product error floors.

A Krylov monitor typically controls the computed linearized residual. If the product approximation acts as $J_k + E_k$, and the computed solve has residual at most $\widehat{\eta}_k \|r_k\|$, then the true tangent residual is bounded by

$$\|J_k s_k + r_k\| \le \widehat{\eta}_k \|r_k\| + \|E_k s_k\| .$$

For direction-dependent or varying product errors, the accumulated Arnoldi relation has to be bounded instead of assuming one fixed matrix E_k. Either way, asking the computed inner residual to fall far below an uncontrolled product error does not establish (4.5).

A full Newton step can leave the local stable region. A line search uses $z_{k+1} = z_k + \lambda s_k$, with $0 < \lambda \le 1$. For $F(z) = \|r(z)\|_2^2/2$, (4.5) gives

$$DF(z_k)[s_k] = r_k^{\mathsf{T}} J_k s_k \le -(1-\eta_k)\|r_k\|_2^2 .$$

Thus the correction is a strict descent direction when $r_k \neq 0$. For any $0 < c < 1$, differentiability implies that sufficiently small positive λ satisfies the Armijo condition

$$F(z_k + \lambda s_k) \le F(z_k) - c\lambda(1-\eta_k)\|r_k\|_2^2 .$$

A practical backtracking search tries $1, 1/2, 1/4, \ldots$ while checking state admissibility and this decrease. This establishes a local acceptance mechanism, not global existence of a physical solution. If finite residual evaluation has a known enclosure, the two sides must be compared with their corresponding allowances.

4.4 Krylov vectors approximate a correction, not a field theory

For a linear correction equation $As = b$ and initial correction zero, a Krylov method searches combinations of $b, Ab, A^2b, \ldots$. GMRES constructs an orthonormal basis V_m with the Arnoldi relation

$$AV_m = V_{m+1}\overline{H}_m,$$

and selects $s_m = V_m y_m$ by minimizing $\left\|\|b\|_2 e_1 - \overline{H}_m y\right\|_2$. The small least-squares residual is the full residual norm in exact arithmetic because V_{m+1} has orthonormal columns. Loss of orthogonality, inexact products, or restarting can change the achieved reduction and must be monitored [17, 38, 2].

A preconditioner approximates the inverse response of the physical operator. With a fixed invertible right preconditioner M, solve $AM^{-1}y = b$ and recover $s = M^{-1}y$. The monitored residual remains $b - As$. Left preconditioning instead minimizes a transformed residual, whose norm must be

converted before applying a physical acceptance test. If inner preconditioner applications vary, a flexible Krylov construction is required to retain the appropriate basis-and-update relation.

Lanczos is available when the matrix is symmetric in the inner product being used. Symmetry collapses Arnoldi to

$$Av_j = \beta_{j-1}v_{j-1} + \alpha_j v_j + \beta_j v_{j+1}.$$

The older coefficients vanish because symmetry transfers each inner product $v_i^{\mathsf{T}} A v_j$ to $(Av_i)^{\mathsf{T}} v_j$, and the latter lies in an already constructed span. Conjugate gradients adds positive definiteness; a symmetric indefinite saddle matrix does not meet that condition. MINRES or an appropriate block formulation can use symmetry without positive definiteness. The convection terms in (4.3) generally make the tangent nonsymmetric, so GMRES is the natural example here. Ordinary Lanczos cannot simply be inserted at that point while retaining its symmetric theory.

Eigenvalues alone also miss nonnormal sensitivity. For

$$A_K = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, \qquad A_K^{-1} = \begin{bmatrix} 1 & -K \\ 0 & 1 \end{bmatrix},$$

both eigenvalues equal one, but a residual $(0, \epsilon)^{\mathsf{T}}$ corresponds to correction $(-K\epsilon, \epsilon)^{\mathsf{T}}$. Its norm is $|\epsilon|\sqrt{K^2+1}$. A residual tolerance selected from the two eigenvalues would miss the amplification by K. Singular values and the chosen physical norm address this issue directly.

4.5 An exact two-step GMRES check

A small nonsymmetric example exposes every part of the Krylov calculation. Choose

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad s_0 = 0.$$

The first basis vector is $v_1 = (1,1)^{\mathsf{T}}/\sqrt{2}$. Multiplication gives $Av_1 = (3,1)^{\mathsf{T}}/\sqrt{2}$, so $h_{11} = v_1^{\mathsf{T}} A v_1 = 2$. Removing this component leaves $v_2 = (1,-1)^{\mathsf{T}}/\sqrt{2}$, with $h_{21} = 1$. The first GMRES least-squares problem is therefore

$$\min_y \left\| \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} y \right\|_2 .$$

Its minimizer is $y = 2\sqrt{2}/5$, producing correction $s_1 = (2/5, 2/5)^{\mathsf{T}}$ and residual $b - As_1 = (-1/5, 3/5)^{\mathsf{T}}$. The residual norm is $\sqrt{2/5}$, a factor $1/\sqrt{5}$ smaller than the initial norm $\sqrt{2}$. This is the actual residual reduction, rather than a prediction from an eigenvalue plot.

For the second product, $Av_2 = v_2$. Thus the projected two-dimensional operator is

$$H_2 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \qquad y_2 = (1/\sqrt{2}, -1/\sqrt{2})^{\mathsf{T}}.$$

It satisfies $H_2 y_2 = (\sqrt{2}, 0)^{\mathsf{T}}$, and $[v_1\ v_2]y_2 = (0,1)^{\mathsf{T}}$ solves the original equation exactly. There is no third independent vector: the Krylov space already fills the two-dimensional state space. Roundoff and inexact products would turn this exact termination into a finite residual check. This example is also an appropriate regression calculation for a GMRES implementation: the basis, projected matrix, first correction, and final answer are all known without a tolerance-dependent reference run.

The same two-dimensional arithmetic can describe scaled spring coefficients, temperatures, or a selected tangent block from a flow mesh. Its exact termination supplies no criterion for choosing among those physical readings. That choice is carried by the original equations, units, and map from coefficients to an observable, not by the Arnoldi or Lanczos recurrence.

4.6 Why multiple grid scales help

Consider the dimensionless one-dimensional Dirichlet Laplacian

$$(A_h e)_j = (2e_j - e_{j-1} - e_{j+1})/h^2, \qquad e_0 = e_{n+1} = 0.$$

Weighted Jacobi updates an error by $e^{\text{new}} = (I - \omega D_h^{-1} A_h)e$, where $D_h = 2I/h^2$. On the sine mode $e_j = \sin(j\theta)$,

$$e_j^{\text{new}} = \mu(\theta) e_j, \qquad \mu(\theta) = 1 - \omega(1 - \cos\theta). \tag{4.7}$$

For the Dirichlet problem $\theta = k\pi/(n+1)$. Choosing $\omega = 2/3$ gives $|\mu(\theta)| \le 1/3$ for the high-frequency range $\pi/2 \le \theta \le \pi$. A sweep reduces those mode amplitudes by at least a factor of three. For low frequency, $\mu(\theta) = 1 - \theta^2/3 + O(\theta^4)$, so the longest waves hardly change on a fine grid. Smoothing is effective at removing rapid oscillations, not at transporting the longest-scale correction through the whole domain.

A coarse grid represents those long waves with fewer unknowns. Let P interpolate a coarse correction to the fine grid, let R restrict a fine residual, and define the Galerkin coarse operator $A_H = RA_hP$. One coarse correction acts on the error as

$$e^{\text{new}} = (I - PA_H^{-1}RA_h)e. \tag{4.8}$$

If $e = Pe_H$, the error is annihilated exactly when the coarse solve is exact. A two-grid cycle with pre- and post-smoothing has error matrix

$$E_{2g} = S_{\text{post}}(I - PA_H^{-1}RA_h)S_{\text{pre}}.$$

Recursively approximating the coarse inverse gives a multigrid V-cycle. Each level performs a stated number of smoothing sweeps, restricts the remaining residual, receives an interpolated correction, and smooths again. The coarsest system needs a declared solution tolerance; the word multigrid alone specifies none of these operators.

For symmetric positive-definite A_h with $R = P^{\mathsf{T}}$, the coarse correction is an energy-orthogonal projection. Indeed its remainder $e' = e - Pc$, with $c = (P^{\mathsf{T}}A_hP)^{-1}P^{\mathsf{T}}A_he$, satisfies $P^{\mathsf{T}}A_he' = 0$. Therefore

$$e^{\mathsf{T}}A_he = (e')^{\mathsf{T}}A_he' + (Pc)^{\mathsf{T}}A_h(Pc).$$

The correction cannot increase this energy norm. A full mesh-independent convergence bound still requires compatible smoothing and approximation estimates. Convection, anisotropy, strong coefficient contrasts, or indefiniteness can require different smoothers, transfer operators, and coarse spaces [2].

A V-cycle can have linear arithmetic cost under stated mesh assumptions. With a bounded number of stencil operations and smoothing sweeps per unknown, a one-dimensional hierarchy with $n_\ell \le n/2^\ell$ unknowns costs a constant times $\sum_{\ell \ge 0} n/2^\ell \le 2n$, plus its bottom solve. Uniform full coarsening in three dimensions gives the analogous geometric factor $\sum_{\ell \ge 0} 8^{-\ell} = 8/7$. These counts describe one

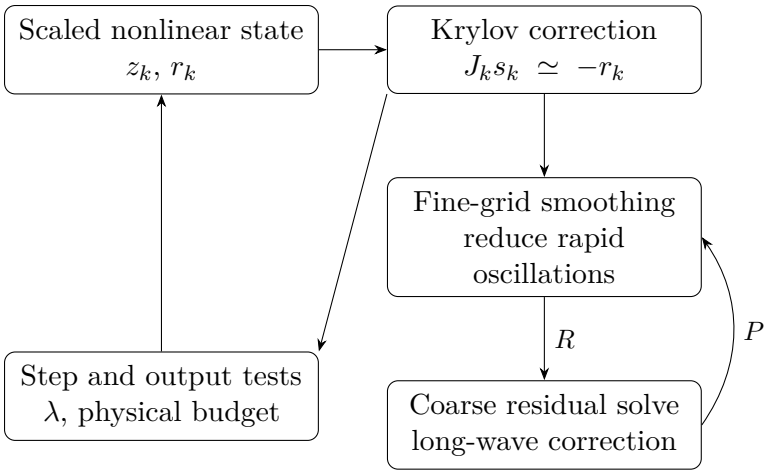


Figure 4.1: Newton controls nonlinear curvature; Krylov methods approximate the tangent solve; multigrid approximates an inverse over several spatial scales. The output budget remains an external acceptance requirement.

cycle, not the number of cycles needed for a requested physical tolerance. Dense coarse operators, costly constitutive evaluations, communication, or a bottom solve whose size grows with the fine problem can change the cost model.

Anisotropy illustrates why a smoother is a physical choice. In a diffusion operator $-\partial_x(a_x\partial_x u) - \partial_y(a_y\partial_y u)$, a large ratio a_x/a_y couples neighboring coefficients much more strongly along one direction. Pointwise Jacobi may then fail to reduce an error that is slowly varying in the strongly coupled direction but rapidly varying in the weak direction. Solving whole lines or planes, semi-coarsening, or using an operator-aware coarse space can restore the missing response. The appropriate calculation depends on the coefficients and boundary conditions; the isotropic one-dimensional factor in (4.7) is not a universal smoothing certificate.

4.7 An exact three-unknown coarse-grid calculation

All the preceding operations can be checked without floating point on a small matrix. Suppress the common h^{-2} factor and take

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \qquad P = \begin{bmatrix} 1/2 \\ 1 \\ 1/2 \end{bmatrix}, \qquad R = P^{\mathsf{T}}.$$

Direct multiplication gives

$$AP = (0, 1, 0)^{\mathsf{T}}, \qquad A_H = RAP = 1.$$

For $b = (0, 1, 0)^{\mathsf{T}}$ and initial approximation zero, the restricted residual is $Rb = 1$. The one-unknown coarse solve gives correction one, and interpolation gives $u = P$. Since $AP = b$, this correction solves the fine problem exactly. The result is special to this right-hand side, which lies in the coarse represented response; it is not a claim that one coarse unknown solves every three-unknown load.

The highest sine mode has vector $v = (\sqrt{2}/2, -1, \sqrt{2}/2)^{\mathsf{T}}$ and eigenvalue $2 + \sqrt{2}$. For $\omega = 2/3$, one weighted Jacobi sweep multiplies it by $(1 - \sqrt{2})/3$, whose absolute value is below $1/3$. A

represented smooth error is removed by coarse correction, while this rapidly alternating mode is strongly reduced by smoothing. These two exact calculations explain the separate jobs in the V-cycle. Their combination must still be checked for the actual discretized operator and its boundary conditions.

For the flow tangent, a physics-based block preconditioner commonly approximates both the momentum response and the pressure Schur complement from (3.2). A velocity-only diffusion multigrid solve leaves continuity unenforced. Constraint-compatible coarse spaces and the pressure reference remain necessary. The Schrödinger operator, a gauge-fixed field equation, and the first-order gravitational system have different signs, fields, and null spaces, and a preconditioner must respect the operator it approximates.

4.8 A finite solution and acceptance procedure

The equations above assemble into one nested computation.

Procedure (Newton–Krylov solution and acceptance).

Inputs. A state z_0 with admissible material parameters and boundary data; the scales of (3.1) and the declared pressure reference; a selected physical output; separate allowances for algebraic, discretization, data, and reporting errors; and a work limit for each linear solve.

Outputs. Either an accepted state with its reported output and the output allowance of (3.7), or a failure status naming the condition below that failed.

Iteration. For $k = 0, 1, 2, \ldots$:

1. Assemble and scale the momentum and constraint residuals. Record both the combined norm and separate continuity or gauge norms. Remove a known pressure null mode by the declared reference condition.
2. Choose a forcing fraction η_k above the estimated tangent and residual evaluation floor. Build or update the approximate momentum and Schur responses used by the preconditioner. A stale preconditioner can be retained if the measured linear convergence remains adequate.
3. Apply right-preconditioned GMRES to the tangent equation, checking the untransformed residual needed in (4.5). Bound any finite-difference product error in that check.
4. Test the proposed state and the line-search decrease. Reduce the step when required by admissibility or nonlinear curvature. After an accepted update, reevaluate the full nonlinear residual rather than reusing the final linear Krylov residual as its replacement.
5. Convert the final residual and constraint allowances to the selected output allowance using established sensitivity bounds, items (ii) and (iii) of Definition 3.1.

Acceptance conditions. The reevaluated nonlinear residual and the separate constraint norms meet the thresholds that step 5 derives from the algebraic allocation, and the mesh, time, input, and reporting contributions, checked separately, meet their own allocations. Only then is the physical output declared accepted.

Failure conditions. The allowed linear work is exhausted before (4.5) is met: return an incomplete-solve status, since a fixed iteration count alone is not convergence. The requested forcing lies below the product-error floor (4.4), so the computed inner residual cannot establish (4.5). The line search finds no admissible step with the required decrease. A separately checked contribution exceeds its allocation.

A small update is not an alternative stopping proof. In a badly scaled or nearly singular problem the step can become small because arithmetic has stalled, because the line search repeatedly rejects motion, or because the preconditioner has hidden an important component. Conversely a large update in an unobserved null direction can leave a selected physical reading unchanged. Monitoring the actual balance, constraints, and output separates these cases.

For a linear output $q = c^{\mathsf{T}} z$ of a fixed invertible tangent, solving $J^{\mathsf{T}}\lambda = c$ gives exactly $q(z) - q(z_*) = \lambda^{\mathsf{T}} r$ for the associated linear equation. Then $\|\lambda\|_2 \, \|r\|_2$ is a direct output allowance. If a $0.02\,\mathrm{Pa}$ algebraic allocation is available and the calibrated scaled sensitivity is at most $4\,\mathrm{Pa}$ per residual unit, a sufficient residual threshold is 0.005. A nonlinear use needs an additional remainder allowance as in the preceding chapter. This makes the stopping criterion a statement about the intended pressure reading, rather than an arbitrary number of digits in a solver log.

4.9 Verification by a prescribed exact flow

A reproducible verification problem separates solution error from an unknown experimental answer. For this benchmark all positions, time, velocity, pressure, viscosity, and force are dimensionless, obtained using fixed length, speed, advective-time, and density scales. On a 2π-periodic square, choose unit density, dimensionless viscosity $\nu > 0$, pressure zero, and stream function $\psi = e^{-t} \sin x \sin y$. Then

$$u = e^{-t}(\sin x \cos y, -\cos x \sin y), \qquad \nabla \cdot u = 0, \quad u_t = -u, \quad \Delta u = -2u.$$

Differentiation gives $(u \cdot \nabla)u = e^{-2t}(\sin x \cos x, \sin y \cos y)$. Consequently the exact forcing

$$f = (-1 + 2\nu)u + e^{-2t}(\sin x \cos x, \sin y \cos y) \tag{4.9}$$

makes the momentum equation hold pointwise. All derivatives, initial values, and periodic boundaries are known. This tests the implemented balance rather than fitting its force to computed errors.

For this exact exponential time dependence, the backward-Euler temporal consistency defect is

$$\left[\frac{1 - e^{\Delta t}}{\Delta t} + 1\right] u(t).$$

Since $0 \le e^a - 1 - a \le a^2 e^a/2$ for $a \ge 0$, its norm is at most $\Delta t \, e^{\Delta t} \, \|u(t)\| \, /2$. This is a local residual defect, not yet a global state-error estimate; stability and accumulation over time steps supply that conversion. It provides a concrete first-order quantity to check when the spatial error has been made smaller.

Compute the selected velocity and pressure norms on meshes $h, h/2, h/4$ and time steps small enough to isolate spatial error, then repeat with fixed fine space and time steps $\Delta t, \Delta t/2, \Delta t/4$. For exact-solution errors $E_h, E_{h/2} > 0$, report $p_{\mathrm{obs}} = \log_2(E_h/E_{h/2})$. A value near the method's expected order is evidence that the tested range has reached its asymptotic behavior. It is not an

enclosure for another problem. Include the constraint norm and the selected output error, because a small velocity RMS error can hide a poor pressure output or a local conservation defect.

Finally lower the nonlinear tolerance by a factor of ten while keeping mesh, time step, and inputs fixed. If the output change is much smaller than its allocated algebraic allowance, further nonlinear work is unlikely to improve the intended reading. If it is larger, tighten the solve or revisit the residual-to-output sensitivity estimate. Then refine the mesh and time step separately and compare with their allocated allowances. For a 0.10 Pa total deterministic pressure-output allowance, an example allocation is 0.02 to the algebraic solve, 0.03 to spatial and temporal approximation, 0.02 to calibrated input data, 0.02 to model discrepancy, and 0.01 Pa to evaluation and reporting. Each term needs its own bound or supporting measurement. Summing these allowances is transparent; running a Krylov loop below machine precision would supply none of the missing physical information.

The procedure accepts a computed state against its allowances only if the implemented residual is the declared one. Chapter 5 supplies exact benchmarks that test this for each of the four field systems.

Chapter 5

Verification against physical field equations

A tolerance becomes an acceptance criterion only after the physical quantity, units, boundary data, and numerical representation have been fixed. A gravity calculation may report tidal curvature; a wavefunction calculation may report a detector probability; a fluid calculation may report velocity or dissipated energy; an electromagnetic calculation may report electric field or radiated energy. The same dimensionless threshold cannot be attached to these readings until their dimensional scales and error propagation have been specified.

The four benchmarks of this chapter are analytical: exact solutions or exact identities of the four action presentations, with computed analytical bounds, and no run of a production PDE code is reported. Each tests equations, constraints, phase, dissipation, or readout accuracy against a known answer. Verification of a discretization in this sense is distinguished from validation against physical data, and manufactured solutions are a standard verification technique [32]. The fourth-variation identity is an additional check on assembling an action in its declared independent fields. The protocol for running a code against the benchmarks is Section 5.6.

5.1 Norms and reference quantities

A useful normalized spatial norm is the root-mean-square norm

$$\|w\|_{\mathrm{rms}} = \left(\frac{1}{|\Omega|} \int_{\Omega} |w(x)|^2 \, \mathrm{d}x \right)^{1/2} .$$

Its units are the units of the field. For a probability-normalized wavefunction we instead use its ordinary L^2 norm, because its integral already has a specified physical normalization. Numerical quadrature must approximate whichever integral has actually been selected.

A residual can be measured at collocation points, in a quadrature-weighted norm, or as the defect of a continuous reconstruction. A vanishing residual at finitely many points need not bound the defect between those points. Fixed discretization means that the field interpolation and quadrature are fixed while the nonlinear iteration runs; changing the mesh begins a new comparison.

5.2 Gravity: a nonconstant flat-coordinate benchmark

For the first-order gravitational presentation, assemble both

$$\mathcal{R}_{\mu\nu} = R_{(\mu\nu)}(\Gamma), \qquad \mathcal{C}_\lambda{}^{\mu\nu} = \nabla_\lambda h^{\mu\nu}.$$

On a length scale L, compare $L^2\mathcal{R}$ and $L\mathcal{C}$ in fixed positive component norms. These are numerical norms in a declared coordinate frame; contracting with the Lorentzian metric is not a positive error norm. A small Ricci residual without compatibility need not describe the intended metric connection.

A useful exact test has nonconstant fields but zero curvature. Starting from Minkowski coordinates, replace the spatial coordinate by

$$X = x + \frac{a}{k}\sin(kx), \qquad Y = y, \quad Z = z, \quad X^0 = x^0, \qquad J(x) = 1 + a\cos(kx), \quad |a| < 1.$$

Take x periodic with period $2\pi/k$. The Jacobian J is positive, so this is a smooth periodic coordinate change. The metric and its densitized inverse are

$$g_{\mu\nu} = \mathrm{diag}(1, -J^2, -1, -1), \qquad h^{\mu\nu} = \mathrm{diag}(J, -J^{-1}, -J, -J).$$

The only nonzero connection coefficient is $\Gamma^x_{xx} = J'/J$. Every Ricci component vanishes: in the only potentially nontrivial component, the two derivative terms cancel and the two connection-product terms cancel. Metric compatibility also holds. For instance,

$$\begin{aligned}
\nabla_x h^{00} &= J' - \frac{J'}{J}J = 0, \\
\nabla_x h^{xx} &= \frac{J'}{J^2} + 2\frac{J'}{J}\left(-\frac{1}{J}\right) - \frac{J'}{J}\left(-\frac{1}{J}\right) = 0.
\end{aligned}$$

The final terms in these expressions include the weight-one density connection term.

Choosing $a = 1/4$ gives $3/4 \le J \le 5/4$ and $9/16 \le J^2 \le 25/16$. Also $|\Gamma^x_{xx}| \le k/3$ follows directly from $|J'| \le k/4$. Thus the test has finite, declared field and conditioning scales. Curvature in every observer frame is exactly zero.

Coverage. The benchmark tests derivatives, connection products, density weights, and coordinate handling, without matter or a gravitational-wave approximation. It is not a free test of an incompatible coordinate driver. Outside its coverage lies the residual-to-observable bound for a general gravitational calculation, which needs the gauge-fixed evolution or boundary operator's stability estimate; coordinate freedom would otherwise supply zero modes. A finite matrix inverse bound at one resolution controls that matrix; mesh-independent convergence additionally requires consistency and a uniform bound in the stated continuum norm. Neither the flat benchmark nor the cubic action establishes such a bound for every vacuum spacetime.

5.3 Schrödinger: phase, spectral truncation, and detector error

For a continuous numerical reconstruction $\psi_h(t)$ in the Hamiltonian domain, define the dimensional defect

$$d_h = i\hbar\partial_t\psi_h - H\psi_h.$$

Self-adjoint evolution and variation of constants give the explicit bound

$$\|\psi_h(T) - \psi(T)\|_{L^2} \le \|\psi_h(0) - \psi(0)\|_{L^2} + \frac{1}{\hbar}\int_0^T \|d_h(t)\|_{L^2}\,\mathrm{d}t. \tag{5.1}$$

Indeed, multiplying the error equation by the inverse unitary propagator leaves the integral of $d_h/(i\hbar)$, and unitarity preserves its norm. This estimate has no growing stability factor. Its hypotheses include a common self-adjoint Hamiltonian and compatible boundary data. A piecewise reconstruction with jumps requires the jump magnitudes in the right-hand side as well. A discrete algebraic residual becomes this defect only after reconstruction and consistency errors are included.

For an explicit computed phase test, take a free particle on a periodic interval of length L. Put

$$\phi_n(x) = L^{-1/2}e^{2\pi i n x/L}, \qquad \omega_n = \frac{\hbar}{2m}\left(\frac{2\pi n}{L}\right)^2, \qquad \psi(t) = \frac{\phi_0 + e^{-i\omega_1 t}\phi_1}{\sqrt{2}}.$$

The state is normalized. Its relative phase controls interference, so an error in that phase has an observable consequence even if the computed norm is exactly one. A Fourier representation retaining these two modes has zero spatial truncation error for this initial state.

Crank–Nicolson time stepping multiplies the nonconstant coefficient by

$$g = \frac{1 - i\omega_1\Delta t/2}{1 + i\omega_1\Delta t/2} = \exp\left[-2i\arctan(\omega_1\Delta t/2)\right].$$

It preserves amplitude but changes frequency. Let $T = N\Delta t$ be one exact cycle, $\omega_1 T = 2\pi$. The accumulated phase error and state error are exactly

$$\delta_N = 2\pi - 2N\arctan(\pi/N), \qquad e_N = \sqrt{2}\,|\sin(\delta_N/2)|.$$

For $z \ge 0$, differentiating $z - 2\arctan(z/2)$ gives $z^2/(4+z^2) \le z^2/4$. Integration therefore proves

$$0 \le z - 2\arctan(z/2) \le z^3/12, \qquad e_N \le \frac{(2\pi)^3}{12\sqrt{2}\,N^2}.$$

With $N = 400$, the exact bound is approximately $9.135312 \cdot 10^{-5} < 10^{-4}$; the exact error formula gives approximately $9.134974 \cdot 10^{-5}$. These numbers are evaluations of the displayed analytical expressions. They verify the promised state threshold in exact time-stepping arithmetic. For normalized states, the probability error for any detector region is consequently below $2 \cdot 10^{-4}$ using the bounded-observable estimate of Chapter 2.

A general free initial state $\sum_n a_n\phi_n$ adds a transparent spatial budget. Discarding modes outside the retained set I gives exactly

$$\epsilon_{\text{space}}^2 = \sum_{n\notin I} |a_n|^2,$$

and this tail norm is constant under exact free evolution. The retained mode phase errors are weighted by their $|a_n|^2$ in the orthogonal L^2 sum. An unrecorded tail is not bounded by showing that retained modes conserve probability. Renormalizing a truncated state also changes the initial error and must be accounted for explicitly.

Coverage. The benchmark tests the time integrator's phase, the spectral truncation budget, and the conversion to a detector probability. Floating-point and linear-solve errors lie outside the exact time-stepping arithmetic and need an additional allowance, as does any Hamiltonian other than the free one, whose defect enters only through (5.1) under its stated hypotheses.

5.4 Navier–Stokes: an exact viscous-decay experiment

Take the periodic square $(0, 2\pi L)^2$, zero forcing, and define $\xi = x/L$, $\eta = y/L$, and $b(t) = e^{-2\nu t/L^2}$. The fields

$$u_x = Ub(t)\sin\xi\cos\eta, \qquad u_y = -Ub(t)\cos\xi\sin\eta,$$
$$p = \frac{U^2 b(t)^2}{4}(\cos 2\xi + \cos 2\eta)$$

are an exact Taylor–Green solution with mean-zero kinematic pressure. The divergence vanishes. The Laplacian is $\Delta u = -2u/L^2$, so $\partial_t u = \nu\Delta u$. Direct differentiation gives $(u\cdot\nabla)u = (U^2b^2/(2L))(\sin 2\xi, \sin 2\eta)$, which is canceled by ∇p. Thus the momentum and incompressibility residuals vanish exactly, not merely to first order in U.

Spatial averaging gives

$$\begin{aligned} \|u\|_{\mathrm{rms}} &= \frac{Ub}{\sqrt{2}}, & K = \tfrac{1}{2}\|u\|_{\mathrm{rms}}^2 &= \frac{U^2b^2}{4}, \\ \nu\|\nabla u\|_{\mathrm{rms}}^2 &= \frac{\nu U^2 b^2}{L^2} = -\frac{\mathrm{d}K}{\mathrm{d}t}. \end{aligned} \tag{5.2}$$

At $\nu T/L^2 = 0.1$, the velocity amplitude ratio is $e^{-0.2} \simeq 0.8187307531$ and the energy ratio is $e^{-0.4} \simeq 0.6703200460$. The response fields can be set to $v = q = 0$, providing the exact stationary lift of the same solution.

A conditional error estimate can also be derived rather than assumed. Let a smooth reconstructed candidate w be exactly divergence-free and periodic, with momentum defect d and the same viscosity. Put $e = w - u$. Subtracting the two momentum equations, pairing with e, and integrating cancels the pressure and the transport by w. The result is

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|e\|_{\mathrm{rms}}^2 + \nu\|\nabla e\|_{\mathrm{rms}}^2 \le \|D(u)\|_\infty \|e\|_{\mathrm{rms}}^2 + \|d\|_{\mathrm{rms}}\|e\|_{\mathrm{rms}}.$$

Here the matrix norm in $\|D(u)\|_\infty$ is the operator norm of the symmetric velocity gradient. The scalar differential inequality, with a limiting argument at zero error, yields

$$\begin{aligned} \|e(T)\|_{\mathrm{rms}} &\le e^{A(T)}\left[\|e(0)\|_{\mathrm{rms}} + \int_0^T \|d\|_{\mathrm{rms}}\,\mathrm{d}t\right], \\ A(T) &= \int_0^T \|D(u)\|_\infty\,\mathrm{d}t. \end{aligned} \tag{5.3}$$

For this exact flow, $\|D(u)\|_\infty \le Ub/L$, so $A(T) \le \mathrm{Re}(1 - b(T))/2$ with $\mathrm{Re} = UL/\nu$. At $\mathrm{Re} = 10$ and $\nu T/L^2 = 0.1$, the amplification bound is approximately $2.475262 < 2.476$. Zero initial error and $\int_0^T \|d\|_{\mathrm{rms}}\,\mathrm{d}t \le 4\cdot 10^{-5}U$ therefore imply $\|e(T)\|_{\mathrm{rms}} < 10^{-4}U$. The residual allowance has units of velocity because an acceleration defect has been integrated over time.

The same velocity bound also controls a specified energy readout. If $K(w) = \|w\|_{\mathrm{rms}}^2/2$, then

$$|K(w) - K(u)| \le \|u\|_{\mathrm{rms}}\|w-u\|_{\mathrm{rms}} + \tfrac{1}{2}\|w-u\|_{\mathrm{rms}}^2.$$

This follows by expanding $|u+e|^2$ and bounding the cross term. At the selected time, $\|u\|_{\mathrm{rms}}/U = e^{-0.2}/\sqrt{2}$. The velocity tolerance $10^{-4}U$ therefore implies an energy-per-mass error below $5.790\cdot$

$10^{-5}U^2$, approximately $3.455 \cdot 10^{-4}$ of the exact remaining kinetic energy. For this two-dimensional field, multiplication by $\rho|\Omega|$ gives kinetic energy per unit out-of-plane depth. A uniform layer of thickness H and depth-independent velocity has total energy $\rho H|\Omega|K$. This dimensional conversion illustrates why a velocity tolerance and an energy tolerance cannot be interchanged simply because the same flow generated them.

An uncertainty in ν changes the physical decay: a relative viscosity perturbation δ changes the amplitude ratio at this time by $|e^{-0.2\delta} - 1| \le 0.2|\delta|e^{0.2|\delta|}$. For $|\delta| \le 10^{-3}$ this relative amplitude allowance is below $2.001 \cdot 10^{-4}$. The allowed perturbation $\delta = -10^{-3}$ produces an rms velocity change approximately $1.158 \cdot 10^{-4}U$, which exceeds the preceding solver target. Reducing algebraic residuals cannot remove this input uncertainty.

Coverage. The benchmark tests the nonlinear transport, the pressure gradient, viscous decay, the energy readout, and the stationary lift, all with exact references. Outside its coverage: a candidate with a divergence defect, which requires additional constraint terms or a quantitatively bounded divergence-free reconstruction before (5.3) applies; an unmeasured boundary flux, which invalidates the periodic cancellation; and every flow other than this smooth finite-time benchmark with compatible candidates. The estimate is not a global regularity theorem for every three-dimensional flow.

5.5 QED: electromagnetic constraints and fermion block checks

An exact coupled-action benchmark is available in the vacuum sector: set $\psi = \bar{\psi} = 0$ and, in natural units, choose

$$A^0 = 0, \qquad \boldsymbol{A} = (0, A_* \cos(kx - |k|t), 0).$$

It satisfies Lorenz gauge and $\Box A^\mu = 0$. For $k > 0$, $E_y = -kA_* \sin(kx - kt)$ and $B_z = -kA_* \sin(kx - kt)$; all other components vanish. Both divergence constraints are zero, and the mean electromagnetic energy density on a period is $k^2 A_*^2/2$. These equalities check phase, relative electric/magnetic amplitude, constraints, and energy. The zero spinors make both Dirac equations and the charge source vanish exactly.

There is also an energy estimate for the vacuum Maxwell block. For periodic fields with defects $\partial_t E - \nabla \times B = r_E$ and $\partial_t B + \nabla \times E = r_B$, subtract an exact solution. Define

$$W(t) = \Big(\|e_E(t)\|_{\mathrm{rms}}^2 + \|e_B(t)\|_{\mathrm{rms}}^2 \Big)^{1/2},$$
$$r(t) = \Big(\|r_E(t)\|_{\mathrm{rms}}^2 + \|r_B(t)\|_{\mathrm{rms}}^2 \Big)^{1/2}.$$

Curl integration by parts cancels the cross terms, giving $\frac{1}{2}\partial_t W^2 \le Wr$. Integration with a limiting argument at $W = 0$ yields

$$W(T) \le W(0) + \int_0^T r(t)\,\mathrm{d}t.$$

A conductor or radiation boundary adds its actual boundary contribution. Restoring SI units requires the energy weights ϵ_0 and $1/\mu_0$, rather than adding raw electric and magnetic field numbers with different units.

For the free Dirac operator block with prescribed zero potential, $H_D = \boldsymbol{\alpha} \cdot \boldsymbol{k} + \beta m$ on a Fourier mode. The anticommutation relations give $H_D^2 = (|\boldsymbol{k}|^2 + m^2)I$, so the frequencies are $\pm\sqrt{|\boldsymbol{k}|^2 + m^2}$.

This provides independent tests of matrix signs, dispersion, and unitary propagation of coefficient spinors. A nonzero charged spinor on a zero potential is not automatically a solution of the coupled Maxwell equation; this is an operator-block test.

Coverage. The vacuum-sector solution verifies a genuine sector of the displayed action, and the Maxwell energy estimate and Dirac block test the two field blocks separately. Outside their coverage lie the nonzero interaction, which additionally requires current and gauge-consistency tests, for example manufactured coefficient fields with analytically calculated source terms; and any quantum observable, since a coefficient residual does not bound a renormalized cross section.

5.6 Protocol for runs

A run of a code against these benchmarks records the dimensional constants and reference scales, initial data, spatial and time boundaries, gauge or coordinate conditions, mesh or spectral cutoff, quadrature, time integrator, nonlinear tolerance, and linear stopping criterion, and states which residual measurement of Section 5.1 is used. The quadrature weights are not changed when two resolutions are compared.

For an exact reference solution, report the measured numerical error in the selected norm. For a general solution, report the estimator and the assumptions that make it an upper bound. A mesh-refinement ratio is evidence about observed discretization order; it is not an unconditional bound on the finest-grid error. Input uncertainty, unresolved boundary flux, and model discrepancy require their own allowances.

Per benchmark:

1. *Gravity.* Monitor nondegeneracy of the reconstructed metric independently, together with the chosen coordinate condition and boundary residuals. To use the flat chart as an evolution benchmark, prescribe coordinate source functions and initial/boundary data consistent with it.
2. *Schrödinger.* Fit floating-point and linear-solve errors into a separate allowance beside the exact phase bound, and record the discarded spectral tail.
3. *Navier–Stokes.* Reproduce both the field and its decay to the declared tolerance, and record the viscosity allowance beside the solver target.
4. *QED.* Separate the bare field-equation verification from a quantum observable calculation. For the first, record the fermion ordering, coefficient basis or regulator, gauge treatment, spatial domain, and boundary flux, and test the Maxwell residual, both Dirac residuals, Gauss's law, and current continuity; for a finite ordered fermion basis these are coefficient array tests whose norm refers to that declared basis and its weights. Check the initial divergence constraints and their propagation in addition to the evolution estimate. Manufactured sources belong to the verification problem and are removed when solving the original source-free model. For a quantum observable, also record the quantum state, renormalization prescription, ultraviolet cutoff, finite-volume error, and statistical or quadrature uncertainty.

Before accepting a run, keep a separate allowance for evaluation and reporting. Analytic benchmarks can be evaluated more accurately than the target discretization, but their rounded printed decimal is not the exact reference field. Numerical differentiation of reference data can also contaminate

a residual test; the expressions above permit analytic derivatives instead. A check of the action's fourth amplitude response should use the declared independent fields and fixed quadrature. Gauge fixing, field elimination, or changing normalization during that check changes the calculation being tested. The resulting acceptance record should name the physical observable, the norm, and every retained error allowance alongside the final number.

The numerical constants in these analytical examples, together with the beam, Krylov, multigrid, coefficient-sensitivity, and instrument examples, are independently reproduced by the script and ledger listed in Appendix A. Finite matrix and coefficient calculations use exact rational arithmetic; exponential acceptance thresholds use bounded Taylor remainders.

Each benchmark verifies a solver in its own field's vocabulary: curvature and compatibility, phase and probability, momentum and viscous energy loss, electric and magnetic energy and charge continuity. Verification establishes that a code computes its declared residual; it leaves open how a computed root becomes a reported digit. Chapter 6 resolves that dependency for one scalar residual, carrying all five items of Definition 3.1 to a certified reporting cell.

Chapter 6

A worked scalar reading

The scalar fixture makes three questions separately decidable: which root the iteration selects, how fast an exact update approaches it, and which finite decimal an independent interval certifies. All coefficients, input values, and arithmetic conventions are fixed below. The calculation uses ordinary integers and rational numbers. It reproduces the numerical expressions of the source construction without running that construction's own checker or measuring its cost. In the terms of Definition 0.1, the conditions of this reading are the polynomial, the branch, the arithmetic, the scale, and the reporting cell. Its comparison with the measured fine structure constant is stated in Section 6.7.1, and only there.

6.1 The residual, its branch, and its units

In normalized coordinates the residual and its first two derivatives are

$$r(y) = 13y^2 - 36y + 18, \qquad r'(y) = 26y - 36, \qquad r''(y) = 26.$$

Expansion gives the exact identity

$$(r'(y))^2 = 360 + 52r(y). \tag{6.1}$$

Thus at either root the derivative magnitude is $6\sqrt{10}$. The first sensitivity is $Dr(y)[v] = (26y - 36)v$ for an input increment v; the mixed second sensitivity is $D^2r(y)[v, w] = 26vw$. The constant second derivative is the curvature of a scalar response; reading it as an acceleration would require a variable, a time scale, and a response law.

Write the roots as

$$\alpha = \frac{18 + 3\sqrt{10}}{13}, \qquad \beta = \frac{18 - 3\sqrt{10}}{13}, \qquad d = \alpha - \beta = \frac{6\sqrt{10}}{13} > 0.$$

The vertex is 18/13. Both roots are positive, so positivity does not choose between them. We select the increasing branch $y > 18/13$.

Where the branch comes from. Turn the middle coefficient on with a parameter $s \in [0, 1]$:

$$r_s(y) = 13y^2 - 36s\,y + 18, \qquad y_\pm(s) = \frac{18s \pm \sqrt{324s^2 - 234}}{13}.$$

The product of the roots is $18/13$ for every s, so each root fixes the other. At $s = 0$ the roots are $\pm i\sqrt{18/13}$, and the matrix of trace zero and determinant 234 that they diagonalize squares to -234 times the identity: a quarter turn, scaled. For $0 \le s < s_0 = \sqrt{26}/6 \simeq 0.8498$ the roots are complex conjugates on the circle $|y| = \sqrt{18/13}$, at angles $\pm\theta$ with $\cos\theta = s/s_0$. They meet on the positive axis at $s = s_0$, and for $s > s_0$ they are real, $y_\pm = \sqrt{18/13}\, e^{\pm\varphi}$ with $\cosh\varphi = s/s_0$. At $s = 1$ they are α and β. *Proved here*, by expansion.

Two operations are distinct here. At $s = 1$ the roots are real, and the order on $\mathbb{R}$ selects the larger directly from the coefficients; that is the selection $y > 18/13$ above. Along the arc the two roots are exchanged by complex conjugation, the only field automorphism of $\mathbb{C}$ fixing $\mathbb{R}$, so no real coefficient distinguishes them there; carrying one particular complex root through the meeting point onto the increasing branch requires a label, the sign given to $\sqrt{-1}$. Circling s_0 in the complex s-plane exchanges the two roots, so no continuation of a label around that loop is single-valued. *Standard.* The monodromy concerns continuation and does not obstruct the ordered selection at $s = 1$. That the reported root is the continuation of the label $+\sqrt{-1}$ from the quarter turn is a reading of the ordered selection. *Proposed.*

The reporting scale. The conversion to the reported-output coordinate is

$$x = g_x y, \qquad g_x = \frac{324}{5}. \tag{6.2}$$

Both coordinates are dimensionless, and g_x is a declared numerical gain. Consequently $g_x\alpha \simeq 137.01129055$ and $g_x\beta \simeq 42.43486330$. The output-coordinate residual is

$$q(x) = 325x^2 - 58320x + 1889568 = 324^2 r(5x/324).$$

This equality verifies the coefficient conversion, and it shows that Newton and Halley updates commute with this linear coordinate conversion when performed exactly.

The seed and the rule of admission. The comparison starts at the jar's lower wall $x_0 = 1296/10 = 129.6$, or $y_0 = 2$. This seed is above the vertex and below α. Its first Newton image is $y_1 = 17/8$, giving $x_1 = 137.7$. These two values enclose the selected root. This is the source's jar-seeded comparison; it is distinct from the source's separate pair-valued crank, whose initial reading is taken from its strip rather than from the jar wall. Retaining the input prevents their iteration counts from being interchanged.

A value of the unknown enters the record of this chapter only as the declared seed x_0, as the image of an admitted value under one of the displayed update maps, or as a wall of a sign-straddling bracket; the branch selection $y > 18/13$ is part of this rule. Every value of the unknown printed below, seed, iterate, bracket wall, and the truncation reported from the walls, traces back through the rule to the seed. The declared coefficients, the reporting scale, and the illustrative coefficient allowances of Section 6.6 are inputs to the record, not values admitted by it.

The manuscript inherits a few figurative names from the source, each defined here: the *jar* is the initial interval $[129.6, 137.7]$ in output coordinates and its *walls* are the two endpoints; a *turn* is one application of an integer update of Section 6.3; a *crank* is the integer update machine itself; a *heartbeat* is the work unit of Section 7.7; a *blank* is the subtracted reference sample of the same section.

6.2 What the exact updates prove

The two real-arithmetic maps are

$$N(y) = y - \frac{r(y)}{r'(y)} = \frac{13y^2 - 18}{26y - 36}, \tag{6.3}$$

$$H(y) = y - \frac{2r(y)r'(y)}{2(r'(y))^2 - r(y)r''(y)}. \tag{6.4}$$

The second is Halley's update, although the source's integer implementation of it carries Lagrange's name; the appendix records the declaration. The second derivative enters a rational correction; simply replacing the residual by its quadratic Taylor polynomial and solving that polynomial is a different operation. Halley's historical method provides context [19]; the convergence orders required here follow from direct identities.

Proposition 6.1 (Exact error laws for this quadratic)**.** *For $e = y - \alpha$, at points where the maps are defined,*

$$N(y) - \alpha = \frac{e^2}{2e + d}, \qquad H(y) - \alpha = \frac{e^3}{3e^2 + 3de + d^2}. \tag{6.5}$$

Proof. Factor $r(y) = 13e(e + d)$ and write $r'(y) = 13(2e + d)$. Substitute in the Newton formula and subtract α. For Halley, cancel the common factor $2 \cdot 13^2$ in its correction. Its remaining denominator is $(2e + d)^2 - e(e + d) = 3e^2 + 3de + d^2$. Subtracting the correction from e leaves exactly e^3 over that denominator. These are algebraic equalities, with no omitted remainder. □

As $e \to 0$, the Newton error divided by e^2 tends to $1/d$, and the Halley error divided by e^3 tends to $1/d^2 = 169/360$. This proves local quadratic and cubic convergence, respectively, in exact arithmetic near this simple root; it assigns no universal number of iterations to either method.

The same identities establish the chosen trajectories. On the increasing branch $2e + d > 0$, so Newton's first step from below the root overshoots it. Thereafter $0 < N(y) - \alpha < e$ whenever $e > 0$; the upper sequence decreases, and its only possible limit in this sector is α. Halley's denominator is $3(e + d/2)^2 + d^2/4 > 0$. For $-d/2 < e < 0$, its new error remains negative and has smaller magnitude: subtracting e^2 from that denominator gives $(e + d)(2e + d) > 0$. Its sequence increases from the seed toward the same root. Neither argument crosses the vertex or silently switches to the smaller root.

Error order also has a coordinate scale. If $E = x - g_x\alpha = g_x e$, the output-coordinate identities are

$$E_N = \frac{E^2}{2E + g_x d}, \qquad E_H = \frac{E^3}{3E^2 + 3g_x dE + (g_x d)^2}.$$

Their limiting coefficients are $1/(g_x d)$ and $1/(g_x d)^2$. The orders survive the conversion, while their numerical constants change. An error reported in normalized coordinates must therefore be converted before comparison with an output tolerance. The plot and the printed table both use reported-output coordinates throughout.

For this fixture the second derivative requires no estimation: its constant value follows from the declared coefficients, which makes a second-order response particularly convenient to retain. The work ledger that separates iteration count from computational cost in a field problem is set out in Section 6.8.

6.3 The integer updates actually compared

Now retain the source's finite arithmetic. Let $S = 10^{24}$ and store $X = Sx$ as an integer. Put $A = 325$, $B = 58320$, $C = 1889568$ and

$$F(X) = AX^2 - BXS + CS^2, \quad F_p(X) = 2AX - BS, \quad F_{pp} = 2A.$$

The source's two integer updates, named in the appendix, become

$$X^N_{k+1} = \left\lfloor \frac{A(X^N_k)^2 - CS^2}{2AX^N_k - BS} \right\rfloor, \tag{6.6}$$

$$X^H_{k+1} = X^H_k - \left\lfloor \frac{2F(X^H_k)F_p(X^H_k)}{2F_p(X^H_k)^2 - F(X^H_k)F_{pp}} \right\rfloor. \tag{6.7}$$

The Newton numerator is nonnegative on the reproduced orbit, so natural subtraction does not clip it. Both denominators are positive. The signed division in the Halley expression therefore uses Euclidean floor even when its numerator is negative. These facts are checked at every generated step by the document-local script.

Turn	Newton, floor of updated value	Halley, floor of correction
0	129.600000000000000000000000	129.600000000000000000000000
1	137.700000000000000000000000	136.953191489361702127659575
2	137.016233766233766233766233	137.011290527013905233309938
3	137.011290807319124540443729	137.011290548979457087737301
4	137.011290548979457793403451	137.011290548979457087737301
5	137.011290548979457087737300	137.011290548979457087737301

Table 6.1: Exact reproduction at scale 10^{24}, with the seed counted as turn zero. Entries are integers with an inserted decimal point; they are not floating-point approximations.

The two floors occupy different positions. Newton takes the floor of its updated value. Since X is integral, Halley takes the ceiling of its updated value: $X - \lfloor X - SH(X/S) \rfloor = \lceil SH(X/S) \rceil$. Their one-step rounding errors consequently lie in $(-S^{-1}, 0]$ and $[0, S^{-1})$, respectively. Once this arithmetic scale is reached, local convergence order cannot decide the last stored digit.

All products and differences inside these expressions are exact integers. The reproduction introduces rounding only at the displayed division, rather than rounding the residual, derivative, and quotient separately. This matters most in the first Halley step: the residual is negative, so flooring its negative correction and then subtracting it rounds the new value upward. Replacing that operation by division truncated toward zero would implement a different machine and would change the table. The script uses Python's integer floor division only after verifying that the divisor is positive.

Halley first reaches its fixed scaled integer at turn three and remains there through turn six; Newton first reaches its fixed integer at turn five. The former is one unit in the last place above the latter, so stabilization at turn three does not certify that Halley's displayed final digit is the root's truncation. The source's four-turn budget, declared for this comparison, distinguishes the trajectories: Newton is still approaching its fixed cell at four, while Halley is already stationary.

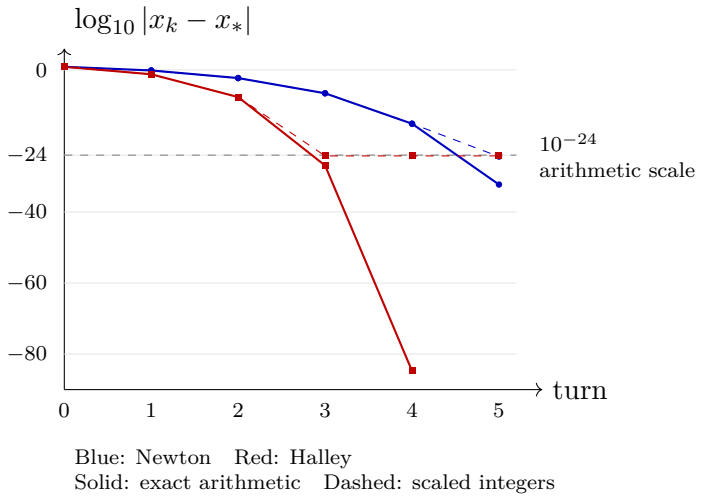


Figure 6.1: Error in reported-output coordinates. Exact-arithmetic curves follow (6.5); dashed curves use the reproduced integer updates. Halley's exact turn-five error is below the plotted range. Decimal calculations with 180-place working precision locate the plotted points; they do not certify the final reporting cell.

6.4 A certificate obtained without either update

An independent bisection starts with $L_0 = 2$, $U_0 = 17/8$. The residual signs are $r(L_0) = -2$ and $r(U_0) = 13/64$. At each step form the exact rational midpoint and replace the wall having its sign. This computation uses neither Newton's derivative nor Halley's second derivative, and takes no iteration output from Table 6.1 as an input.

Because $r' > 0$ throughout the initial interval, continuity and the sign straddle give exactly one enclosed root. Induction preserves that straddle and halves the width. After 108 halvings the normalized bracket endpoints are

$$L_{108} = \frac{5489222943682627729547976100959957}{2596148429267413814265248164610048},$$
$$U_{108} = \frac{2744611471841313864773988050479979}{1298074214633706907132624082305024}.$$

The exact width after conversion is

$$g_x(U_{108} - L_{108}) = \frac{81}{10 \cdot 2^{108}} < 2.50 \cdot 10^{-32}. \tag{6.8}$$

The 108 steps are this volume's own budget. The source runs a separate midpoint budget of one halving per cell of its description, currently 106, and its bracket after those halvings is likewise narrower than one printed place; neither budget is the four jar turns. Directed decimal enclosure of the two rational walls yields

$$\begin{aligned} 137.011290548979457087737300387516\overline{95} \;&\le g_x L_{108} < g_x\alpha < g_x U_{108} \\ &\le 137.011290548979457087737300387516\overline{98}. \end{aligned}$$

An arbitrarily short interval can still cross a decimal-cell boundary, so narrowness alone does not settle the printed value. The decisive integer checks are

$$\lfloor S g_x L_{108} \rfloor = \lfloor S g_x U_{108} \rfloor = 137011290548979457087737300.$$

These endpoint tests can themselves be stated without decimals. For a normalized bracket endpoint N/D, $D > 0$, and proposed reporting integer K, the required inequalities are

$$5KD \le 324SN < 5(K+1)D.$$

For a residual-sign test, the sign of $r(N/D)$ is exactly the sign of $13N^2 - 36ND + 18D^2$. The midpoint ledger and the reporting test therefore reduce to products, sums, differences, and comparisons of integers. The JSON retains the rational walls as numerator–denominator pairs, so another implementation can check the certificate without trusting its decimal display or its root approximation.

Monotonicity of floor then certifies the 24-place truncation

$$\boxed{137.011290548979457087737300} \tag{6.9}$$

Newton's turn-five integer equals this certified integer. Halley's stationary integer exceeds it by one. Both are excellent approximations, but only the first gives the declared truncation cell. A distinct rounding protocol would need its own cell test.

6.5 Residual size, sensitivity, and reproducibility

For any sign-straddling bracket $[L, U]$ above the vertex, define $m = 26L - 36 > 0$. For a tested point y in that bracket,

$$|r(y)| = \left| \int_\alpha^y r'(t)\,\mathrm{d}t \right| \ge m|y - \alpha|, \qquad |g_x y - g_x \alpha| \le \frac{g_x}{m}|r(y)|.$$

Thus a residual becomes a value-error bound only after retaining a positive derivative margin and the conversion factor. The initial bracket already gives $m = 16$; its refined lower wall gives a stronger bound. Near the vertex the inverse factor would deteriorate. The exact discriminant identity explains why the selected simple root itself has a nonzero response, while the bracket establishes which root is being tested.

The finite numeral remains rational, whereas the selected root is irrational: $\alpha = (18 + 3\sqrt{10})/13$ is finitely represented by its coefficients, and it has no terminating decimal. Indeed, the quadratic discriminant is 360, strictly between 18^2 and 19^2; an integer that is a rational square must be an integer square after cancelling the fraction. The quadratic formula therefore excludes a rational root. A successful decimal certificate establishes membership in a reporting cell, with a controlled nonzero residual: an exact algebraic value and a usable finite indication remain distinct.

A small successive change answers another question. Either rounded iteration eventually has zero successive change in its integer state, although its stored value still has nonzero polynomial residual. Its stopping observation certifies a fixed point of the rounded map. The sign-straddling interval, derivative margin, and endpoint-cell inequalities supply the additional statements about the exact root. Stable iteration, small residual, and stable displayed digits thus have three different acceptance calculations.

The same derivative margin can carry a model-error contribution. Suppose a perturbed residual $\widetilde{r}$ has a root $\widetilde{\alpha}$ in this bracket and $|\widetilde{r} - r| \le \eta$ there. Then $|r(\widetilde{\alpha})| \le \eta$, and the preceding integral estimate gives $|g_x\widetilde{\alpha} - g_x\alpha| \le g_x\eta/m$. Both the perturbation envelope and preservation of the branch are hypotheses. The calculation enters an instrument's error budget once coefficient uncertainties have supplied such an envelope.

6.6 Coefficient uncertainty and calibrated parameter sensitivity

The preceding digit certificate treats every coefficient as exact. To make the role of coefficient uncertainty quantitative, write the residual as

$$r(y; a, b, d_0) = ay^2 - by + d_0, \qquad (a, b, d_0) = (13, 36, 18).$$

Implicit differentiation at a simple root gives the local sensitivities

$$\frac{\partial y}{\partial a} = -\frac{y^2}{2ay - b}, \qquad \frac{\partial y}{\partial b} = \frac{y}{2ay - b}, \qquad \frac{\partial y}{\partial d_0} = -\frac{1}{2ay - b}.$$

These are derivatives of the root with respect to input coefficients, not extra terms in Newton's correction. They quantify how a calibrated input allowance propagates into the calculated quantity. A standard uncertainty calculation could combine their products with a coefficient covariance matrix. A deterministic acceptance calculation instead needs bounded coefficient intervals and a branch-preservation check [22, 24].

Suppose each coefficient changes in absolute value by at most 10^{-8}. On the complete initial bracket $2 \le y \le 17/8$, the residual perturbation is bounded by

$$\eta \le 10^{-8}\left[(17/8)^2 + 17/8 + 1\right] = \frac{489}{64}\,10^{-8}.$$

This bound is much smaller than either endpoint sign margin: the unperturbed values are $r(2) = -2$ and $r(17/8) = 13/64$. The perturbed residual therefore still changes sign. Its derivative is at least $16 - (2 \cdot 17/8 + 1)10^{-8} > 0$, so it has exactly one root in this bracket. At that perturbed root the original residual has magnitude at most η. The original derivative margin $m = 16$ then yields the finite output allowance

$$|\widetilde{x} - x_*| \le \frac{g_x \eta}{16} = 3.094453125 \times 10^{-7}.$$

This is a rigorous bounded-input example, not a linearized estimate; the 10^{-8} allowances are illustrative assumptions. They show how an explicit parameter uncertainty can dominate a 24-place arithmetic certificate by many orders of magnitude while both calculations remain correct for their stated inputs.

Changing the output gain supplies another contribution. If the physical indication is $Y = gx + b_0$, and its calibrated gain and offset have bounded errors ε_g and ε_b, then a calculated input error $|\delta x| \le \varepsilon_x$ gives

$$|\delta Y| \le |g|\varepsilon_x + |x|\varepsilon_g + \varepsilon_g\varepsilon_x + \varepsilon_b.$$

The gain has the units of Y per unit of x, and the offset has the units of Y. A quantizer adds its own output allowance. This is how a root certificate could enter a length, voltage, or phase measurement; without such a measurement equation, the number printed here is a dimensionless result of the specified computation.

6.7 Where the coefficients come from, and their neighbours

The previous section perturbed the coefficients as if they were calibrated inputs. In the source construction they are not inputs. They are four readings the construction takes of its own structure, and the argument of the middle card combines them into one polynomial.

Name the four models t, c, x and r. Their combination is

$$t^2(c-t)\,x^2 - 2ctr^2\,x + cr^4 = 0, \tag{6.10}$$

linear in c, quadratic in x, cubic in t and quartic in r. With $t = 5$, $c = 18$ and $r = 18$ it is exactly $q(x) = 325x^2 - 58320x + 1889568$: $5^2 \cdot 13 = 325$, $2 \cdot 18 \cdot 5 \cdot 324 = 58320$ and $18 \cdot 18^4 = 1889568$. *Proved here*, by the three products. In the normalized coordinate the same three readings appear as lengths of records. The source reads its argument as a list of tapes and cuts that list at two depths taken from two of its own readers, four and five. The first cut has length $13 = c - t$, the second length $18 = c$; the leading and constant coefficients of $r(y) = 13y^2 - 36y + 18$ are those two lengths, and the middle coefficient is twice the second, the factor two being the length of the source's basis reading. The gain $g_x = 324/5$ is the second length squared over the second depth. *Measured*, extracted from the source's declarations.

The four readings as calibrations. Read as a measurement, the quartic fixes three conventions and one scale, each against a standard internal to the source. The weight c is what an agreeing comparison counts, and fixes which comparison is named TRUE. The weight t is carried only by a step from one state to the other, and its place in the transition table of Section 7.6 fixes a direction. The quartic does not record that direction, since a matrix and its transpose have the same characteristic polynomial. The unknown x is reported on the increasing branch (Section 6.1), which fixes up from down. The fourth reading, r, fixes the reporting scale through $g_x = r^2/t$. The right-hand side is a declared zero: a stored value is declared a zero of the residual at the resolution of the reporting cell when it satisfies the stated stopping predicate, here the same-bin endpoint test of Section 6.4. The declaration does not assert that the residual vanishes at the stored value, which it does not, since the root is irrational; a finite computation reaches the root only as a bracket whose walls straddle it. Relabelling the two states or transposing the table leaves every count $\mathrm{tr}(M^n)$ unchanged, and the counts grow at the larger root, the one the increasing branch reports. *Proved here*, by the trace and the eigenvalues. The assignment of name, direction and orientation to c, t and x is *Proposed*.

The three conventions close inside the construction. The one comparison that does not close is with the measured constant.

6.7.1 The comparison with the measured constant

The certified reading (6.9) is $137.011\,290\,5\ldots$. The recommended value of the inverse fine structure constant is $\alpha^{-1} = 137.035\,999\,177(21)$ [31], with relative standard uncertainty 1.5×10^{-10}. The reading lies below it by $0.024\,708\,6\ldots$, a relative difference of 1.80×10^{-4}, about 1.2×10^{6} times the standard uncertainty of the recommended value. Measurement therefore excludes the reading as a value of α^{-1}. *Standard* for the cited value; *Proved here* for the difference, by subtraction.

The relative difference is not removed by relabelling, transposing, or choosing the branch, and so can be carried only by the scale. Under the SI the scale-bearing electromagnetic constants are the vacuum permeability and impedance; the reading puts the impedance at $376.798\,\Omega$ against the recommended $376.730\,\Omega$. *Standard* for the difference against the cited value; *Proposed* for its location in the scale.

A second measurement fixes the same scope directly. The electron's anomalous magnetic moment is predicted by quantum electrodynamics as a series in α whose leading term is $\alpha/2\pi$, and it is

measured as $a_e = 1.159\,652\,180\,59(13) \times 10^{-3}$ [13]. *Standard.* Replacing the measured coupling by the reading shifts α by 1.80 parts in ten thousand, and so shifts the predicted anomaly, to leading order, by about 2.1×10^{-7}: some 1.6×10^{6} times the measurement's uncertainty. *Proved here*, to leading order. As a value of the fine structure constant in every circumstance, the reading is therefore excluded. The companion cards offer it only as a prediction for one stated arrangement, tested there. The exclusion concerns this quantitative identification; the certificate, the exact error laws, and the derivations of Chapter 2 stand on their own hypotheses. Elsewhere this volume cites this section rather than restating the comparison.

6.7.2 Neighbouring constructions

A construction with this many lengths might be tuned to land near any target. The test varies the construction where it can vary and asks how often the reading lands near the measured value of Section 6.7.1. Keep the source's list of tapes and its two rules, and let the two cut depths range over $1, \dots, 12$, the basis length over $1, 2, 3$, and the root over both branches: 412 neighbouring constructions with a real reading. Exactly one lands within 10^{-3} of the measured value, the source's own, at 1.80×10^{-4}; the nearest other is 3.67×10^{-3} away, at depths nine and ten; two lie within 10^{-2}. *Measured*, by direct evaluation of every neighbour in a run of the enumeration.

first depth	second depth	basis	reading	relative distance
4	5	2	137.011 29	1.80×10^{-4}
9	10	2	136.533 33	3.67×10^{-3}
10	11	2	138.740 40	1.24×10^{-2}
8	9	2	134.807 02	1.63×10^{-2}

Table 6.2: The closest neighbouring constructions to the measured value.

Within this enumerated family, the constructions reachable by changing where the source cuts its own records, the source's cut is isolated near the measured value: the next nearest lies 3.67×10^{-3} away. *Measured*, by evaluating every member. The scan concerns the family, not how the construction was found. It does not vary the contents of the records themselves, the list of tapes and their lengths, which are fixed by the definitions of the middle card; a scan over those is a scan over other constructions altogether. Isolation is not agreement: the source's reading still differs from the measured value by the 1.80 parts in ten thousand of Section 6.7.1. *Proposed*: that this difference is the residue the companion cards locate, rather than an error in the four models.

6.8 Resolution, stopping, and computational work

The stored scale $S = 10^{24}$ sets a uniform output increment $\Delta_x = 10^{-24}$. Truncation of a positive exact value has one-sided error in $[0, \Delta_x)$; nearest rounding has error at most $\Delta_x/2$ with a declared tie rule. The source's integer Newton and Halley formulas implement different positions of the floor operation. Their fixed integer states consequently differ by one increment even though both approximate the same simple root extremely closely. That difference is completely explained by the arithmetic rules; it is not an observed fluctuation in a physical constant.

For a required output tolerance ε_x, an interval solver can stop when its converted interval has width at most $2\varepsilon_x$ and report its midpoint with that error allowance. A decimal truncation decision

needs the stronger same-bin endpoint test already used above. The stopping rules answer different questions: one bounds a numerical error, while the other fixes one particular display code. Repeated displayed digits supply neither the sign margins nor the interval inclusion. A repeated display is the arithmetic's slogan: it may appear in an exposition, and it certifies nothing.

Newton's exact local error is proportional to the square of the previous error; Halley's is proportional to its cube. Computational cost still depends on the implementation. For this scalar quadratic, $r'' = 26$ is a stored constant, so its evaluation is negligible compared with a general field problem. A multidimensional second response may require additional residual evaluations, derivative storage, tangent solves, or contractions with a large field increment. The total work is therefore recorded as residual evaluations, linear iterations, preconditioner applications, memory, and wall time for a fixed output tolerance; iteration counts alone omit those different costs. The operator costs of the four model presentations are evaluated on their own stated domains.

The 108 bisections supply an independent verification route. They require only exact residual signs at rational points; they use neither Newton's tangent nor Halley's second response. Their work is deliberately redundant with the fast updates. Agreement of the independent bracket and the printed Newton integer is an arithmetic consistency check with a specified branch and rounding convention. The certificate is the bracket of Section 6.4. A program that halts with the printed digits has established that it halted with them.

The reproducible input is `scalar_reading.py`. Run it to regenerate the table, vector figure, certificate macros, and `scalar-reading.json`; `-check` verifies the retained products. The ledger records every rational bracket and iterate, integer state, coefficient, seed, rounding choice, plotting precision, and inspected source hash. Its assertions check denominator signs, example rows, endpoint truncations, and the stationary integers.

The reading is calibrated as a computation. A computed number becomes a reading of a physical quantity only through an instrument chain, and the source delivers its coefficients through channels of its own; Chapter 7 treats both.

Chapter 7

Calibrated instruments and common readings

Chapter 3 set out the general framework: residuals scaled to compatible units, a stability estimate that turns a residual into a state error, an error budget for an observable (Section 3.7), and the comparison of two readers against a reporting cell (Section 3.8). By Definition 0.1, a calibrated reading reports a quantity with its admitted inputs, its operation of reading, its reference, and the allowance governing the comparison; for an instrument these conditions are the measurand, the response model, the reference, the operating range, and the uncertainty. Chapter 6 supplied all of them for one computation, a dimensionless number whose single comparison with the measured constant is stated in Section 6.7.1. The first three sections below supply those conditions for an instrument chain; the remaining four find several of them supplied exactly by the source's own channels.

7.1 The measurement chain and the quantity being measured

A numerical result becomes a physical reading through a measurement chain. The chain starts with a specified specimen, field, or operating condition; a sensor responds to a quantity of interest; signal conditioning applies a gain and an offset; a digitizer supplies a finite record; and a calibration converts that record to the reported units. A simulation has a parallel chain: material and boundary data, a field equation, a discretization, an iterative solution, and an output functional. Agreement at the end of two chains is a statement about those complete chains, including their tolerances; it does not require their internal mechanisms to be identical.

Let s denote the operating state, and let $Q(s)$ denote the scalar or vector measurand, in declared units. For instrument i, write $E_i(s)$ for its ideal internal indication and r_i for its calibration curve. A bound

$$\|r_i(E_i(s)) - Q(s)\| \le \eta_i$$

allows for the discrepancy of the complete idealized instrument from the measurand. This can include probe placement, a sensor model, or a difference between a spatial average and a point value. If the acquired indication $\widehat{z}_i$ satisfies $\|\widehat{z}_i - E_i(s)\| \le \delta_i$, and the calibration has sensitivity bounded by L_i, then

$$\|r_i(\widehat{z}_i) - Q(s)\| \le \eta_i + L_i\delta_i.$$

For a scalar differentiable calibration, one can take $L_i = \sup |r_i'|$ over the admitted indication interval. A steep inverse calibration amplifies small indication errors. A vector calibration uses the induced matrix norm of its sensitivity matrix.

Applying this estimate to two instruments gives the instrument form of the two-reader bound of Section 3.8:

$$\|r_1(\widehat{z}_1) - r_2(\widehat{z}_2)\| \le \eta_1 + \eta_2 + L_1\delta_1 + L_2\delta_2. \tag{7.1}$$

To prove it, add and subtract each calibrated ideal indication and $Q(s)$, then use the triangle inequality. The estimate compares the same operating state and the same measurand. If the probes sample different times or positions, their timing and placement errors belong in η_i; using the same unit on both displays does not remove those errors.

An exact common reading has $\eta_i = \delta_i = 0$ for the selected numerical chain. A physical calibration generally carries uncertainty. A worst-case bound such as (7.1) adds admitted error magnitudes. Standard uncertainties, coverage intervals, and probabilistic confidence statements require the additional statistical assumptions of the measurement model [22, 24]; a root-sum-of-squares calculation replaces this hard bound only with a justified treatment of correlations and error distributions.

7.2 A quantitative calibration and digitization budget

Consider two hypothetical displacement instruments observing the same motion x, with $|x| \le 1\,\mathrm{mm}$. Their nominal voltage sensitivities are $a_1 = 100\,\mathrm{V\,m^{-1}}$ and $a_2 = 200\,\mathrm{V\,m^{-1}}$. Within their calibrated range, suppose the actual conditioned voltage obeys

$$v_i = a_i(1 + \gamma_i)x + b_i + \Delta b_i + n_i.$$

Here b_i is the nominal offset, γ_i the fractional gain error, Δb_i the uncorrected offset, and n_i the admitted voltage noise. The reported displacement is $(\widehat{v}_i - b_i)/a_i$, followed by decimal rounding. The model fixes what each error acts on.

Take a uniform 12-bit digitizer over a $4\,\mathrm{V}$ span. Its ideal code spacing is

$$\Delta v = \frac{4}{4096}\,\mathrm{V} = 0.9765625\,\mathrm{mV}.$$

For nearest-code conversion away from saturation, quantization contributes at most $\Delta v/2$. The corresponding displacement errors are $4.8828125\,\mu\mathrm{m}$ and $2.44140625\,\mu\mathrm{m}$. These values use a declared ideal digitizer: transition-level nonuniformity, reference drift, and amplifier nonlinearity would require their own specifications, which a finite code count does not bound.

Assume $|\gamma_i| \le 2 \times 10^{-4}$, $|\Delta b_i| \le 0.2\,\mathrm{mV}$, $|n_i| \le 0.3\,\mathrm{mV}$, and reporting to the nearest $0.01\,\mu\mathrm{m}$. Let the model and placement errors be $\eta_1 = 1\,\mu\mathrm{m}$ and $\eta_2 = 1.5\,\mu\mathrm{m}$. Direct conversion to displacement gives the following hard budget.

Their readings can differ by at most $17.73421875\,\mu\mathrm{m}$ under these assumptions. A larger disagreement rejects at least one premise of the measurement chain. A smaller disagreement is compatible with the budget; it cannot identify which individual errors occurred. In particular, the instruments may share a reference error or the same mistaken mechanical model. Repeating a measurement can reduce an estimated random component while leaving a common calibration error unchanged.

Additional printed digits need not improve a physical tolerance. Reducing the final decimal step by a factor of ten barely changes a budget dominated by sensor noise and digitization. Increasing

Error contribution	Instrument 1 μm	Instrument 2 μm
Model and placement	1	1.5
Residual offset	2	1
Admitted voltage noise	3	1.5
Ideal code quantization	4.8828125	2.44140625
Gain over the declared motion range	0.2	0.2
Final decimal rounding	0.005	0.005
Maximum total	11.0878125	6.64640625

Table 7.1: An illustrative deterministic displacement budget. The numbers are assumed specifications of the example, rather than measurements of an existing apparatus.

the analog gain improves displacement resolution only while the full signal, offset, and transient excursions remain within the digitizer range. A complete operating specification therefore declares both a smallest resolvable increment and a largest unsaturated input.

7.3 A common indication and an independent reference

A practical comparison excites more than one operating point. Zero input tests residual offset; two separated reference displacements test gain; additional points test nonlinearity. Reversing the direction of motion can reveal hysteresis, while a hold at fixed displacement can reveal drift. Each test compares a specified observable over a specified time and bandwidth. A static calibration cannot establish the response to an arbitrarily rapid transient.

Any comparison must allow a simple common-mode failure. If two calibrated outputs are both $x + b$, their difference is exactly zero for every x, although both have error b. A shared reference or shared processing model can produce such a correlated bias. An independently established reference constrains that otherwise invisible component. Likewise, running the same polynomial through two storage formats verifies agreement of the processing paths, while measuring an independently prepared specimen tests the additional physical calibration.

For a time-dependent measurement the acquisition specification also declares the sample times and the analog bandwidth. Two digitized records can agree at every sample and differ between samples. Bounding those unobserved excursions requires a stated smoothness, frequency, or dynamics assumption. The associated interpolation tolerance then belongs beside the quantization tolerance, rather than being hidden inside the count of reported digits.

7.4 Exact count channels and storage consistency

The source delivers its coefficients through a count channel, with an allowance of exactly zero. For a finite tape t, let $c(t)$ be its number of cells. A computation of the form $F(t) = G(c(t))$ receives only this integer. Replacing every cell's payload while retaining the same number of cells gives $c(t') = c(t)$ and therefore $F(t') = F(t)$. This is an exact input-output invariance of the implemented channel, analogous to a pulse counter that records the number of accepted pulses while discarding their individual amplitudes.

The source's selected tape readers count facts: each returns the cell count of a stored tape. The source also constructs a second stored form of the same tape, and its reading of that form preserves the count. Two named results compare the resulting scalar calculation. The first checks that another implementation of the reading pair delivers the same pair; the second overwrites cell contents while retaining the count and checks that the same pair is produced. A third statement accepts an alternative counter only when it agrees with tape length on every admitted tape, and under that premise produces the same result. These are the source's declarations, inspected here; the appendix names them.

A detector that reads a cell's payload is a different measurement channel. It may distinguish the two tapes even when the count channel cannot. Similarly, two pulse trains can have the same count while differing in arrival times, energy, spectrum, or polarization. Exact equality of the count is valuable because its scope is fixed: the two selected implementations deliver the same integer input to the same scalar computation.

Consequently a polynomial coefficient can be inserted explicitly and the two count-based implementations can still agree exactly. The agreement establishes computational consistency for those inputs; experimental validation would additionally require an independently calibrated link between the coefficients, the apparatus, and the physical measurand, of the kind in Section 7.1. The reported truncation rests on the bracket certificate of Section 6.4, which a reader can recompute without the source or its checker.

7.5 An erased channel and the auxiliary record

Some processing rules deliberately merge all inputs into one reported class. The source contains an exact example. It compares pairs of natural numbers using

$$(a, b)\ \mathcal{R}\ (c, d) \quad \Longleftrightarrow \quad ad \leq cb.$$

For any two pairs, either this comparison or the reverse comparison holds. The implemented class-identification rule treats either direction as sufficient to merge the two records. All pairs are therefore merged. The source records this as a theorem that the quotient has exactly one point; the appendix names it. Any output computed from this single merged class alone is constant.

In measurement language this is a completely erased output channel. The surrounding apparatus may still store the original numerator, denominator, sign choice, or interval. Those fields constitute an auxiliary record. A familiar example is reporting a squared voltage while separately retaining its polarity bit: the squared channel alone loses the sign, and the augmented record recovers it. The separate specification manuscript constructs two different binary auxiliary records and states which measurement each one completes.

The scalar calculation retains a numerator, a positive denominator, the selected upper-root interval, and a bracket certificate. Its decimal reading is computed from those retained data, not recovered from the erased channel. Every recovered quantity has a named source in the stored measurements; a calculation cannot reconstruct a discarded distinction unless an additional measurement, prior constraint, or retained record supplies it.

7.6 A finite transition network and the scalar root

The polynomial itself has a second exact operation of reading: closed-walk counts on a finite network. Let

$$M = \begin{pmatrix} 18 & 5 \\ 18 & 18 \end{pmatrix}.$$

Interpret M_{ij} as the number of distinct allowed transitions from state i to state j during one clock step. This is a multiplicity matrix, rather than a probability matrix: its rows need not sum to one. The number of closed transition sequences of length n is $N_n = \operatorname{tr}(M^n)$. Its characteristic polynomial is

$$\chi_M(\lambda) = \lambda^2 - 36\lambda + 234, \qquad \chi_M(\lambda) = 13r(\lambda/13), \quad r(y) = 13y^2 - 36y + 18.$$

The two growth multipliers $\lambda_\pm = 18 \pm 3\sqrt{10}$ therefore give exactly the two roots $y_\pm = \lambda_\pm/13$ used in Chapter 6. The output conversion in that chapter adds its separately declared scale.

Cayley–Hamilton gives

$$N_0 = 2, \qquad N_1 = 36, \qquad N_{n+2} = 36N_{n+1} - 234N_n.$$

For example $N_2 = 828$; both direct matrix multiplication and the recurrence give that integer. Moreover $N_n = \lambda_+^n + \lambda_-^n$, so

$$\lim_{n\to\infty} \frac{\log N_n}{n} = \log \lambda_+.$$

This is a logarithmic transition-count growth rate, in nats per clock step. If one clock step lasts Δt, the corresponding rate per second is $(\log \lambda_+)/\Delta t$. The counts specify no energy scale, temperature, or probability distribution; a thermodynamic entropy production rate, measured in $\mathrm{J\,K^{-1}\,s^{-1}}$, is a different calibrated quantity.

The entire sequence also satisfies the generating identity

$$\exp\left(\sum_{n\geq 1} \frac{N_n z^n}{n}\right) = \frac{1}{\det(I - zM)} = \frac{1}{1 - 36z + 234z^2}.$$

For this matrix, expanding $-\log(1-\lambda_+ z) - \log(1-\lambda_- z)$ proves the identity coefficient by coefficient. Its singularities occur at the reciprocals of the multipliers.

The source derives the table from one rule on the arms of its comparison: an arm carries the thirteen-cell coefficient tape exactly when its stored fact can click, and every arm carries the coordinates. The arm that stores a false comparison carries the coordinates alone, which gives the entry 5; the other three give 18. The table so obtained equals the written table, and the recurrence counts equal the closed-walk counts on the two-state graph. *Measured*, extracted from the source's declarations, inspected and not re-run here. The remaining obligation is narrower: the reason for the arm rule itself. A matrix with the same characteristic polynomial has the same $N_n = \operatorname{tr}(M^n)$ for every $n \geq 1$, since these traces are power sums of the eigenvalues; what it need not share is the state-to-state counts $(M^n)_{ij}$, the graph, or a physical realization.

7.7 Software instruments, blanks, and independent checks

For a reading of computational work, *measured* means observed in a run of the source's meter, in its own work units. The source's structural cost experiments measure computational work under

a stated protocol. They retain both a raw sample and a blank, then subtract in signed integer arithmetic. If $H(n)$ is the sample and $B(n)$ the blank, the corrected reading is $C(n) = H(n) - B(n)$; a negative result remains visible. Truncating the subtraction at zero would destroy the evidence of a sample below its blank.

The source's walk command visits the same nesting depths in ascending and descending order before publishing intermediate declarations. Its pass gap compares corrected readings at matching depths. This removes an identified change in the intervening declaration environment, though not every source of run-to-run variation. A reproducible performance study would retain software versions, configuration, input fixture, meter definition, warm-up policy, repeated runs, and the corresponding blanks. Heartbeats are the implementation's work units, rather than an SI time or energy measurement.

A second fixture constructs a pure chain and a branched chain of variation records. On the pure chain the selected Newton and Krylov traversals have the same output length. On the branched fixture, one retains a single branch and the other retains both. Their specified lengths are $n + 1$ and $1 + n(n + 3)/2$. The measured work difference therefore tests the cost of those implemented traversals.

Interpreting this as a measurement of fluid convection would require a further engineering model: which stored quantity represents velocity, which operation represents spatial differentiation, which clock represents time, and how the measured work relates to force, power, or error. The actual Newton linearization of $(u \cdot \nabla)u$ already contains $(u \cdot \nabla)h + (h \cdot \nabla)u$. A difference in traversal cost cannot replace a check of those physical terms.

Finally, an iterative solver does not uniquely identify the field equation that produced it. If an operator has independent blocks $A = T \oplus C$ and the initial vector excites only T, every observed Krylov vector remains in that block. Changing C changes no observed recurrence. Lanczos additionally assumes an appropriate self-adjoint problem and inner product; convection Jacobians are generally nonsymmetric. Arnoldi/GMRES and an explicitly specified preconditioner preserve that distinction [25, 38]. Multigrid also requires its coarse operators, transfers, smoothers, and boundary treatment.

Numerical verification checks that the implemented calculation solves its declared mathematical problem to its declared tolerance. Instrument validation checks that this problem and its calibration describe the physical measurement in the stated operating range. A measured value enters this volume only through a calibrated chain of the kind in Section 7.1; Equation (7.1) states the error budget through which independent calibrated instruments can test a physical identification. The middle card's review asks for facts; each answer must be located in one of these readings and carry its calibration. Chapter 8 keeps that record.

Chapter 8

The record, the thirty-six replies, and the press check

The certified reading of Chapter 6, and the channels of Chapter 7 through which the middle card counts, are publishable only in a record that retains the datum licensing each.

The middle card's last two acts define two roles. The first is a *review*: thirty-six demands, indexed by pages from thirty-six down to one, each requiring a claim to show the fact it rests on by pairing the claim as found with the same claim as handed over by name. The pages of the review are an index, not a count: they say where the review is pointing. The second is a *typeset record*: a publication that carries one quantity unchanged, counts its own entries, accumulates a trace, and is read once at the end. Either surrounding card may play either role, and this chapter makes this manuscript play both. The companion volume fixes what must be retained by a record of the whole: the fourth coordinate, the sheet sign, the Clifford seam, the certified faces. This volume fixes what may be claimed from a reading: units, scales, stability, an error budget, a reporting cell, each computed as the necessary condition recovered from the middle construction's sufficient one. The record checks that both are carried; it is a mechanical audit of retention, and each entry cites the place where its mathematics is proved, cited, reproduced, or proposed.

8.1 The record

Every entry carries a calibrated reading in the sense of Definition 0.1, with the calibrations *proved here*, *standard*, *measured*, and *proposed*. In this chapter *measured* always means extracted from the inspected declarations of the middle card, as in Section 7.4, without running it, except in entry 33, where it means work counts the source records from its own runs, which this manuscript does not rerun. The fifth kind, *presence*, is used for exactly three entries: the review's pages 24, 30, and 31 extract no fact by construction, and the corresponding entries mark where rhetoric, universality, and deduction are present in this manuscript and record nothing else.

Definition 8.1 (The register)**.** The register is the certified twenty-four-place truncation

$$137.011290548979457087737300$$

of equation (6.9). Its digits are printed from the generated certificate of Chapter 6 and appear otherwise only as the scaled integer of Section 6.4. The selected root $\alpha = (18 + 3\sqrt{10})/13$, whose

scaled value $g_x\alpha$ the register truncates, reappears in Chapter 7 as $\lambda_+/13$, with $\lambda_+ = 18 + 3\sqrt{10}$ the growth multiplier of Section 7.6 and $\chi_M(\lambda) = 13\, r(\lambda/13)$; the register is carried unchanged to this chapter, and Chapter 10 does not restate it. At the top of the record it is reported as the *value*. The verdict reads readings of the register, never the register itself.

Definition 8.2 (The counter)**.** The counter is the reply index $k = 1, \ldots, 36$ in the typeset order of Section 8.2. Entry k answers review page k. The counter increases by exactly one per entry and is never skipped, reordered, or batched: entry k is written only after entry $k - 1$ exists. At the top of the record the counter is reported as the *velocity*, and its value there is 36.

Remark 8.3 (Counting is free over the field with one element; writing is not)*.* The field with one element has one element; every map is the identity on it; counting it costs nothing. That is an assumption, and this record names it as one: the field with one element is, as this record reads it, the assumption that counting its elements takes zero steps instead of one. The positional reading $2^n - 1$ of n unit cells in Chapter 10 (Remark 10.2) inherits it, since it treats the n cells as already counted. Writing is not free. Each entry of this record is written in one step, the record grows by exactly one entry per step, and the counter of Definition 8.2 is the tally of those steps and nothing else. The velocity reported by the press check, 36, is therefore the cost of writing the record, not of counting it; over the field with one element the latter is zero. Appending to a deterministic record can leave its Shannon uncertainty unchanged while its length, and the number of steps taken to write it, increase by one per entry. The retained count that Chapter 10 assigns to the typeset record is that length; Remark 10.2 refers back here for the distinction.

Definition 8.4 (Record entry)**.** A *record entry* with index k has four fields.

(i) The *obligation*: the demand of review page k, restated in the vocabulary of this manuscript.

(ii) The *located result*: the propositions, equations, tables, and sections of this manuscript that answer the obligation, cited by number.

(iii) The *calibration*: the calibrated reading of the located result (Definition 0.1), given as its calibration label followed by what the result licenses and under which conditions.

(iv) The *correspondence*: the reading c_k constructed by this manuscript from the located result, the reading s_k handed over by name on the same point, and the proposition

$$c_k \iff s_k,$$

which the entry writes and does not decide.

The reading s_k carries its provenance. Either it is accompanied by a statement quoted exactly from the middle card's patter or from another named source, with the source named, or it is a *comparison statement constructed for the review*, which states in the vocabulary of this manuscript the reading that the obligation puts in question. In a *presence* entry the correspondence pairs the carrier's symbol with itself and extracts nothing.

Definition 8.5 (The trace)**.** The trace is the sequence of the thirty-six record entries of Section 8.2. What accumulates is quoted; no entry on the trace is re-decided or edited after it is written. The calibration column alone is read again, as a whole, by the separate demand-level reading of Section 8.4.

Definition 8.6 (The press check)**.** The press check reads the register once at the top of the record, through readings that Chapter 6 already proves, and nothing beneath it. For two readings X and

Y of the register, with ϕ_X the decidable proposition that X equals the certified integer, the verdict *slipped* for the pair is the decidable proposition

$$\neg(\phi_X \leftrightarrow \phi_Y),$$

decided from the two integer comparisons themselves and never by an appeal to choice. The complete verdict is the triple (slipped, value, velocity).

The slip bit compares agreement with the certified integer K, not agreement of X with Y: when one reading is K, the pair slips exactly when the other differs from it; two readings that both differ from K, even from each other, give *slipped* false. A disagreement between two readings is therefore detected only when one of them is certified, and every pair checked in this chapter has one certified member. A procedure that returns no integer gives *no reading*, not a slip.

8.2 The thirty-six replies

The entries follow Definition 8.4; the proposition $c_k \leftrightarrow s_k$ is written in every entry and decided in none.

Entry 1

Obligation Exhibit the single distinguishable mark on which the publication rests and compare it with the mark the record started from; the descent stops here.
Located result The seed $x_0 = 1296/10$ of Section 6.1 and the bracket start $L_0 = 2$ of Section 6.4; the distinction between deterministic enclosure and statistical confidence, Sections 3.7 and 7.1.
Calibration Proved here: the certified digits rest on a deterministic enclosure from the stated seed and bracket, not on a confidence level.
Correspondence c_1: the record starts from $x_0 = 1296/10$, and every later value of the unknown is admitted by the rule of Section 6.1. s_1: the announced digits rest on confidence in a repeated display (constructed for the review).

Entry 2

Obligation State the local rule under which the mark was admitted and compare the carried fact with the symbol so admitted.
Located result The rule of admission of Section 6.1; the admissibility checks of Sections 4.3 and 4.8.
Calibration Proved here: a value is admitted exactly when the stated rule, branch $y > 18/13$ included, produces it.
Correspondence c_2: a value of the unknown enters only as the seed, an update image, or a bracket wall, on the stated branch. s_2: a number is admitted because the process that produced it was named (constructed for the review).

Entry 3

Obligation Counting must count a concrete indexed carrier, not merely name a class.
Located result The count channel $c(t)$ of Section 7.4; the merged class of Section 7.5.
Calibration Measured, from the inspected declarations: the delivered integer is a tape's cell count, and an alternative counter delivers it only under the premise that it agrees with tape length.

Correspondence c_3: the integer delivered to the scalar computation is the cell count of a stored tape, and two stored forms deliver the same count. s_3: the count is the name of a class of tapes rather than the length of one of them (constructed for the review).

Entry 4

Obligation Encoding must preserve a read path back to the original mark, through the limit process and its index.

Located result The rational walls retained as numerator–denominator pairs, Section 6.4; the named-source requirement of Section 7.5.

Calibration Proved here: the printed decimal follows from the retained integer walls by exact integer inequalities.

Correspondence c_4: the printed decimal is computed from retained integer pairs by $5KD \le 324SN < 5(K+1)D$. s_4: the printed decimal is read back from its own display (constructed for the review).

Entry 5

Obligation Residue must be a limit record indexed down to the counted carrier, not a leftover narrative.

Located result The counterexample $1 + 1/n$ of Section 3.8; the derivative margin $|r(y)| \ge m|y - \alpha|$, $m = 16$, of Section 6.5; the width (6.8), indexed by the halving count; Proposition 2.2, where the cosmological constant is the residue of an integration, fixed by data and carried unchanged.

Calibration Proved here: on the stated bracket the residual is controlled by the derivative margin and the width by the halving count; on the unimodular sector the cosmological residue is fixed by data.

Correspondence c_5: the residual at the certified cell is nonzero and bounded through the derivative margin, and the width is $81/(10 \cdot 2^{108})$. s_5: a small residual is a leftover that further iteration will remove (constructed for the review).

Entry 6

Obligation Record where the binary split is written after execution, reducing both bits, initial condition and signal response, to the still-readable carrier fact, and restore the exhibit the submission constructed and then dropped.

Located result The polarity bit and auxiliary record of Section 7.5, with the companion manuscript's two binary auxiliary records; the sign test of $13N^2 - 36ND + 18D^2$ in Section 6.4.

Calibration Proved here: each halving's bit is the sign of an exactly computed integer, recoverable because it is retained with its wall.

Correspondence c_6: the bit written after each halving is the sign of an integer, retained with the wall it selected. s_6: the sign is recoverable from the merged class alone (constructed for the review).

Entry 7

Obligation Repetition must re-run the experiment, an expectation with a signal response, not restate the sentence.

Located result Section 7.2 (random component and common calibration error); Section 7.3 (operating-point tests); Section 4.5 (regression calculation); Section 7.7 (reproducible-study list).

Calibration Proved here (the budget and the regression check): repetition reduces only the random

allowance, and the regression check compares against a known exact answer; standard (the common-mode failure): a shared calibration error survives repetition.

Correspondence c_7: a repeat changes the random allowance and not the shared bias. s_7: a repeated display of the same digits is a repeated experiment (constructed for the review).

Entry 8

Obligation Show what the number counts: trace both numeric carriers down to the fact each was packaged with, hypothesis or tally.

Located result The count channel of Section 7.4; the floor checks of Section 6.4; the transition counts $N_n = \operatorname{tr}(M^n)$ of Section 7.6, whose matrix entries the source derives from one arm rule.

Calibration Proved here (the floor checks and the recurrence), for the stated scale and matrix; measured, from the inspected declarations (the count channel; the table read off the arm rule; the recurrence agreeing with closed walks). The reason for the arm rule is the remaining obligation.

Correspondence c_8: $N_0 = 2$, $N_1 = 36$, $N_2 = 828$ count closed transition sequences of the stated matrix, and the reporting integer counts cells of width 10^{-24}. s_8: the entries of M were derived from the larger apparatus rather than supplied to it. The middle card's patter states that the table "is now a definition from the arms", and in the same margin that the arm rule "is transcribed, not derived".

Entry 9

Obligation A representation must have a carrier state at the end of the representation chain, not merely a notation.

Located result The integer state $X = Sx$, $S = 10^{24}$, of Section 6.3; the second stored form of Section 7.4.

Calibration Proved here (the integer state): the carrier at the end of the chain is X at the stated scale; measured, from the inspected declarations: the second stored form preserves the count.

Correspondence c_9: the stationary integers of Table 6.1 are states of the carrier X. s_9: the representation is the decimal notation with an inserted point (constructed for the review).

Entry 10

Obligation The physical claim must exhibit its own threshold fact, halting or non-halting, rather than inherit authority from higher levels.

Located result The latched threshold reader of Section 9.6 (the count g_σ following equation (9.6)); the model-discrepancy allowance of Section 3.7; the Burgers gradient calculation of Section 9.7.

Calibration Proved here (the threshold bound and the gradient calculation): the count is bounded for the stated energy supply and threshold, and a speed limit alone bounds no gradient; the closing physical thesis is the author's, calibrated proposed.

Correspondence c_{10}: the count is at most $\lfloor G(0)/(T\sigma) \rfloor$, eighty-three in the example, and a speed limit alone does not bound a spatial gradient. s_{10}: the regularity of a relativistic fluid follows from the speed limit. The closing thesis of Section 9.7 states that "relativistic material effects bound the gradient before the continuum description fails".

Entry 11

Obligation Comparability requires a common physical value process, a base and a step, relating the two quantities; naming them alike is not comparing them.

Located result The closing paragraph of Section 2.5 after Theorem 2.4; the two-reader bound (7.1);

the characteristic-polynomial caveat of Section 7.6; Section 2.6, where the one derivative the four hierarchies share has one reading (Proposition 2.5).

Calibration Proved here: the four actions share the degree bound on their stated field spaces, and two readings are comparable only through a declared common measurand and its allowances.

Correspondence c_{11}: the four actions share $D^4 S_i = 0$, and that degree identifies nothing else. s_{11}: the four models yield a common physical result because they share the closure; in its sharpest form, electrodynamics is the derivative of the modified field equations. The first clause is constructed for the review; the sharpest form is the identification proposed in the remark *Electrodynamics as the derivative of the modified field equations*, Section 2.6.

Entry 12

Obligation An observation must be a projection with a stated dimension, what was projected onto what, not a report of seeing.

Located result The stationary projections of Propositions 2.1 and 2.3; the bounded-observable estimate of Section 2.2; the opening of Chapter 2, where projection can preserve the solution set without preserving polynomial degree.

Calibration Proved here, under the hypotheses of the cited propositions and estimate.

Correspondence c_{12}: each observation is a stated projection, the connection eliminated, the response fields set to zero, or a bounded operator paired with a normalized state, with its dimension declared. s_{12}: an observation is the report that a solution was seen (constructed for the review).

Entry 13

Obligation State where instantaneous presence accumulates into a record, the increment tree or the integration over $\mathrm{d}t$.

Located result The time integral ΔS and the count g_σ of Section 9.6, following equation (9.6); the time-integrated acceleration defect of Section 5.4.

Calibration Proved here, for the stated dissipation model.

Correspondence c_{13}: presence accumulates by the integral $\int_{t_0}^{t_1} \int_0^L |q_t|^2 \,\mathrm{d}x \,\mathrm{d}t$ and by the nondecreasing floor $g_\sigma = \lfloor \Sigma/\sigma \rfloor$. s_{13}: an instantaneous reading is already a record (constructed for the review).

Entry 14

Obligation Measurability must be anchored to an observable clock, a gauge process with a stated reference, rather than asserted.

Located result Section 7.6, where one clock step Δt converts a rate per step to a rate per second; the sample times and bandwidth of Section 7.3; Chapter 9, where the stationary equation specifies no clock.

Calibration Proved here: a rate is per second only through a declared clock step; without one it is per step.

Correspondence c_{14}: every rate in this manuscript carries its clock as a stated reference, and the transition counts carry none. s_{14}: the counting of work units is a measurement of time. The middle card's patter calls its work count "the resolution in time".

Entry 15

Obligation Even the auxiliary or comic concept class must account for its evidence, reduced to the fact it reports.

Located result The figurative names (jar, wall, turn, crank, heartbeat, blank) defined in Section 6.1; the two-mode superposition of Section 5.3, whose only observable is a relative phase.

Calibration Proved here: each name denotes its defined object, and the phase error is bounded for the stated discretization.

Correspondence c_{15}: each figurative name denotes one defined object, and the superposition contributes the phase error $e_N \le (2\pi)^3/(12\sqrt{2}\,N^2)$ and nothing else. s_{15}: a concept named for a color, a bang, or a superposition carries evidence by its name (constructed for the review).

Entry 16

Obligation Identify which fact at the source survives transmission through the channel: the physical and the zero-like datum carry a fact; the unit-like branch carries none.

Located result Section 7.4, where the count survives and the payload does not; Section 7.5, where the merged class carries nothing.

Calibration Measured, from the inspected declarations: the cell count passes the channel unchanged, and the merged class transmits no fact.

Correspondence c_{16}: the cell count passes unchanged, a payload-reading detector is a different channel, and the quotient by $ad \le cb$ has one point. s_{16}: the channel transmits the payload because it transmits the count (constructed for the review).

Entry 17

Obligation Execution must leave a record of the elementary operations performed; a gesture without a record is not execution.

Located result The exact integer operations and Euclidean floor of Section 6.3; the ledger paragraph of Section 6.8.

Calibration Proved here, for that machine: division truncated toward zero would be a different machine with a different table.

Correspondence c_{17}: every operation is an integer product, sum, difference, comparison, or floor, and the ledger records the state at every turn. s_{17}: the plotted curve is the record of the execution (constructed for the review).

Entry 18

Obligation Separate a value from the process that produced it, whatever mode (described, compiled, or executed) the process was recorded in.

Located result The three acceptance calculations of Section 6.5; the stopping rules of Section 6.8.

Calibration Proved here: the certified cell does not depend on the process, provided each process passes its own acceptance calculation.

Correspondence c_{18}: the certified cell is the same whether read from the Newton orbit, the bisection, or the exact root. s_{18}: the value is the fixed point of whichever process printed it (constructed for the review).

Entry 19

Obligation Show the step at which magnitudes are actually added; the additive identity is not a sufficient claim.

Located result Equations (3.6) and (3.7); Table 7.1.

Calibration Proved here: the totals are additive bounds whenever each contribution has a deterministic enclosure.

Correspondence c_{19}: the totals 11.0878125 and 6.64640625 μm are sums of named allowances, and the output bound $6.1 \cdot 10^{-6}$ is $L_Q K_s \sum_j \epsilon_j + \epsilon_Q$. s_{19}: a zero residual is already an additive error budget (constructed for the review).

Entry 20

Obligation A scale must be exhibited as the product that produces it, not by naming the unit.
Located result The conversion (6.2) with $q(x) = 324^2 r(5x/324)$; the scale matrices D_Z and D_R of Section 3.2.
Calibration Proved here: the gain converts the coefficients exactly, between two dimensionless coordinates.
Correspondence c_{20}: $g_x = 324/5$ is the product that converts the coefficients. s_{20}: the gain 324/5 is the speed of light (constructed for the review).

Entry 21

Obligation Before any comparison, exhibit the basis or the null space in which the datum is expressed.
Located result The three-unknown coarse grid of Section 4.7; the two rigid modes of Section 1.6; the Krylov span of Section 4.4.
Calibration Proved here: the coarse solve is exact for the stated right-hand side, whose solution lies in the coarse trial space, and for no other load by that argument.
Correspondence c_{21}: $AP = b$, so P lies in the coarse trial space and the coarse solve is exact. s_{21}: one coarse unknown solves every three-unknown load (constructed for the review).

Entry 22

Obligation State whether finiteness is imposed exactly, as a multiplier or boundary condition satisfying the inf-sup condition, or only penalized by a relaxation parameter; record the finite expansion at its fact-bearing terms.
Located result The saddle block (3.2), its Schur complement, and the inf-sup condition in Section 3.4; the exact lift of Proposition 2.3 against the penalized action of Section 2.3.
Calibration Proved here: in the example the multiplier enforces the constraint exactly, and the penalized objective asserts no limit; standard: the inf-sup condition.
Correspondence c_{22}: the constraint is enforced by the multiplier p, with $S = 1$, $p = 1$, $u = (1/2, -1/2)^{\mathsf{T}}$ in the example, and the penalized reduced objective is quartic. s_{22}: a penalty parameter enforces the constraint (constructed for the review).

Entry 23

Obligation A smooth curve through data is evidence only as an explicit interpolant with its knots and error.
Located result The Hermite certificate (3.5) of Section 3.6; the spline reactions of Section 1.6.
Calibration Proved here: the bound holds on each interval of width h for a function with bounded fourth derivative, and its constant is sharp.
Correspondence c_{23}: the cubic Hermite interpolant has error at most $h^4 \left\|\phi^{(4)}\right\|_\infty /384$, and x^4 attains the constant. s_{23}: a smooth curve through the plotted points is a derivation (constructed for the review).

Entry 24

Obligation Rhetorical framing may appear in the exposition but is granted no evidential content.
Located result Section 6.8, where a repeated display is the arithmetic's slogan; Section 7.2, where additional printed digits need not improve a physical tolerance.
Calibration Presence; no fact extracted.

Correspondence The carrier's symbol against itself: $c_{24} \leftrightarrow s_{24}$ has the same symbol on both sides.

Entry 25

Obligation Adherence to a canonical method is not itself a result: reduce the appeal to canon to the theorem actually used.

Located result Equation (7.1), by the triangle inequality; the bound $|\widehat{q}_1 - \widehat{q}_2| \leq E_1 + E_2$ of Section 3.8.

Calibration Proved here: the bound holds whenever each reading's error is within its stated allowance.

Correspondence c_{25}: the theorem used is the triangle inequality, and its content is the displayed bound. s_{25}: the method is authoritative because it is canonical (constructed for the review).

Entry 26

Obligation The invariant must be stated as a theory with a checkable remainder that could fail, not as a hypothesis or a repeated observation.

Located result The exact remainder (3.3) of Section 3.5; the Hermite remainder of Section 3.6; the controlled nonzero residual of Section 6.5.

Calibration Proved here: the remainder is exact for the stated closure, so its size can be computed and found too large.

Correspondence c_{26}: the closure has the remainder $B(H, H)$, which could fail to be small. s_{26}: the invariant is a hypothesis confirmed by repetition (constructed for the review).

Entry 27

Obligation A proposition called true must come with the path by which it became known.

Located result Section 6.4, where the walls are retained for checking by another implementation; Section 3.8, where a repeated display is not a certificate.

Calibration Proved here: the digits hold for any implementation that rechecks the retained walls.

Correspondence c_{27}: the path is the hundred-and-eight-step ledger of sign tests and the two endpoint inequalities. s_{27}: the digits are true because the file that printed them says so (constructed for the review).

Entry 28

Obligation Report the content of the observation rather than the credential of the observer: reduce testimony to the ledger entry it records.

Located result The acceptance record at the close of Section 5.5; the record of Section 5.6.

Calibration Proved here, as a specification of the record: a number is reported only with its observable, norm, and allowances.

Correspondence c_{28}: the testimony reduces to observable, norm, allowances, number. s_{28}: the observation is the citation of a proof (constructed for the review).

Entry 29

Obligation Separate the statement that the quantity is real from the announcement that it is: the status counts only when the doubt-resolution chain ends in a recorded ledger entry.

Located result The model-discrepancy allowance ϵ_{model} of Section 3.7; the independent reference of Section 7.3.

Calibration Proved here: the status *real* requires a calibration against an independent reference, which mesh refinement does not supply; standard: the definition of calibration.

Correspondence c_{29}: the status *real* is a ledger entry recording a calibration against an independent reference, and this manuscript records none. s_{29}: the quantity is real because its residual is at machine precision (constructed for the review).

Entry 30

Obligation Every universally quantified statement must come with an admissible local instance, a witness at a point; absent one it is recorded as present and introduces no global theorem.

Located result Proposition 3.2 with its witness; Section 5.2, where no bound is claimed for every vacuum spacetime; Section 5.4, which is not a global regularity theorem; Proposition 2.2, the witness for the cosmological-constant extension of the gravity model: de Sitter space on the unimodular sector, with the constant the residue of the Bianchi integration.

Calibration Presence; no fact extracted: universal statements are located with their local witnesses and license nothing beyond them.

Correspondence The carrier's symbol against itself: $c_{30} \leftrightarrow s_{30}$ has the same symbol on both sides.

Entry 31

Obligation Logical validity may be recorded as present, with the inference rules used, but is granted no empirical content.

Located result The closing paragraph of Section 7.7, where a measured value enters only through a calibrated chain; Section 9.7, where polynomial degree supplies no regularity bound.

Calibration Presence; no fact extracted.

Correspondence The carrier's symbol against itself: $c_{31} \leftrightarrow s_{31}$ has the same symbol on both sides.

Entry 32

Obligation Replace the assertion that the procedure terminates by an exhibited finite trajectory: the intended path and the path taken, compared as data.

Located result The acceptance procedure of Section 4.8; Table 6.1; the stopping observation of Section 6.5.

Calibration Proved here: termination is licensed by the exhibited finite orbit and its acceptance tests, not by the turn budget.

Correspondence c_{32}: the submitted route is the exact orbit of Proposition 6.1, the route taken is the integer orbit of Table 6.1, and Figure 6.1 compares them turn by turn. s_{32}: the procedure terminated because its turn budget was reached (constructed for the review).

Entry 33

Obligation Show the measured quantity is invariant under re-presentation at a different level of description.

Located result Section 7.7: the corrected readings $C(n) = H(n) - B(n)$, the pass gap, and heartbeats as work units; the work ledger of Section 6.8.

Calibration Measured, as work counts recorded in the source's own runs and not rerun here; in work units, not SI time or energy.

Correspondence c_{33}: the value on file is the signed corrected reading at each depth, compared across depths through the pass gap. s_{33}: the heartbeat record is a measurement of time or energy. For time, the middle card's patter calls heartbeats "the resolution in time"; the energy reading is constructed for the review.

Entry 34

Obligation Distinguish "the computation type-checks and terminates" from "the computation establishes the claim".

Located result Section 7.4, where two implementations deliver the same integer and the truncation rests on the bracket; Section 6.8, where the bracket is the certificate; the blanks of Section 7.7.

Calibration Proved here (the certificate): the integer inequalities establish the digits independently of any build; measured, from the inspected declarations (the agreement of the two implementations).

Correspondence c_{34}: the checkable proposition is the pair $5KD \le 324SN < 5(K+1)D$ at both walls. s_{34}: the computation establishes the digits because it compiled and halted (constructed for the review).

Entry 35

Obligation Exhibit the initial datum from which the announced value was computed, and show the final number is a function of it.

Located result Equation (6.9), from the bracket $[2, 17/8]$ by a hundred and eight halvings; the branch selection of Section 6.1; the named-source requirement of Section 7.5.

Calibration Proved here: on the stated branch the reading is a function of the seed, the coefficients, and the halving count.

Correspondence c_{35}: the announced value is $\lfloor Sg_x L_{108} \rfloor = \lfloor Sg_x U_{108} \rfloor$. s_{35}: the announced value is a free-standing announcement (constructed for the review).

Entry 36

Obligation State the submitted claim once, verbatim, as the fixed object of custody; nothing in it may be amended retroactively.

Located result Section 5.6; the opening of Chapter 6; the ledger paragraph of Section 6.8.

Calibration Proved here, as the fixed record: every earlier entry answers the inputs fixed in Chapter 6, unamended.

Correspondence c_{36}: the object of custody is $r(y) = 13y^2 - 36y + 18$, the gain $324/5$, the seed $1296/10$, the scale 10^{24}, and the branch $y > 18/13$. s_{36}: the submitted claim is admitted unchanged; the review's page 36 states that "the submitted manuscript is admitted unchanged". The proposition $c_{36} \leftrightarrow s_{36}$ has the admitted claim on both sides.

8.3 The lifts

Definition 8.7 (Lift)**.** Entry $k+1$ of the typeset record carries the register of entry k lifted to a universe at least as high, and lowering it returns the register unchanged. A *lift* is that passage: it adds no datum, and it points to the step taken between the two entries, whose mathematics is performed at the pointer. Thirty-five lifts separate the thirty-six entries. The geometric card performs lifts 1 to 20, from the fact to the coupling $e^2/4\pi = 1/137.0112905\ldots$; this section performs lifts 21 to 35. What a lift is, a level set once and carried, is fixed in *The Infinite Loop Considered Harmless*, section "The search outside the record".

Throughout, $D_\mu = \partial_\mu + ieA_\mu$, the convention of Section 2.4.

Lift 21: the coupling to a phase-carrying state. Let the state carry a phase at every point, $\psi(x) \mapsto e^{i\chi(x)}\psi(x)$ with χ arbitrary. The naive comparison is not covariant:

$$e^{i\chi(y)}\psi(y) - e^{i\chi(x)}\psi(x) \neq e^{i\chi(x)}(\psi(y) - \psi(x))$$

unless $\chi(y) = \chi(x)$. Two points cannot be compared without a rule. *Proved here*, by substitution.

Lift 22: the rule to transport. Let $W(y, x)$ be a phase that transforms as $W(y, x) \mapsto e^{i\chi(y)}W(y, x)e^{-i\chi(x)}$. Then $W(y, x)\psi(x)$ transforms as $\psi(y)$, and $\psi(y) - W(y, x)\psi(x)$ is covariant. The rule of comparison is transport. *Proved here.*

Lift 23: transport to the connection. For a step ε along μ, write $W(x + \varepsilon e_\mu, x) = 1 - ie\varepsilon A_\mu(x) + O(\varepsilon^2)$. The transformation law of W gives $A_\mu \mapsto A_\mu - e^{-1}\partial_\mu\chi$, and then

$$D_\mu(e^{i\chi}\psi) = e^{i\chi}(\partial_\mu\psi + i\partial_\mu\chi\,\psi + ieA_\mu\psi - i\partial_\mu\chi\,\psi) = e^{i\chi}D_\mu\psi.$$

Transport taken small is the connection A, the potential of light. *Proved here.*

Lift 24: the connection to curvature. Transport around the boundary of a small square of side ε in the (μ, ν) plane fails to come home by

$$1 - ie\varepsilon^2 F_{\mu\nu} + O(\varepsilon^3), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

and F is unchanged by $A \mapsto A - e^{-1}\,\mathrm{d}\chi$, since mixed partials of χ commute. The failure is the field. *Proved here*, from Stokes's theorem; *standard*.

Lift 25: curvature to the commutator.

$$D_\mu D_\nu\psi = \partial_\mu\partial_\nu\psi + ie(\partial_\mu A_\nu)\psi + ieA_\nu\partial_\mu\psi + ieA_\mu\partial_\nu\psi - e^2A_\mu A_\nu\psi.$$

Every term but the second is symmetric in (μ, ν), so

$$[D_\mu, D_\nu]\psi = ie\,F_{\mu\nu}\,\psi.$$

The commutator of the two derivatives is the charge times the field: the commutator of representation. *Proved here.*

Lift 26: the commutator to the ingredients. Added from outside: a background with a light cone, the metric $\eta = \mathrm{diag}(1, -1, -1, -1)$; the spinor representation, $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$; the kinetic terms; and a mass m. Each is named where it enters. Together they are a bent card in the sense of *The Infinite Loop Considered Harmless*: adjoined, checked where used, not computed. *Standard.*

Lift 27: the ingredients to the action.

$$S[\psi, \bar\psi, A] = \int \left(\bar\psi\,(i\gamma^\mu D_\mu - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}\right)\mathrm{d}^4x,$$

the action (2.10), minimally coupled: the only appearance of A outside F is inside D. A definition; *standard*.

Lift 28: the action to the stationary system. Varying $\bar{\psi}$ gives $(i\gamma^\mu D_\mu - m)\psi = 0$, and varying A_ν gives $\partial_\mu F^{\mu\nu} = e\bar{\psi}\gamma^\nu\psi$, equations (2.11) and (2.12). The divergence of the second vanishes identically, so $\partial_\nu(\bar{\psi}\gamma^\nu\psi) = 0$: the current is conserved. Its $\nu = 0$ component is Gauss's constraint $\partial_i F^{i0} = e\psi^\dagger\psi$. *Proved here*, Section 2.4.

Lift 29: the system to the knower's choice. Under $\psi \mapsto e^{i\chi}\psi$, $\bar{\psi} \mapsto \bar{\psi}e^{-i\chi}$ and $F \mapsto F$, the term $\bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}$ is invariant as well, as is $(F_{\mu\nu}F^{\mu\nu})^2$. Covariance does not choose the action; the selection of the minimal one is by degree and renormalizability. *Proved here*, by substitution; the selection is *standard*.

Lift 30: the choice to the photon. Without a source and in a Lorenz gauge, $\Box A^\nu = 0$. A plane wave $\varepsilon^\nu e^{-ik\cdot x}$ solves it exactly when $k \cdot k = 0$; the gauge condition requires $k \cdot \varepsilon = 0$, and $\varepsilon \mapsto \varepsilon + ck$ is a change of representative. Two polarizations remain. The field propagates. *Proved here*, by substitution; calling it the photon is *proposed*.

Lift 31: the photon to the electron. The electron is the Dirac field of mass m and charge e, with $e^2/4\pi = 1/137.0112905\ldots$, solving the system of Lift 28. Every bilinear $\bar{\psi}\Gamma\psi$ is even in ψ, so ψ and $-\psi$ give one current, one density and one field: the sign is a coin the record cannot see. *Proved here*; the record's blindness to it is *recorded*.

Lift 32: the electron to quantum electrodynamics. Added from outside: quantization, by canonical anticommutation of ψ and commutation of A on a Fock space, and renormalization of the charge and mass. Neither is in the deck. A bent card. *Standard*.

Lift 33: quantum electrodynamics to the laboratory. The laboratory's value is $\alpha^{-1} = 137.035\,999\,177(21)$ [31], a relative standard uncertainty of 1.5×10^{-10}. *Standard*, a measurement in the literature.

Lift 34: the laboratory to the miss.

$$\frac{137.035\,999\,177 - 137.011\,290\,5\ldots}{137.035\,999\,177} = 1.80 \times 10^{-4},$$

about 1.2×10^6 times the laboratory's standard uncertainty. No reader, convention or declaration absorbs it; the quantitative identification of Lift 20 is excluded. *Proved here*, by arithmetic on a cited value.

Lift 35: the miss to the press. The press check lowers the register thirty-five times, reads it once, and returns the triple of Section 8.4: the pair (A, B) not slipped, the pair (B, C) slipped by one unit of 10^{-24}, Proposition 8.8. *Proved here.*

8.4 The press check

The press check of Definition 8.6 uses three readings that Chapter 6 has already produced.

- Reading A is Newton's integer at its fifth turn, computed by the integer update (6.6) and recorded in Table 6.1. Section 6.4 states that it equals the certified integer, and the ledger of Section 6.8 checks the equality.
- Reading B is the hundred-and-eight-halving bisection truncation of equation (6.9), obtained without either update. It is the register of Definition 8.1.
- Reading C is Halley's stationary integer, reached at its third turn by the integer update (6.7). Section 6.3 and Section 6.4 state that it exceeds the certified integer by one unit in the last place, the ledger of Section 6.8 checks the stationary integers, and Section 6.3 explains the unit: that update takes a ceiling where the certificate takes a truncation.

Let K denote the certified integer, and for each reading X let ϕ_X be the proposition $X = K$, decided by comparing two integers.

Proposition 8.8 (The two verdicts)**.** *The pair (A, B) is not slipped: $\neg(\phi_A \leftrightarrow \phi_B)$ is false. The pair (B, C) is slipped: $\neg(\phi_B \leftrightarrow \phi_C)$ is true.*

Proof. ϕ_B holds by the definition of K. ϕ_A holds by the integer equality stated after equation (6.9): Newton's turn-five integer equals the certified integer. Hence $\phi_A \leftrightarrow \phi_B$ holds and its negation is false. ϕ_C fails, since $C = K + 1$ by the same passage; hence $\phi_B \leftrightarrow \phi_C$ fails and its negation holds. Both decisions are integer comparisons already stated in Section 6.4 and checked by the ledger of Section 6.8; no further computation and no appeal to choice is involved. □

The slip between B and C is one unit of 10^{-24}, explained entirely by the position of the floor operation (Section 6.8): not a disagreement about the root, but the smallest difference the stored scale can represent, and decidable.

The demand-level reading. A second reading, separate from the press check and outside its rule of reading the register once, reads the calibration column of the trace as a whole: not the propositions $c_k \leftrightarrow s_k$, which remain undecided, but the calibration each entry assigns to its located answer. It records whether the reading s_k handed over by name is established anywhere in this manuscript. At the following rungs it is not: s_k says strictly more than the reading derived here, and the entry names the hypothesis it lacks:

- entry 11, a common physical result across the four models, which Theorem 2.4 does not identify, and in its sharpest form the identification of electrodynamics with the derivative of the modified field equations, which Section 2.6 records as proposed and does not derive;
- entry 10, the relativistic regularity thesis, which the Burgers calculation of Section 9.7 shows a speed limit alone does not supply;
- entry 33 (and entry 14), heartbeats read as a time or an energy, which Section 7.7 declines;
- entry 8, the transition-matrix weights of Section 7.6: the source derives the table from one arm rule and the closed-word counts agree with closed walks, and the reason for the arm rule is the remaining obligation.

For each listed rung the manuscript proves c_k, and s_k is not a consequence of c_k under the stated hypotheses. The demand-level reading records s_k as *not established here* at these five entries; it

refutes no s_k, each of which remains a proposed reading with a named obligation. These five are also the entries whose s_k carries, in whole or in part, a named source (Definition 8.4); the only other such entry is entry 36, whose source is the review's own admission of the manuscript.

Value and velocity. The value is the register, 137.011290548979457087737300, read back unchanged from Definition 8.1. The velocity is the counter at the top, 36. The press check compares two pairs of readings and reports one decided verdict for each, a triple (slipped, value, velocity):

$$(A, B): \ (\text{false},\ 137.011290548979457087737300,\ 36),$$
$$(B, C): \ (\text{true},\ 137.011290548979457087737300,\ 36).$$

The demand-level reading is reported beside the triples and is a component of neither: s_k not established here at entries 11, 10, 33, 14, and 8.

Inhabitation is not acceptance. Thirty-six entries, each carrying a located answer, show that the record is well formed. A record whose readings all agreed would be exactly as well formed as this one, whose two numerical readings of the register slip at one unit. Acceptance is a reading of the trace at the top, not a clause of the record, and the record is not accepted by being complete.

8.5 The hypothesis the apparatus tests

The review and the typeset record are the apparatus for testing one hypothesis, stated in the margins of the middle card's eighth act and restated here. It is calibrated *proposed*.

> Names are required by a proof and never produced by it, one name per rung. A representation has an optimal partition, below which further symbols are required by the proof itself. There is a representational noise floor separating a falsifiable repeat of an experiment from an unfalsifiable exact replay of its record, so that a falsifiable hypothesis requires noise in order not to return exactly to an earlier observable state.

This manuscript's proposed realization of the floor is the one unit in the last place by which Reading C exceeds Reading B. Two procedures with the same stored scale, the same coefficients, and the same root produce integers that differ by exactly the smallest representable amount, and Section 6.8 explains the difference completely by the position of the floor. Below that unit the two procedures cannot be told apart by their printed output; at it they can, and the distinction is a decidable integer fact rather than a fluctuation. A repeat of the calculation by another implementation, Section 7.4, is falsifiable at that unit; an exact replay of the same ledger is not. The identification of that unit with the floor of the hypothesis is calibrated proposed. The companion monograph proposes a realization of the same floor in its own geometry.

The counter is reported as a velocity for a reason that Section 3.4 makes available. A constraint can be enforced exactly by a multiplier or relaxed by a penalty parameter. The record enforces one entry per page exactly and pays one increment per entry; the count at the top is the cost of carrying the constraint exactly rather than relaxing it. This reading of the counter as a stiffness is calibrated proposed.

Chapter 9

Dissipation, entropy, and a retained count

Universe numbers only increase.

Chapter 1 derived the clamped beam's stationary shape: with zero load and constant rigidity the balance is $q^{(4)} = 0$, and Proposition 1.3 makes the cubic q_* the unique minimum-energy deflection for its clamp data. A stationary equation fixes a shape. It supplies no clock, no drag law, no temperature, and therefore no direction in which the shape is approached. Adding them to the same beam, with the conventions and declared parameters of Chapter 1, gives a calibrated chain in which each link is a dependency of the next: mechanical assumptions, power balance, decay estimate, discrete dissipation, reservoir entropy, measurement uncertainty, and a retained threshold count. A mechanical cycle can close its displacement reading while retaining a new event; comparing that count with entropy requires the clock, energy, temperature, and calibration that the force balance and a thermal measurement equation supply.

9.1 The physical beam and its dissipative equation

Let $x \in [0, L]$ measure position along the slender beam of Chapter 1, and let $q(x, t)$ be its small transverse displacement. Both x and q are measured in metres. For constant Young's modulus E and second moment of cross-sectional area I_b, the flexural rigidity $EI = EI_b$ has units $\mathrm{N\,m^2}$. The small-slope Euler–Bernoulli bending energy is

$$U[q] = \frac{EI}{2} \int_0^L |q_{xx}|^2 \, \mathrm{d}x. \tag{9.1}$$

Since curvature q_{xx} has units $\mathrm{m^{-1}}$, this expression has units $\mathrm{N\,m} = \mathrm{J}$. The engineering assumptions are those of Chapter 1: a linear elastic bending law and negligible shear deformation in the selected regime [41]. They are part of the material model.

Suppose distributed viscous drag opposes the transverse velocity with force per length $-\zeta q_t$, where $\zeta > 0$ is constant and has units $\mathrm{N\,s\,m^{-2}}$. In the overdamped approximation, force balance is

$$\zeta q_t + EI q_{xxxx} = f(x, t), \qquad [f] = \mathrm{N\,m^{-1}}. \tag{9.2}$$

Each term is force per length. Inertial bending dynamics would also contain μq_{tt}, where μ is mass per length. Omitting that term requires the time scale of interest to make inertia small relative to drag; it is an additional approximation, not a consequence of stationarity. The constant mobility of a dimensionless bending formulation corresponds here to EI/ζ, with units $\mathrm{m}^4\,\mathrm{s}^{-1}$.

Take $f = 0$ and fixed endpoint displacements and slopes. Their equilibrium q_* is the cubic of Proposition 1.3. The deviation $v = q - q_*$ satisfies homogeneous clamps $v = v_x = 0$ at both ends, and $\zeta v_t + EIv_{xxxx} = 0$. The equilibrium equation $q_*^{(4)} = 0$ is unchanged; the evolution equation adds the rule by which that equilibrium is approached.

9.2 Power, boundary work, and the decay estimate

The drag law makes the energy of the previous section a function of time. For a sufficiently smooth displacement, differentiating the bending energy and integrating twice by parts gives the full mechanical power balance

$$\begin{aligned}\dot{U} &= \left[EIq_{xx}q_{xt} - EIq_{xxx}q_t\right]_0^L + \int_0^L EIq_{xxxx}q_t \,\mathrm{d}x \\ &= P_{\text{supports}} + \int_0^L fq_t \,\mathrm{d}x - \zeta\int_0^L |q_t|^2 \,\mathrm{d}x. \end{aligned} \tag{9.3}$$

The first bracket is power delivered through the ends with the displayed sign convention. Its moment multiplies endpoint angular velocity, and its shear force multiplies endpoint transverse velocity. Fixed clamps make both endpoint velocities zero. Applied distributed loads supply the second work term. The last term is nonnegative viscous loss, in watts.

With fixed clamps and $f = 0$, equation (9.3) becomes

$$\dot{U} = -\zeta\int_0^L |q_t|^2 \,\mathrm{d}x = -\frac{(EI)^2}{\zeta}\int_0^L |q_{xxxx}|^2 \,\mathrm{d}x \le 0. \tag{9.4}$$

For a nonzero time-independent conservative load, the decreasing quantity is instead the total potential $\Pi = U - \int_0^L fq \,\mathrm{d}x$ of Chapter 1. A time-dependent load adds $-\int f_t q \,\mathrm{d}x$ to $\dot{\Pi}$. These distinctions matter when interpreting a driven cycle: a decrease in elastic energy and an increase in dissipated heat need not be equal if the supports or actuator supply work.

For the unforced relaxation define the physical energy gap $G(t) = U[q(t)] - U[q_*] = (EI/2)\,\|v_{xx}\|_2^2$, which is the gap of Proposition 1.3 evaluated along the motion. Homogeneous clamps give the elementary estimates

$$\|v\|_2 \le (L/\pi)^2\,\|v_{xx}\|_2\,, \qquad \|v_{xxxx}\|_2 \ge (\pi/L)^2\,\|v_{xx}\|_2\,.$$

The first follows by applying the interval Poincaré inequality to v and then to v_x, whose mean is zero. For the second, integrate to obtain $\|v_{xx}\|_2^2 = \int v_{xxxx}v \,\mathrm{d}x$ and apply Cauchy–Schwarz. Substitution in (9.4) proves

$$G'(t) \le -\kappa G(t), \qquad G(t) \le G(0)e^{-\kappa t}, \qquad \kappa = \frac{2EI}{\zeta}\left(\frac{\pi}{L}\right)^4. \tag{9.5}$$

The exponent is dimensionless because κ has units s^{-1}. This gives a directly usable settling-time estimate: $G(t) \le \varepsilon_G$ is guaranteed once $t \ge \kappa^{-1}\log(G(0)/\varepsilon_G)$ when $0 < \varepsilon_G < G(0)$.

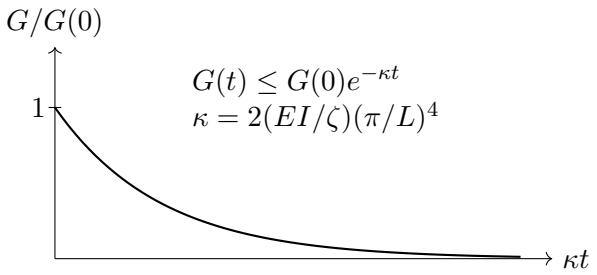


Figure 9.1: The energy gap decreases under a specified drag law and fixed clamps. The exponential estimate sets a physical settling-time tolerance. The plotted curve is the normalized exponential upper bound.

9.3 A mode calculation with declared units

The estimate (9.5) bounds the decay rate from below; a single mode shows the rate attained. The clamped beam has bending modes φ_j with positive stiffness eigenvalues λ_j. Finite integrated initial curvature permits the energy-based evolution [12]. In modal coordinates,

$$v(x,t) = \sum_j a_j \exp\left(-\frac{EI\lambda_j}{\zeta}t\right)\varphi_j(x).$$

The modal solution smooths the displacement at positive times and justifies the energy identities for finite initial bending energy.

For an explicit single-mode example, choose dimensionless φ_1 with $\int_0^L \varphi_1^2\,\mathrm{d}x = L$. Solving $\varphi_1^{(4)} = \lambda_1\varphi_1$ with the four clamp conditions gives $\lambda_1 = \beta_1^4/L^4$, where the first positive bending root satisfies $\cosh\beta_1\cos\beta_1 = 1$ and $\beta_1 \approx 4.73004$. The displacement $v = a\varphi_1 e^{-t/\tau}$ then has

$$\tau = \frac{\zeta L^4}{EI\beta_1^4}, \qquad G(t) = \frac{EI\,a^2\beta_1^4}{2L^3}e^{-2t/\tau}.$$

The displacement amplitude decays with time constant τ, while its quadratic bending energy decays twice as fast in the exponent.

As an illustrative set of model parameters, take $L = 1\,\mathrm{m}$ and $EI = 1\,\mathrm{N\,m^2}$, as in the declared clamped-displacement example of Chapter 1, together with $\zeta = 100\,\mathrm{N\,s\,m^{-2}}$ and $a = 1\,\mathrm{mm}$. Then $\tau \approx 0.200\,\mathrm{s}$ and $G(0) \approx 2.50\times10^{-4}\,\mathrm{J}$. These numbers specify a model calculation, not a characterization of an unmeasured material. If mass is retained, the modal inertial-to-drag ratio on this relaxation scale is $\mu EI\lambda_1/\zeta^2$; the overdamped approximation requires that ratio to be small.

9.4 What a time discretization dissipates

A computed relaxation inherits the continuous energy law only if its time step does; the discrete balance is therefore the next link. In a conforming beam approximation, such as the cubic elements of Section 1.6, let $\mathbf{v}^n$ contain the displacement coefficients at time $n\Delta t$, with homogeneous clamps enforced. Let $\mathsf{M}_{ij} = \int N_i N_j\,\mathrm{d}x$ be its mass matrix and $\mathsf{K}_{ij} = \int N_i'' N_j''\,\mathrm{d}x$ its curvature matrix. Backward Euler gives

$$\frac{\zeta}{\Delta t}\mathsf{M}(\mathbf{v}^{n+1} - \mathbf{v}^n) + EI\,\mathsf{K}\mathbf{v}^{n+1} = 0.$$

Set $\mathbf{d} = \mathbf{v}^{n+1} - \mathbf{v}^n$ and $G_h^n = (EI/2)(\mathbf{v}^n)^T \mathsf{K} \mathbf{v}^n$. Multiplication by $\mathbf{d}^T$ yields the exact identity

$$G_h^{n+1} - G_h^n = -\frac{\zeta}{\Delta t} \mathbf{d}^T \mathsf{M} \mathbf{d} - \frac{EI}{2} \mathbf{d}^T \mathsf{K} \mathbf{d} \leq 0.$$

Thus the fully solved step cannot increase this discrete bending energy for any positive time step. The first loss term is the time-step integral of the resolved viscous power. The second is additional numerical damping from backward Euler.

Assigning all discrete energy loss to the physical reservoir would also assign the algorithm's extra damping to it. Spatial and time refinement must control this error before the result supplies a physical entropy reading. Inexact linear solves contribute residual work to the same energy budget.

9.5 Heat transfer and an entropy calibration

The power balance identifies where the lost energy goes only once a thermal premise is added. Assume the viscous loss is delivered as heat to a large reservoir at fixed absolute temperature $T > 0$, with no applied work in the selected relaxation. The reservoir then receives the power $-\dot{U}$, and

$$\dot{S}_{\text{reservoir}} = -\frac{\dot{U}}{T} = \frac{\zeta}{T} \int_0^L |q_t|^2 \, \mathrm{d}x = \frac{(EI)^2}{\zeta T} \int_0^L |q_{xxxx}|^2 \, \mathrm{d}x \geq 0. \tag{9.6}$$

Entropy is measured in $\mathrm{J\,K^{-1}}$ and its rate in $\mathrm{W\,K^{-1}}$. This equation calculates the reservoir entropy change. A statement about total entropy production must also account for the beam's thermal state, the drag medium, and any other energy stores. An isothermal free-energy model can provide that complete account. The second law permits zero total production in the reversible limit [15, 29].

The entropy transferred during a time interval is

$$\Delta S = \frac{\zeta}{T} \int_{t_0}^{t_1} \int_0^L |q_t|^2 \, \mathrm{d}x \, \mathrm{d}t = \frac{U[q(t_0)] - U[q(t_1)]}{T}.$$

At equilibrium, $q^{(4)} = 0$, the drag power vanishes, and the cubic minimizes bending energy for its fixed clamps. Thus the variational cubic of Chapter 1 is the equilibrium of this explicit dissipative realization. The energy balance, drag coefficient, and reservoir specify the direction of relaxation; the spatial stationary equation supplies the endpoint.

At $T = 300\,\mathrm{K}$, the single-mode example can transfer at most about $8.34 \times 10^{-7}\,\mathrm{J\,K^{-1}}$ to the reservoir. This finite capacity is a useful engineering limit: the initially stored elastic energy cannot supply indefinite heat production. A subsequent cycle requires an actuator or another energy source to restore the displacement, and its work and heat must enter the complete balance.

9.6 A threshold reader and its retained count

The last link converts the calibrated entropy into a count. Let $\sigma > 0$ be the entropy increment assigned to one registered pulse. For the unforced beam define

$$\Sigma(t) = \frac{U[q(0)] - U[q(t)]}{T}, \qquad g_\sigma(t) = \lfloor \Sigma(t)/\sigma \rfloor .$$

Equation (9.6) makes Σ nondecreasing, so this ideal calibrated count is nondecreasing and obeys

$$0 \le \Sigma(t) - \sigma g_\sigma(t) < \sigma, \qquad g_\sigma(t) \le \left\lfloor \frac{G(0)}{T\sigma} \right\rfloor .$$

The first is the quantization tolerance; the second is the finite energy limit. For $\sigma = 10^{-8}\,\mathrm{J\,K^{-1}}$, the illustrative relaxation can register at most 83 such increments.

A physical implementation measures energy loss and temperature with finite errors. Suppose its energy-loss estimate has error at most ε_D, its temperature estimate error at most ε_T, both true and reported temperatures exceed $T_{\min} > 0$, and the true energy loss is at most $D_{\max}$. Direct subtraction of the two ratios gives

$$\left|\widehat{\Sigma} - \Sigma\right| \le \frac{\varepsilon_D}{T_{\min}} + \frac{D_{\max}\varepsilon_T}{T_{\min}^2}.$$

This bound must be combined with threshold quantization and sampling delay. A noisy unguarded display can cross the same threshold in both directions. A retained pulse counter instead latches each accepted crossing; triggering only after a lower error bound passes the next threshold avoids certifying heat that was not established by the measurement. The resulting latency is part of its response specification.

The counter of the typeset record has a simpler update law:

$$g_{n+1} = g_n + r_n, \qquad r_n \ge 0.$$

In the record of Chapter 8 the increment is exactly one per entry, applied as a successor to the count below. It is an accumulating event counter, and its retained count is a count of entries. It is not a count of the universe levels of the source's logic, where lifting a type to a higher level takes the maximum of the two levels, so that nesting alone need not produce a successor. Matching physical increments to $\Delta S/\sigma$ requires the calibration above.

A known deterministic symbol can increase record length without increasing Shannon uncertainty. Reversible computation and irreversible erasure also have different physical requirements [3, 26]. Event count, message uncertainty, and thermodynamic entropy therefore retain their separate units and measurement procedures. The beam construction supplies one explicit calibration to dissipated energy, with a finite passive energy supply.

9.7 The physical thesis and its controlling estimate

The action closure, scalar digit certificate, and beam dissipation estimate hold under their stated operating assumptions. A stronger statement about fluid singularities requires the relevant material equations and a bound on the physical quantity whose divergence is excluded. A bound on velocity is not a bound on its spatial gradient, strain rate, vorticity, or the frequency content of a field.

A short exact calculation isolates this distinction. Inviscid Burgers flow obeys $u_t + uu_x = 0$. Use the dimensional initial profile $u_0(x) = -a\sin(kx)$, where a is a speed and k an inverse length. Along the characteristics $x = \xi + tu_0(\xi)$ the speed remains $u_0(\xi)$, but

$$u_x(0,t) = -\frac{ak}{1-akt}.$$

Choosing $0 < a < c$ keeps the speed below c before the first crossing, while its gradient diverges at $t = 1/(ak)$. This calculation concerns an inviscid one-dimensional model. Its purpose is to identify which estimate a finite propagation-speed argument lacks. Viscous fluid regularity must be assessed using its own dissipation, constraints, boundary conditions, and derivative estimates.

A relativistic or microscopic material description can change the validity range of an incompressible continuum approximation. The established engineering result of this chapter is the calibration of an accumulating count to entropy transfer in a beam relaxation whose stationary shape is a variational cubic.

Universe numbers only increase.

A proof that the Navier–Stokes equations blow up is a statement about the equations. They are not relativistic: their viscous term diffuses, and signals in them propagate without limit. Real fluids are relativistic and molecular. The author's thesis: relativistic material effects bound the gradient before the continuum description fails, so no real fluid follows the Navier–Stokes model to a singularity. It requires that material description, its quantitative fluid limit, and the controlling gradient bound, which polynomial degree does not supply and which is not derived here; the mathematics of the equations themselves remains a separate question, open in the unforced case. *Proposed.*

Chapter 10

The shell, the counter, and the two holographies

Chapter 9 ended with a count calibrated to entropy transfer by a displayed thermal chain: drag law, power balance, reservoir temperature, and threshold. The counter of the typeset record in Chapter 8 increases by the same update law, one per entry, and carries no such chain. This chapter investigates the corresponding question for time: what the counter, the record's two directions, and the companion monograph's enclosing shell become when they are read against a spacetime, and which calibration that reading would need.

10.1 The correspondence investigated

The companion monograph realizes its enclosing whole as the boundary of a convex four-dimensional polytope, normalized to a round three-sphere. Its observation of that shell is an orthogonal projection to a three-ball, with a silhouette two-sphere, two sheets, and a sheet sign; away from two core circles, its paired Hopf observation retains one antipodal bit. The typeset record of Chapter 8 has a strictly increasing counter and two orders: the review's descent from page thirty-six to page one, and the typeset ascent from entry one to entry thirty-six. These are the objects on the record side.

On the spacetime side the objects are those of de Sitter space of radius ℓ in its global slicing (10.1): a constant-time slice, global time, an observer's static patch and its cosmological horizon, the pair of antipodal patches and the elliptic quotient that identifies them, the conformal boundary at future infinity, and, for comparison, the timelike boundary and radial coordinate of anti-de Sitter space.

Definition 10.1 (The record–spacetime correspondence)**.** The *record–spacetime correspondence* $\mathcal{C}$ is the assignment of Table 10.1, row by row, of an object on the record side to an object on the spacetime side. Two statements, one on each side, *correspond under* $\mathcal{C}$ when each is obtained from the other by replacing every object in it by its image in that table.

The spacetime facts below are calibrated *Standard*, each cited; the record side contributes one identity, *Proved here*; and Proposition 2.2 of Chapter 2 proves that de Sitter space is a solution of this manuscript's gravity model.

Record side	Spacetime side	Item
normalized shell of the companion monograph	global slice of (10.1)	(i)
counter of the typeset record	global time on the stack of slices	(ii)
projection to the three-ball; silhouette two-sphere	static patch; cosmological horizon	(iii)
two sheets of the projection; sheet sign	two antipodal static patches; the datum separating them	(iii)
retained antipodal bit of the paired Hopf observation	the datum separating de Sitter space from its elliptic quotient	(iii)
review's descent, page 36 to page 1	anti-de Sitter reconstruction from boundary data	(iv)
typeset ascent, entry 1 to entry 36	de Sitter evolution to future infinity	(iv)
final datum supplied to every stage from outside	boundary data at future infinity	(v)
index n of the positional reading $2^n - 1$	global time, or anti-de Sitter radial depth	Rem. 10.2

Table 10.1: The record–spacetime correspondence $\mathcal{C}$. The last column names the item of Section 10.4 in which each row is proposed. Every row is *Proposed*.

10.2 Standard facts

De Sitter space of radius ℓ is the one-sheeted hyperboloid $-X_0^2+X_1^2+\cdots+X_4^2 = \ell^2$ in five-dimensional Minkowski space. In its global slicing the metric is

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + \ell^2 \cosh^2(t/\ell)\ \mathrm{d}\Omega_3^2, \tag{10.1}$$

so each constant-t slice is a round three-sphere of radius $\ell\cosh(t/\ell)$. The spacetime is maximally symmetric; an observer's static patch carries the time-independent metric $-(1 - r^2/\ell^2)\,\mathrm{d}t_s^2 + (1 - r^2/\ell^2)^{-1}\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_2^2$, bounded by a cosmological horizon at $r = \ell$ with entropy $A/4G$ and temperature $1/(2\pi\ell)$ [16]. The conformal boundary of de Sitter space is spacelike, at future and past infinity; no observer inside the spacetime has access to it. Identifying antipodal points of the hyperboloid gives the elliptic interpretation of de Sitter space, in which the two static patches of antipodal observers are one [33].

Anti-de Sitter space is the other maximally symmetric solution, with a timelike conformal boundary at spatial infinity. The holographic correspondence places a unitary conformal field theory on that boundary and reconstructs the bulk from boundary data [27, 18, 42]; local bulk fields are reconstructed from boundary operators by an explicit smearing [20]. In Euclidean signature the bulk is hyperbolic space and the reconstruction from boundary data is a Dirichlet problem for an elliptic operator: well posed, with the data supplied on a wall that is present at every time. The radial coordinate is a scale, and the warp factor grows exponentially in it.

A de Sitter version of the correspondence places a Euclidean theory on future infinity [35]. It is obtained from the anti-de Sitter dictionary by the continuation $\ell_{\mathrm{AdS}} \to i\ell_{\mathrm{dS}}$, which exchanges the radial coordinate for time and the boundary at spatial infinity for the boundary at future infinity [28]. In that construction a scalar of mass m in $(d+1)$-dimensional de Sitter space of radius ℓ corresponds to a boundary operator of weight $h_\pm = \frac{d}{2} \pm \sqrt{d^2/4 - m^2\ell^2}$, complex when $m^2\ell^2 > d^2/4$; for such field content the boundary theory is not unitary in the usual sense, and the construction supplies no positive inner product. The de Sitter observables it computes are not accessible to any single observer [43, 1].

10.3 What is exact

Remark 10.2 (One exponential, two readings). A positional reading in base two of n identical unit cells is $2^n - 1$. The cell read never changes; the count of readings does. Interpreted as a scale factor indexed by n, the reading grows as $e^{n \log 2}$:

$$2^n - 1 = e^{n \log 2} - 1.$$

The two readings of its index are the last row of Table 10.1. The positional reading also presumes the n cells already counted: over the field with one element that presumption costs nothing, and in a written record it costs one step per cell, which is what the counter of Chapter 8 tallies (Remark 8.3).

10.4 The proposed correspondence

The exponential of Remark 10.2 is read two ways, according to the image of its index in the last row of Table 10.1. If n is assigned to global time, the exponential is the growth of (10.1) at late global time; if n is assigned to radial depth, it is the growth of the anti-de Sitter warp factor. In both readings one function of n is produced by repetition of an unchanged datum, and which spacetime it describes is fixed by the choice of image for the index, time or depth.

Remark 10.3 (Boundary-value and initial-value readings). A record admitted at its top and reconstructed downward, one stage at a time with the datum above each stage already in hand, is a boundary-value problem. Its reconstruction is elliptic in character and terminates: every stage has a condition to satisfy. A record built upward toward a final page that is read once, after every stage below it exists, is an initial-value problem. Its reconstruction is hyperbolic in character and, from inside the record, it does not terminate until the last stage is reached; a reader placed before that stage receives no signal from it. These two descriptions use the same stages in opposite order. The proposal is that the two records of Chapter 8, the review's descent from page thirty-six to page one and the typeset ascent from entry one to entry thirty-six, are of these two kinds respectively.

(i) The normalized shell of the companion monograph is a global slice of (10.1). The apparent expansion of the universe is then the growth of the slicing index, not a change in the hyperboloid.

(ii) The strictly increasing counter of the typeset record is global time on the stack of slices. It increases because the record is extended by one stage per entry, and for no other reason.

(iii) The orthogonal projection of the shell to a three-ball, with its silhouette two-sphere, is the view from a static patch, with the silhouette as the cosmological horizon. The two sheets of the projection are the two antipodal patches, and the sheet sign is the datum that separates them. The elliptic quotient identifies antipodal points of the slice, which the companion monograph realizes, away from its two core circles, as the fiber of its paired Hopf observation; the retained antipodal bit, not the sheet sign, is the datum that separates de Sitter space from its elliptic quotient.

(iv) The review's descent is the anti-de Sitter reading of the record: boundary data given, bulk reconstructed inward, a computation that terminates by search. The typeset ascent is the de Sitter reading: a page at the end of the record, reachable only once the whole record exists, read once.

(v) The continuation that exchanges the two readings costs what the continuation $\ell \to i\ell$ costs. In the middle card's record the ascent resolves only when the final datum is supplied to every stage from outside the construction; in the spacetime the boundary theory at future infinity is non-unitary and its data are unavailable to any observer. The two statements correspond under $\mathcal{C}$, through the row that assigns the final datum supplied from outside to the boundary data at future infinity: on each side the de Sitter boundary condition is not computed from inside, it is chosen.

The cosmological term this reading needs is supplied in Section 2.1. Proposition 2.2 shows that on the unimodular sector the constant is the residue of the Bianchi integration, fixed by data and carried unchanged, so the slice of item (i) is the shell of a solution of this manuscript's gravity model with the degree count unchanged. What the reading still owes is item (ii): the map from the record's steps to the global time of the slicing is proposed, not derived. It is the analogue, for time, of the thermal chain of Chapter 9, which converts heat into the count g_σ; no corresponding chain converts entries into global time, and the record of Chapter 8 carries item (ii) as *Proposed*.

Appendix A

Sources and the review correspondence

A.1 Scope and snapshot

This appendix is the only part of the manuscript that names the source construction's files and declarations. The body chapters are self-contained mathematics and refer to the construction only as "the source" or "the middle card". Paths in this appendix are relative to `CARD_2/Measurement/`. The working tree was inspected on 22 October 2026. Declaration names are given in place of line numbers, because declarations survive edits that move lines; a declaration named here is one that was present and kernel-checked in the inspected tree, and this appendix reports no rebuild. The margins quoted in the record chapter were read again on 3 November 2026 (Section A.7). Each identifier appears below beside the section that describes it in prose.

A.2 The scalar reading (Chapter 6)

The residual coefficients, scale, seed, and conversion of Section 6.1 are read from `Episode08.lean` and `Episode12.lean`.

- `Episode08.lean`: the closure-unit coefficients `closureA`, `closureB`, `closureC`, which give $r(y) = 13y^2 - 36y + 18$; `the_areal_velocity`, the pair $(324, 5)$ behind the gain $g_x = 324/5$; `theSeedPair`, the strip-seeded start of the source's own crank, distinct from the jar seed used in Chapter 6; `theBase`, `thePlaces`, and `theScale`, which give $S = 10^{24}$; `theReadingPair`, `theReadingScaled`, and `theReading`, the source's printed readout; the monomial tapes `the_leading_tape`, `the_middle_tape`, `the_constant_tape` and their lengths `quadA`, `quadB`, `quadC`, from which the output-coordinate coefficients 325, 58320, 1889568 are read. The lengths are forced by the definitions; the source carries no separate receipt for them.
- `Episode12.lean`: `jarLo`, `jarHi`, `jarSeed`, the jar and its lower wall $x_0 = 1296/10$; `jarReadingScaled` and `jarReading`, the integer Newton update (6.6); `lagrangeReadingScaled` and `lagrangeReading`, the integer Halley update (6.7), which the source names after Lagrange and which Section 6.2 identifies as Halley's; `theLowWall` and `theHighWall`, the initial bracket of Section 6.4; `theBisectionLedger` with its receipts `the_bisection_ledger_starts`, `the_bisection_ledger_turns`, `the_bisection_ledger_ends`, `the_bisection_ledger_length`; `theNewtonLedger` with `the_newton_ledger_starts`, `the_newton_ledger_turns`, `the_`

`newton_ledger_ends`, `the_newton_ledger_length`; `theBracket` and `theNewtonPoint`, the final bracket and the final Newton integer; `the_crank_is_its_ledger`, `the_subtraction_is_valid_at_the_reading`, and `the_reading_is_above_the_vertex`. The turn budget of Section 6.3 is the source's declared input `the_orbital_data_points` of `Episode08.lean`.

A.3 Count channels, the erased channel, and the transition matrix (Chapter 7)

Section 7.4 describes, in `Episode12.lean`, `ComputerProgram.ofTape` (the second stored form), `read_ofTape` (its reading preserves the count), `the_other_count` and `the_other_pair` with the receipt `the_other_machine_agrees` (another implementation of the reading pair delivers the same pair), `the_relettered_pair` with the receipt `relettering_is_free` (cell contents overwritten, count retained, same pair), `the_same_natural_both_ways`, and `any_count_of_the_lengths` (an alternative counter accepted under the premise that it agrees with tape length on every tape).

Section 7.5 describes, in `Episode08.lean`, the relation `ReadingOrder` ($ad \leq cb$), the receipt `any_pair_is_its_inferred_integer`, and `theReadingCertified`, the integer retained beside the quotient; and, in `Episode12.lean`, the theorem `the_box_has_one_point`, that the quotient has exactly one point.

Section 7.6 describes, in `Episode12.lean`, `the_click_table`, the 2×2 matrix M, with `the_trace` and `the_determinant` and the receipts `the_trace_is_the_middle_coefficient` and `the_determinant_is_the_outer_coefficients`, which identify the trace with `closureB` and the determinant with `closureA` $\cdot$ `closureC`. The table is read off the arms by `the_click_table_is_derived`, through `the_arm_weight`: an arm carries the coefficient tape exactly when its stored fact can click. Its closed words are closed walks, `the_closed_words_are_closed_walks`. The source's margin leaves the reason for the arm rule as the remaining obligation.

A.4 Work measurements (Section 7.7)

In `Episode12.lean`: the command `monte_walk`, which visits the same nesting depths ascending and descending; `corrected` and `signedSteps`, the signed subtraction of a blank; `theSizesUp`, `theSizesDown`, `theStepsUp`, `theStepsDown`; and `thePassGap`, the pass gap at matching depths. The second fixture is the command `variation_walk` over `theNewtonRung`, `gateauxAt`, and `frechetAt`, the pure and branched chains of `Variation` records whose readers `Variation.newton` and `Variation.krylov` are declared in `Episode08.lean`; its corrected readings are `theSpineN`, `theSpineK`, `theBranchN`, `theBranchK`, with `theNameGap` and `theConvectiveGap`. The two readings of one derivative in Section 2.6 are the two constructors `Variation.the_gateaux` and `Variation.the_frechet` of `Episode08.lean`, instantiated there as `the_bent_card` and `the_fabric_of_spacetime`. Heartbeats are the elaborator's own work count, obtained through `withHeartbeats`; the manuscript does not rerun these measurements.

A.5 The census and register names behind the count channels

Section 7.4 says that the source's selected tape readers count facts. In `Episode08.lean` the elaborator `episode7_elaboration_of` mints four census tapes, one cell per distinct constant reached by its target: `the_causal_prediction`, `the_data_table`, `the_differential_form`, and `the_fit_curve`. Their lengths are the counts `the_orbital_data_points` and `the_dimensions_of_universe`, with `thirteen`, `theCoupling`, `theRadius`, and `thePlaces` built from the walk `from_the_inferred_end` over `the_steps_of_the_argument`; the list-level receipts are `the_causal_prediction_is_the_stokes`, `the_data_table_is_the_gauss`, and `the_differential_form_is_the_coefficient`, and the length-level receipts are the per-step `*_length` theorems. The source's own register is the Horner fold `the_register` at base two, with the receipts `the_other_side_of_the_slip`, `the_once_around_machine_is_all_ones`, `the_calculus_machine_is_all_ones`, `the_ladder_machine_is_all_ones`, and `the_four_monomials`. That register is a reading of a census tape; it is distinct from the register of the typeset record (next section) and the two are not conflated in the body. The walk readers that the count channel factors through are the `ListFACT` projections of `Episode08.lean` (`ThePreprint.the_model`, `Measurement.the_reading`, and their companions), and the source's crank is the section-polymorphic `Crank` instantiated once at natural numbers with `List.length`.

A.6 The review: page to reply map

The review is `Episode10.lean`: the class `REVIEWED`, whose single field `rebuttal` is a preprint record, and thirty-six instances, one per page from 36 down to 1, each consing one methodology cell onto the rebuttal of the page above. Reply k of Section 8.2 answers page k. The table gives, for each page, the instance name and the rung class of the ladder whose reading the page extracts; pages 24, 30, and 31 are presence paragraphs that extract nothing, and page 36 admits the manuscript unchanged.

Page	Instance	Rung class
36	`REVIEWER_2_ASKS_WHERE_THE_RECORD_BEGINS`	seam; manuscript admitted unchanged
35	`REVIEWER_2_ASKS_WHAT_THE_ANSWER_WAS_ANSWERING`	DISTINGUISHABLE with an explicit ledger fact
34	`REVIEWER_2_ASKS_IF_COMPILING_IS_EVIDENCE_OR_A_FILENAME`	COMPILED
33	`REVIEWER_2_ASKS_HOW_THE_NEEDLE_SURVIVED_THE_RETELLING`	MEASURED
32	`REVIEWER_2_ASKS_IF_HALTING_WAS_CHECKED_OR_DECLARED`	HALTED
31	`REVIEWER_2_CATCHES_LOGIC_POSING_AS_AUTHORITY`	LOGICAL (presence)
30	`REVIEWER_2_REFUSES_A_UNIVERSAL_CLAIM_WITHOUT_A_LOCAL_WITNESS`	UNIVERSAL with LOCAL (presence)
29	`REVIEWER_2_DISTINGUISHES_REALITY_FROM_ITS_PRESS_RELEASE`	REAL
28	`REVIEWER_2_ASKS_WHAT_WAS_ACTUALLY_WITNESSED`	WITNESSED
27	`REVIEWER_2_OBJECTS_TO_TRUTH_CITED_BY_NAME_ONLY`	TRUTH

Page	Instance	Rung class
26	REVIEWER_2_REQUIRES_THE_INVARIANT_TO_CARRY_A_REMAINDER	SCIENTIFIC
25	REVIEWER_2_ASKS_IF_DEVOTION_TO_METHOD_IS_A_METHOD	ACOLYTE
24	REVIEWER_2_SEPARATES_PERSUASION_FROM_EVIDENCE	PROPAGANDA (presence)
23	REVIEWER_2_SUSPECTS_THE_SMOOTH_STORY_IS_INTERPOLATION_IN_A_LAB_COAT	BULLSHIT
22	REVIEWER_2_ASKS_IF_THE_CONSTRAINT_WAS_ENFORCED_OR_PENALIZED	OF
21	REVIEWER_2_NOTES_A_LOAD_MUST_LAND_IN_A_BASIS	LOAD
20	REVIEWER_2_REQUIRES_SCALE_TO_SHOW_ITS_MULTIPLICATION	SCALED
19	REVIEWER_2_ASKS_WHERE_MAGNITUDE_BECOMES_ADDITIVE	MAGNITUDE
18	REVIEWER_2_DISTINGUISHES_A_VALUE_FROM_ITS_PROCESS	VALUE
17	REVIEWER_2_ASKS_IF_EXECUTION_LEFT_AN_OPCODE_RECORD	EXECUTED
16	REVIEWER_2_ASKS_WHICH_SOURCE_FACT_SURVIVED_THE_CHANNEL	SOURCE
15	REVIEWER_2_MAKES_THE_JOKE_PAY_ITS_EVIDENTIARY_BILL	GUNGAN
14	REVIEWER_2_ASKS_IF_THE_CLOCK_IS_OBSERVABLE_OR_ASSERTED	MEASURABLE
13	REVIEWER_2_ASKS_WHERE_PRESENCE_ACCUMULATES	PRESENT
12	REVIEWER_2_ASKS_IF_OBSERVATION_HAS_A_RECORDED_DIMENSION	OBSERVED
11	REVIEWER_2_ASKS_WHAT_MAKES_COMPARISON_MORE_THAN_PARALLEL_NAMING	COMPARABLE
10	REVIEWER_2_REQUIRES_THE_PHYSICAL_CLAIM_TO_SHOW_ITS_BASE_FACT	PHYSICAL
9	REVIEWER_2_ASKS_IF_REPRESENTATION_HAS_A_CARRIER_OR_A_NOTATION	REPRESENTABLE
8	REVIEWER_2_ASKS_WHERE_THE_NUMBER_ACTUALLY_COUNTS	NUMERIC
7	REVIEWER_2_ASKS_IF_REPETITION_REPEATS_THE_EXPERIMENT_OR_THE_SENTENCE	REPEATABLE
6	REVIEWER_2_ASKS_WHERE_THE_MISSING_EXHIBIT_WENT	BINARY (the dropped exhibit re-supplied)
5	REVIEWER_2_ASKS_IF_RESIDUE_IS_A_RECORD_OR_A_LEFTOVER_STORY	RESIDUE
4	REVIEWER_2_ASKS_IF_ENCODING_KEEPS_A_READ_PATH_TO_THE_MARK	ENCODED
3	REVIEWER_2_ASKS_IF_COUNTING_COUNTS_A_CARRIER_OR_NAMES_A_CLASS	COUNTABLE
2	REVIEWER_2_ASKS_WHO_ADMITTED_THE_MARK	ADMISSIBLE
1	REVIEWER_2_ASKS_IF_IT_ALL_RESTS_ON_A_MARK_OR_ON_CONFIDENCE	DISTINGUISHABLE

The rung classes are declared in `Episode01.lean` through `Episode05.lean`; the facts the pages

extract are values of `FACT`, a proposition packaged with its decision procedure. The obligation field of each entry restates, without identifiers, the demand recorded in the documentation of that page's instance. The located result and the calibration are this volume's own. The reading c_k is constructed by this volume; the provenance of each reading s_k is given in the next section.

A.7 Provenance of the readings handed over by name

The thirty-six rebuttal paragraphs of `Episode10.lean` each thank the reviewer for a demand and name the gap the page closes; of them, only page 36's states a reading taken up as an s_k. To locate the remaining statements, the inspected tree was searched on 3 November 2026 for the key terms of each s_k (among them confidence, speed of light, heartbeat, time, regularity, derivative, compiled, canonical, penalty), the margins that matched were read in full beside their declarations, and a margin was quoted only where it states the reading itself; the body quotes it without identifiers as "the middle card's patter". Where no margin states the reading, the body labels s_k a comparison statement constructed for the review. Two s_k are located in this volume's own body rather than in the source.

Entry	Source of s_k	Location
8	middle card's patter	`Episode12.lean`, the margin headed "What was not yet derived" above `the_arm_weight` and `the_click_table_is_derived`
10	this volume	the closing thesis, Section 9.7
11	this volume (sharpest form only)	Remark *Electrodynamics as the derivative of the modified field equations*, Section 2.6
14, 33	middle card's patter (time reading only)	`Episode12.lean`, opening margins, the paragraph headed "The one dimensionless measurement" (the ratio it describes is `theCourant`)
36	the review	`Episode10.lean`, documentation of `REVIEWER_2_ASKS_WHERE_THE_RECORD_BEGINS`
all others	comparison statement constructed for the review	—

A.8 The typeset record: field map

The typeset record is `Episode11.lean`: the class `TYPESET` with three fields, and the structure `PressVerdict` with three. The correspondence with Section 8.1 is:

Source field	Section 8.1	Reported at the top as
`TYPESET.register_value`	the register, Definition 8.1	`PressVerdict.value`
`TYPESET.universe_id`	the counter, Definition 8.2	`PressVerdict.velocity`
`TYPESET.the_page`	the trace, Definition 8.5	read once by `pressCheck`
`PressVerdict.slipped`	the verdict, Definition 8.6	`PressVerdict.slipped`

The rung-one instance is `TYPESET_Episode1_FACT`; rungs two to thirty-six are `TYPESET_page_2` through `TYPESET_page_36`. Each rung above the first wraps the register below in one more `ULift`,

applies one successor `.one` to the counter below, and prepends one results cell quoting the head of that page's review record together with the proposition `heads.1.truth` ↔ `heads.2.truth`, undecided. The function `pressCheck` reads the head of the top page once and forms `slipped` as the negation of that biconditional with a decision procedure built from the two readings' own; it unwraps the register to `value` and reports the counter as `velocity`. The record's counter is the constructor `.one` of `Number` (`Episode01.lean`) applied to the counter below, starting from `.zero`; the body's remark that lifting takes the maximum of two levels refers to the source's own margin in `Episode12.lean` on `ULift`.

A.9 The floor built by hand and the press check

`Episode12.lean` supplies the rung-one instances that search cannot reach: `theManuscript` (three cells decoded by `decodeBits`), `reviewedByHand`, `typesetByHand` with register and counter `cookPage`, the instance `typesetBase`, and the parametric instance `reviewedAny`, which returns the same manuscript at every page number and is declared in the source as a gimmick that routes around the review's descent. The top is `theTop`; its trace is `theTopTrace`. The trace's step predicate is `ThePreprint.stepOK`; `traceSat` and the theorem `trace_correspondence` state that a trace satisfies its own step formula exactly when it is well stepped, with the instances `cook_levin_the_description`, `cook_levin_on_the_mechanism`, and, for the swapped trace `theSwappedTrace` decoded from `theSwapBits`, `cook_levin_on_the_meter`. The source does not state the press verdict as a named theorem; Section 8.4 computes its own verdict from readings this manuscript proves and does not cite the source for it.

A.10 The hypothesis

Section 8.5 restates the hypothesis recorded in the margins of `Episode08.lean` beside `Closure.infer`, `natural_zero`, `zeroth_rational`, and `the_register`. The source's own realization of the noise floor is the sign change of the Borwein slack `the_slack` between `once_around_holds`, `it_holds_through_thirteen`, and `twice_around_slips`, with `the_two_slips_meet`; the manuscript's realization in Section 8.5 is its own and is calibrated proposed.

A.11 Reproduction

The document-local scripts and their ledgers are

- `scalar_reading.py`, ledger `scalar-reading.json`;
- `engineering_examples.py`, ledger `engineering-examples.json`.

Each script has a `-check` mode that regenerates its products and compares them byte for byte. The scalar ledger records hashes of the inspected source files beside its rational walls and iterates. The scripts reproduce the printed arithmetic; they do not build the source construction, run a field solver, or measure an apparatus.

A.12 Stanzas and results

The review's thirty-six pages and the typeset record's thirty-six rungs name the same ladder of classes in the same order, so one concordance serves both. Each row gives the rung, the act its typeset verse names, the demand its review page makes, the kind of item that answers it in this volume, and the located items themselves. "Named result" means a theorem, proposition, corollary, definition, or table; "worked calculation" means a displayed equation inside a section; "presence" marks the three pages that extract no fact. The full entries, with their calibrations and correspondences, are in Chapter 8.

Rung	Verse (act)	Page (demand)	Kind	Located items
1	the origin: the carrier's number, zero, the first page	asks if it all rests on a mark or on confidence	section	Section 6.1; Section 6.4; Section 3.7; Section 7.1
2	admits the mark	asks who admitted the mark	section	Section 6.1; Section 4.3; Section 4.8
3	counts the number	asks if counting counts a carrier or names a class	section	Section 7.4; Section 7.5
4	encodes the glyph	asks if encoding keeps a read path to the mark	section	Section 6.4; Section 7.5
5	residues the code	asks if residue is a record or a leftover story	named result	Proposition 2.2; eq. (6.8); Section 3.8; Section 6.5
6	binaries the value	asks where the missing exhibit went	section	Section 7.5; Section 6.4
7	repeats the process	asks if repetition repeats the experiment or the sentence	section	Section 7.2; Section 7.3; Section 4.5; Section 7.7
8	numbers the value	asks where the number actually counts	section	Section 7.4; Section 6.4; Section 7.6
9	represents the code	asks if representation has a carrier or a notation	named result	Table 6.1; Section 6.3; Section 7.4
10	physicals the method	requires the physical claim to show its base fact	worked calculation	eq. (9.6); Section 3.7; Section 9.7
11	compares the process	asks what makes comparison more than parallel naming	named result	Theorem 2.4; Proposition 2.5; eq. (7.1); Section 2.5; Section 2.6; Section 7.6
12	observes the mark	asks if observation has a recorded dimension	named result	Proposition 2.1; Proposition 2.3; Section 2.2; Chapter 2
13	presents the process	asks where presence accumulates	worked calculation	eq. (9.6); Section 5.4
14	measures the step	asks if the clock is observable or asserted	section	Section 7.6; Section 7.3
15	gungans the glyph	makes the joke pay its evidentiary bill	section	Section 6.1; Section 5.3
16	sources the code	asks which source fact survived the channel	section	Section 7.4; Section 7.5
17	executes the code	asks if execution left an opcode record	section	Section 6.3; Section 6.8

Rung	Verse (act)	Page (demand)	Kind	Located items
18	values the function	distinguishes a value from its process	section	Section 6.5; Section 6.8
19	magnitudes the value	asks where magnitude becomes additive	named result	Table 7.1; eq. (3.6); eq. (3.7)
20	scales the value	requires scale to show its multiplication	worked calculation	eq. (6.2); Section 6.1; Section 3.2
21	loads the process	notes a load must land in a basis	section	Section 4.7; Section 1.6; Section 4.4
22	finite-elements the model	asks if the constraint was enforced or penalized	named result	Proposition 2.3; eq. (3.2); Section 3.4; Section 2.3
23	bullshits the model	suspects the smooth story is interpolation in a lab coat	worked calculation	eq. (3.5); Section 3.6; Section 1.6
24	propagandizes the model	separates persuasion from evidence	presence	Section 6.8; Section 7.2
25	acolytes the method	asks if devotion to method is a method	worked calculation	eq. (7.1); Section 3.8
26	sciences the method	requires the invariant to carry a remainder	worked calculation	eq. (3.3); Section 3.5; Section 3.6; Section 6.5
27	truths the code	objects to truth cited by name only	section	Section 6.4; Section 3.8
28	witnesses the mark	asks what was actually witnessed	section	Section 5.5; Section 5.6
29	reals the model	distinguishes reality from its press release	section	Section 3.7; Section 7.3
30	locals the variable	refuses a universal claim without a local witness	presence	Proposition 3.2; Proposition 2.2; Section 5.2; Section 5.4
31	universals the function	catches logic posing as authority	presence	Section 7.7; Section 9.7
32	logics the code	asks if halting was checked or declared	named result	Table 6.1; Proposition 6.1; Figure 6.1; Section 4.8
33	halts the method	asks how the needle survived the retelling	section	Section 7.7; Section 6.8
34	measures the step	asks if compiling is evidence or a filename	section	Section 7.4; Section 6.8; Section 7.7
35	compiles the tape	asks what the answer was answering	worked calculation	eq. (6.9); Section 6.4; Section 6.1; Section 7.5
36	outputs the value	asks where the record begins	named result	Table 6.1; Definition 8.1; Proposition 2.2; Theorem 2.4

Appendix B

The reader's experiment

Every result calibrated *proved here* in this volume is a written record: hypotheses, a sequence of displayed steps, and a conclusion. The same record, carried out, is a run of an apparatus, and the account of that run belongs to the theory of the device as much as the result does. Three objects are distinguished throughout this appendix.

Definition B.1 (Procedure, exhibited record, run)**.** The *procedure* is the finite list of operations a result prescribes, in order, with the observation each operation produces. The *exhibited record* is what the volume prints of the lines the procedure determines: here, an initial excerpt and the final certificate. A *run* is one physical execution of the procedure by an apparatus, which writes one line per observation and changes the state of the apparatus one step at a time, taking the time it takes.

The run has the parts of the measurement chain of Section 7.1. The *setup* is the declared hypotheses and definitions. The *apparatus* is whoever or whatever performs the operations: a reader with paper, an abacus, a computer. The *law* is the rule each operation obeys: an inference rule, or integer arithmetic. Each *observation* is one operation's outcome seen and written: an equality checked, a sign read, an inequality between two integers decided. The *reading* is the conclusion, reported once the last observation is in. Conclusions of two kinds result, and they are kept apart. The observations are facts about the run: this apparatus, on this occasion, saw these signs. The reading, when every observation holds, is a statement about the mathematical object, and its warrant is the procedure, which fixes every line in advance; the run is the occasion on which a particular reader sees that warrant discharged.

The run given here is on the result on which the register of Chapter 8 rests.

B.1 Setup: the certified reading

The result is the certificate of Section 6.4: the twenty-four-place truncation (6.9). Its declared inputs are the residual $r(y) = 13y^2 - 36y + 18$, the walls $L_0 = 2$ and $U_0 = 17/8$, the budget of 108 halvings, the stored scale $S = 10^{24}$, and the gain $324/5$. The apparatus needs only to add, multiply, and compare integers. The law is integer arithmetic, together with the rule of admission of Section 6.1: a value of the unknown enters the record only as a seed, an image under a displayed map, or a wall of a sign-straddling bracket.

B.2 Procedure

The operations performed are integer additions, multiplications and comparisons, and the writing of one line per observation. In order:

1. Observe the initial straddle. With $y = N/D$, the sign of $r(y)$ is the sign of the integer $13N^2 - 36ND + 18D^2$. At $L_0 = 2/1$ it is -2; at $U_0 = 17/8$ it is 13. Observe also $26L_0 - 36 = 16 > 0$, so r increases across the bracket and encloses exactly one root.

2. For $k = 1, \ldots, 108$: form the exact midpoint N/D of the current walls, compute the integer $13N^2 - 36ND + 18D^2$, read its sign, and replace the wall that has that sign. Write one line: k, the midpoint, the sign.

3. At the end, observe the straddle again at both final walls: the integer is negative at the lower wall and positive at the upper wall.

4. Observe the reporting cell at each final wall N/D, with $K = 137011290548979457087737300$: the two integer inequalities $5KD \le 324SN < 5(K+1)D$.

5. Report the reading: the digits of K, with the point after the third digit.

B.3 The exhibited record

The procedure determines every line of the record. The first eight are the following; each entry is an integer computed by the apparatus and a sign read from it.

k	N	D	$13N^2 - 36ND + 18D^2$	wall replaced
1	33	16	-243	lower
2	67	32	-395	lower
3	135	64	-387	lower
4	271	128	877	upper
5	541	256	-1355	lower
6	1083	512	4293	upper
7	2165	1024	-2267	lower
8	4331	2048	29797	upper

Table B.1: The first eight lines of the record. After line eight the walls are 2165/1024 and 4331/2048; one hundred further lines follow by the same rule.

The final certificate is the following. After line 108 the walls have denominators 2^{111} and 2^{110}, both straddle observations of step 3 hold, and both reporting inequalities of step 4 hold with the same K. The record therefore contains $3 + 108 + 2 + 4 = 117$ observations, each a sign or an integer comparison, and one reading:

$$137.011290548979457087737300.$$

The record grows by one line per observation; no earlier line is re-read or revised. The count of lines is the counter of the run, and the digits are its value.

Three conclusions follow, of different kinds. That the procedure writes these lines and this reading is *proved here*: the procedure is deterministic and its every line is fixed by the declared inputs. That a given run wrote them is an observation of that run, made by the apparatus that performed it; the reader who carries out the procedure has observed it. That the root α of r in $[L_0, U_0]$ lies between the final walls, and therefore in the reporting cell of K, is a statement about the mathematical object; it follows from the straddle observations and the monotonicity observed at step 1, and it holds whether or not any particular run is performed.

B.4 What the run detects

The error model is a single wrong-wall update, with every later computation and the final checks performed correctly. Under it, the wrong wall moves, the bracket stops straddling the root, and the remaining halvings converge on a point that is not α; the final straddle observations of step 3 fail at one wall, and the run reports no reading. A misread reporting inequality at step 4 is outside that model: the walls do not detect it, and its detection requires a stated redundancy, an independent re-evaluation of that inequality. The check is not that the digits look right; it is that every observation holds. A reading is reported only from a run whose observations all hold, which is the reporting rule of the reading note: a certified reading with its receipt, or no reading at all.

A second run, by another reader or another machine, is a repeat. Its reading either equals the first or it does not, and the comparison is the press check of Section 8.4: with the first reading certified, slipped exactly when the second differs from it. Two runs agree here because both observed the same integers, not because one copied the other's output. An exact replay of the first run's record, line by line, observes nothing; it is the distinction between a falsifiable repeat and an unfalsifiable replay stated in Section 8.5.

B.5 A bench counterpart

The same protocol has a bench form for the clamped-beam results of Chapter 1. Clamp a slender strip at both ends with zero slope, impose an end displacement d by a micrometer stage, and read the transverse reaction on a force gauge. Proposition 1.3 and the clamped-displacement calculation that follows it give the reaction magnitude $12EI\,|d|\,/L^3$, linear in d. Measure EI independently, for instance by a three-point bending test, so that the comparison is not circular. Step d through several values in both directions. Each force reading is an observation; the law is the Euler–Bernoulli model with its stated assumptions; the reading is the slope of force against displacement, compared with $12EI/L^3$ through the two-instrument budget (7.1). A disagreement larger than that budget rejects at least one premise: the clamps, the small-slope assumption, the independent EI, or the gauge calibration. No record is exhibited and no measured value is reported; the run belongs to whoever performs it. Its observations, unlike the integer run's, are not fixed in advance by the procedure, so a reading it produced would be calibrated *measured* in the sense of observed in a run.

B.6 Every proof as a run

For any entry calibrated *proved here*, the experiment is read off its record in the same way: the setup is its hypotheses, the procedure is its displayed steps in order, each observation is one displayed

identity or inequality checked by the reader, and the reading is its conclusion, reported only if every observation holds. Where the conclusion concerns a continuum, as in Proposition 2.5, the run checks the finite record of the argument, one line at a time; the continuum is what the conclusion is read as being about, and that reading is the reader's. The record supplies the procedure. The experience, every observation in it, happens in the one who runs it.

Bibliography

[1] Dionysios Anninos. De Sitter musings. *International Journal of Modern Physics A*, 27(13):1230013, 2012.

[2] Richard Barrett, Michael Berry, Tony F. Chan, James Demmel, June M. Donato, Jack Dongarra, Victor Eijkhout, Roldan Pozo, Charles Romine, and Henk van der Vorst. *Templates for the Solution of Linear Systems: Building Blocks for Iterative Methods*. SIAM, Philadelphia, 2 edition, 1994.

[3] Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.

[4] Z. Bern, J. J. M. Carrasco, and H. Johansson. New relations for gauge-theory amplitudes. *Physical Review D*, 78:085011, 2008.

[5] Z. Bern, J. J. M. Carrasco, and H. Johansson. Perturbative quantum gravity as a double copy of gauge theory. *Physical Review Letters*, 105:061602, 2010.

[6] Zvi Bern, John Joseph Carrasco, Marco Chiodaroli, Henrik Johansson, and Radu Roiban. The duality between color and kinematics and its applications. arXiv:1909.01358, 2019.

[7] Zvi Bern, John Joseph M. Carrasco, Wei-Ming Chen, Henrik Johansson, and Radu Roiban. Gravity amplitudes as generalized double copies of gauge-theory amplitudes. *Physical Review Letters*, 118:181602, 2017.

[8] Daniele Boffi, Franco Brezzi, and Michel Fortin. *Mixed Finite Element Methods and Applications*, volume 44 of *Springer Series in Computational Mathematics*. Springer, 2013.

[9] Carl de Boor. *A Practical Guide to Splines*. Springer, New York, 1978.

[10] Carl de Boor and Robert E. Lynch. On splines and their minimum properties. *Journal of Mathematics and Mechanics*, 15(6):953–969, 1966.

[11] Stanley Deser. Self-interaction and gauge invariance. *General Relativity and Gravitation*, 1:9–18, 1970.

[12] Lawrence C. Evans. *Partial Differential Equations*. American Mathematical Society, Providence, 2 edition, 2010.

[13] X. Fan, T. G. Myers, B. A. D. Sukra, and G. Gabrielse. Measurement of the electron magnetic moment. *Physical Review Letters*, 130(7):071801, 2023.

[14] Charles L. Fefferman. Existence and smoothness of the Navier–Stokes equation. In James Carlson, Arthur Jaffe, and Andrew Wiles, editors, *The Millennium Prize Problems*, pages 57–67. Clay Mathematics Institute and American Mathematical Society, 2006.

[15] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics*, volume 1–3. Addison-Wesley, Reading, MA, 1965.

[16] G. W. Gibbons and S. W. Hawking. Cosmological event horizons, thermodynamics, and particle creation. *Physical Review D*, 15(10):2738–2751, 1977.

[17] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.

[18] S. S. Gubser, I. R. Klebanov, and A. M. Polyakov. Gauge theory correlators from non-critical string theory. *Physics Letters B*, 428:105–114, 1998.

[19] Edmond Halley. Methodus nova accurata et facilis inveniendi radices aequationum quarum-cumque generaliter, sine praevia reductione. *Philosophical Transactions*, 18(210):136–148, 1694.

[20] Alex Hamilton, Daniel Kabat, Gilad Lifschytz, and David A. Lowe. Local bulk operators in AdS/CFT: a boundary view of horizons and locality. *Physical Review D*, 73:086003, 2006.

[21] Marc Henneaux and Claudio Teitelboim. The cosmological constant and general covariance. *Physics Letters B*, 222(2):195–199, 1989.

[22] Joint Committee for Guides in Metrology. Evaluation of measurement data: Guide to the expression of uncertainty in measurement. Technical Report JCGM 100:2008, Bureau International des Poids et Mesures, 2008.

[23] Joint Committee for Guides in Metrology. International vocabulary of metrology: Basic and general concepts and associated terms (VIM). Technical Report JCGM 200:2012, Bureau International des Poids et Mesures, 2012. Third edition, 2008 version with minor corrections.

[24] Joint Committee for Guides in Metrology. Guide to the expression of uncertainty in measurement: Part 6: Developing and using measurement models. Technical Report JCGM GUM-6:2020, Bureau International des Poids et Mesures, 2020.

[25] Dana A. Knoll and David E. Keyes. Jacobian-free Newton–Krylov methods: a survey of approaches and applications. *Journal of Computational Physics*, 193(2):357–397, 2004.

[26] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

[27] Juan Maldacena. The large N limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2:231–252, 1998.

[28] Juan Maldacena. Non-Gaussian features of primordial fluctuations in single field inflationary models. *Journal of High Energy Physics*, 2003(05):013, 2003.

[29] Massachusetts Institute of Technology. Concept and statements of the second law. Unified Engineering thermodynamics notes, Section 5.1. Accessed 21 October 2026.

[30] Norman G. Meyers and James Serrin. $H = W$. *Proceedings of the National Academy of Sciences of the United States of America*, 51(6):1055–1056, 1964.

[31] Peter J. Mohr, David B. Newell, Barry N. Taylor, and Eite Tiesinga. CODATA recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics*, 97(2):025002, 2025.

[32] Kambiz Salari and Patrick Knupp. Code verification by the method of manufactured solutions. Technical Report SAND2000-1444, Sandia National Laboratories, June 2000.

[33] Erwin Schrödinger. *Expanding Universes*. Cambridge University Press, Cambridge, 1956.

[34] Gilbert Strang and George J. Fix. *An Analysis of the Finite Element Method*. Prentice-Hall, Englewood Cliffs, NJ, 1973.

[35] Andrew Strominger. The dS/CFT correspondence. *Journal of High Energy Physics*, 2001(10):034, 2001.

[36] David Tong. Quantum mechanics. University of Cambridge lecture notes. Author-hosted course notes; accessed 21 October 2026.

[37] David Tong. Quantum field theory. University of Cambridge Part III lecture notes, 2006. Chapter 6: Quantum Electrodynamics.

[38] Lloyd N. Trefethen and David Bau, III. *Numerical Linear Algebra*. SIAM, Philadelphia, 1997.

[39] W. G. Unruh. Unimodular theory of canonical quantum gravity. *Physical Review D*, 40(4):1048–1052, 1989.

[40] Steven Weinberg. The cosmological constant problem. *Reviews of Modern Physics*, 61(1):1–23, 1989.

[41] Tomasz Wierzbicki. Structural mechanics: Beam deflections and energy methods. MIT OpenCourseWare, course 2.080J, Lectures 5 and 8, 2013. Fall 2013; lecture notes.

[42] Edward Witten. Anti de Sitter space and holography. *Advances in Theoretical and Mathematical Physics*, 2:253–291, 1998.

[43] Edward Witten. Quantum gravity in de Sitter space. Strings 2001 lecture, arXiv:hep-th/0106109, 2001.